

# FORGE: CHEMISTRY-INFORMED GENERATIVE DESIGN OF IONIZABLE LIPIDS WITH DISCRETE FLOW MATCHING

**Anonymous authors**

Paper under double-blind review

## ABSTRACT

Lipid nanoparticles (LNPs) are a central platform for RNA delivery across vaccines, protein replacement, and genome editing, yet ionizable lipid design remains challenging. Small structural changes can produce large, context-dependent shifts in delivery, while experimental data are sparse. Existing generative approaches remain limited and largely operate over predefined building-block spaces, leaving open-ended molecular generation for ionizable lipids largely unexplored. We introduce **FORGE** (**Flow-matched Open-ended Reaction-grounded Generation and Exploration**), a chemistry-informed discrete flow-matching framework for *de novo* molecular design of modular molecular families, instantiated here on ionizable lipids assembled by the Ugi three-component reaction. FORGE embeds known reaction chemistry directly into the learned flow by assigning each generated molecular position to the amine-, aldehyde-, or isocyanide-derived region specified by the Ugi reaction and using those roles to condition denoising, while the model generates the detailed atom-, bond-, head-, linker-, and tail-level structure. We demonstrate that correct role assignments reduce denoising loss, especially at highly corrupted states, and that contextual flow learning recovers lipid-native molecular organization not reproduced by a role-marginal generator with the same chemical constraints and role-specific source distributions. FORGE generates novel ionizable-lipid chemistry, with 90% of generated lipids not seen during structural training. Because biological measurements cover only part of this expanded space, we tilt sampling toward regions supported by measured data, increasing supported yield by 2.23-fold. FORGE then identifies synthesis routes for selected lipids, tracing noncommercial components to commercially available starting materials through forward-verified chemistry. **[Prospective experimental results.]**

## 1 INTRODUCTION

Lipid nanoparticles (LNPs) have made it possible to deliver nucleic-acid therapeutics *in vivo*, from RNAi and mRNA vaccines to *in vivo* gene editing (Adams et al., 2018; Polack et al., 2020; Baden et al., 2021; Gillmore et al., 2021), with the ionizable lipid playing a central role in the resulting particle’s delivery performance (Cullis & Hope, 2017; Hou et al., 2021). Designing these molecules at scale remains difficult. Ionizable lipids combine a polar, ionizable head with hydrophobic regions whose architecture varies through chain length, branching, asymmetry, unsaturation, linker chemistry, and degradable functionality (Semple et al., 2010; Jayaraman et al., 2012; Whitehead et al., 2014; Hajj et al., 2019; Sabnis et al., 2018; Lee et al., 2021). Small local changes along these axes can substantially alter delivery, and their effects depend on the surrounding molecular context rather than following monotonic design rules (Cheng et al., 2020; Han et al., 2021; Hatit et al., 2023). The challenge is therefore not only to construct a chemically valid lipid, but to connect small structural changes to meaningful functional consequences.

How a generative model represents a molecule determines how much of that chemistry it has to learn. Many *de novo* graph, string, diffusion, and flow models treat a molecule as a collection of atoms and bonds, leaving domain-specific organization to be inferred from data (Austin et al., 2021; Vignac et al., 2023; Lipman et al., 2023). Fragment-level models place chemical structure directly into

the generative representation (Lee et al., 2026), and combinatorial or synthesis-grounded discovery defines molecular choices through reactions, fragments, or building-block inventories (Bradshaw et al., 2020; Gao et al., 2022; Cretu et al., 2025; Koziarski et al., 2024; Luo et al., 2024). The latter yields more experimentally interpretable candidates, but bounds structural diversity: heads, linkers, and tails are drawn from a predefined vocabulary. This motivates a broader question:

*Can known chemistry organize an open-ended generative model without reducing molecular design to a finite set of components or to independently generated parts?*

We introduce **FORGE** (Flow-matched Open-ended Reaction-grounded Generation and Exploration), a discrete flow-matching framework for *de novo* molecular design that leverages known modular chemistry. We instantiate it on ionizable lipids assembled through the Ugi three-component reaction (Ugi-3CR), which partitions every product into amine-, aldehyde-, and isocyanide-derived regions of known chemical origin. FORGE conditions the flow on those regions throughout denoising, while the structures occupying the regions are generated rather than selected from a predefined component vocabulary. One distribution covers the whole molecule: FORGE neither generates three reactants independently and combines them afterwards, nor receives named precursors, component identities, or a stored fragment catalogue. A shared contextual discrete flow instead generates the complete molecular realization at atom-and-bond resolution, recovering catalogue- and fragment-constrained generation as the special case in which the structure occupying each region is selected from a finite vocabulary rather than generated.

Because FORGE is not limited to known building blocks, it can invent molecules for which we lack biological evidence and readily available starting materials. We therefore add two safeguards: the activity predictor ranks only molecules near measured examples, and the synthesis layer checks whether the required components can be made.

We make three main contributions.

- **Chemistry-informed discrete flows.** A discrete flow in which each atom’s precursor region conditions denoising while the structure occupying it is generated, with matched ablations showing that correct roles reduce denoising loss, that the gain is 7.7–11.3 times larger at high corruption, that a frequency-matched permutation recovers none of it, and that contextual learning reproduces lipid-native structure a chemistry-constrained marginal baseline does not.
- **Discovery under asymmetric supervision.** A separation between the space FORGE can generate and the space where measured data support ranking, with sampling concentrated in the latter: 2.23-fold higher supported yield from the same trained generator, without retraining or predictor-guided transitions.
- **Synthesis grounding.** A layer that earns experimental accessibility instead of inheriting it, projecting each generated lipid to its exact components and reducing every component to purchasable material through forward-verified transformations, which resolves for 95.0% of assessed designs.

Across matched semantic ablations, distributional evaluation against held-out lipids, evidence-aware deployment, and prospective synthesis, we establish FORGE as a single framework in which known reaction chemistry organizes open-ended molecular generation. We first formalize the generation problem and the limitation of independent role generation, then introduce the chemistry-informed discrete flow that addresses it, and finally separate structural generation from biological ranking and experimental actionability.

## 2 BACKGROUND AND PRELIMINARIES

**Discrete flow matching** Let  $x = (x^1, \dots, x^d)$  take values in a the space  $\mathcal{S} = \mathcal{V}^d$  with per-variable vocabulary  $\mathcal{V}$ . Discrete flow matching models generation as a continuous-time Markov chain  $\{Z_t\}_{t \in [0,1]}$  whose transition rates  $u_t(y, z)$  transport an initial distribution  $p_0$  to the data distribution  $p_1$  (Campbell et al., 2024; Gat et al., 2024; Lipman et al., 2024). Let  $p_t$  denote the marginal

at time  $t$ , and its evolution is governed by the Kolmogorov forward equation

$$\frac{d}{dt} p_t(y) = \sum_{z \in \mathcal{S}} u_t(y, z) p_t(z), \quad (1)$$

so that for small  $h > 0$  the one-step kernel is  $p_{t+h|t}(y | z) = \delta(y, z) + h u_t(y, z) + o(h)$ . Rates factorize across variables,

$$u_t(y, z) = \sum_i \delta(y^{\bar{i}}, z^{\bar{i}}) u_t^i(y^i, z^i), \quad (2)$$

where  $\bar{i}$  denotes all indices other than  $i$ : a transition changes one variable at a time, while each per-variable rate  $u_t^i$  may depend on the entire state. The rates are learned by prescribing a conditional path from a tractable source to an observed datum and marginalizing over data. FORGE uses the mixture path

$$q_t(z^i | x^i) = t \delta_{x^i}(z^i) + (1 - t) p_0(z^i), \quad t \in [0, 1], \quad (3)$$

which resamples a variable from the source with probability  $1 - t$ , so  $t = 0$  is the source and  $t = 1$  the observed state, and small  $t$  is a strongly corrupted state. The marginal rate generating this path is determined by the posterior over clean states, so it suffices to learn an endpoint ( $x$ -) predictor  $p_\theta(x | z_t, t)$  rather than the rate itself (Campbell et al., 2024; Gat et al., 2024). Sampling alternates two operations: predict the clean categorical state, then take a discrete step under the rate that prediction induces. Because the model emits a distribution over clean values at every position, structural masks can act on the induced transitions rather than being learned as molecular-validity predictors. Related categorical denoising and flow formulations have been applied to graph generation (Austin et al., 2021; Vignac et al., 2023; Qin et al., 2025) and molecular generation (Eijkelboom et al., 2024).

**Ionizable lipids and the Ugi three-component reaction** Ionizable lipids combine an amine-containing, ionizable region with hydrophobic regions whose structure influences amphiphile organization, nanoparticle properties, and intracellular delivery (Cullis & Hope, 2017; Hou et al., 2021). Delivery is sensitive to head-group chemistry, tail length and asymmetry, branching, unsaturation, linker identity, and degradable functionality (Semple et al., 2010; Jayaraman et al., 2012; Whitehead et al., 2014; Hajj et al., 2019; Lee et al., 2021). The effects of these variables are strongly context dependent, which makes the design problem poorly described by independent or monotonic structure-activity rules.

Much of ionizable-lipid discovery uses modular chemistry: a fixed reaction is applied across inventories of molecular components and the resulting product library is synthesized or enumerated (Akinc et al., 2008). This work uses the Ugi three-component reaction (Ugi-3CR), an isocyanide-based multicomponent condensation joining an amine, an aldehyde, and an isocyanide (Dömling & Ugi, 2000; Dömling, 2006), and used here as the construction chemistry for ionizable lipids (Wang et al., 2025; Vlasova et al., 2025). The reaction supplies a known atom correspondence between precursor-derived regions of the final product together with a shared Ugi core, providing an auditable semantic organization rather than one inferred from the completed molecule. At the same time, an existing finite enumeration over fixed precursor inventories (Xu et al., 2024) provides a concrete reference against which generation beyond a predefined component vocabulary can be measured. The acidic phosphorus reagent used experimentally is a catalyst and is therefore not treated as a fourth structural component.

### 3 PROBLEM FORMULATION: OPEN-ENDED REACTION-GROUNDED GENERATION

Modular reactions provide prior knowledge about molecular organization: they identify the precursor-derived regions of a product and the interfaces through which those regions are assembled. Existing reaction-based design typically operationalizes this knowledge by selecting components from finite inventories. Our objective is instead to use reaction structure to organize a joint distribution over complete molecules while leaving the identity of each precursor-derived region open-ended.

Formally, let  $\mathcal{R}$  be a modular reaction with  $K$  precursor roles  $\mathcal{K}_{\mathcal{R}}$ , let  $C$  denote a coarse architectural context, and let  $X = (X_1, \dots, X_K)$  denote the precursor-derived regions of a complete

product  $\mathcal{G}$ . Given a structural corpus of products assembled by  $\mathcal{R}$ , our primary objective is to learn one conditional distribution  $p_\theta(\mathcal{G} \mid C)$  over complete molecular structures. Crucially, biological measurements are not used to define or train the generator.

We require  $p_\theta(G \mid C)$  to satisfy three properties:

1. **Reaction grounding.** Every generated molecule must respect the structural interfaces imposed by  $\mathcal{R}$  and project to components whose forward assembly reconstructs the generated product.
2. **Open-ended component identity.** Each precursor-derived region is generated at atom- and bond resolution rather than selected from a finite component inventory.
3. **Joint contextual generation.** The model must represent dependencies among all precursor-derived regions conditional on the architectural context, rather than generating those regions independently.

Existing model classes generally provide only part of this combination. Atom-level de novo models can be open-ended and contextual but do not, by construction, provide reaction-compatible components. Inventory-based reaction models provide compatibility and immediate component identity, but their molecular support is bounded by the inventories supplied to them. Generating each reaction role separately can remove that inventory restriction, but introduces a different limitation: it discards dependencies among the regions once the context is fixed.

## 4 CHEMISTRY-INFORMED DISCRETE FLOWS

FORGE realizes the three requirements above by using the roles supplied by a modular reaction to organize one discrete molecular state. The roles structure the representation, source distribution, and learning objective, while a shared contextual denoiser generates the complete molecule and chemical constraints preserve reaction compatibility. We instantiate the construction on Ugi-3CR ionizable lipids.

[FIGURE IN PROGRESS]

Figure 1: **Semantic organization without generative factorization.** The design context  $c$  induces a deterministic role map  $r_c$  over a single molecular state; a shared contextual denoiser generates the complete graph, and  $D_{\mathcal{R}}$  projects the terminal state back to amine-, aldehyde-, and isocyanide-derived components for forward-assembly qualification.

### 4.1 CHEMISTRY-INFORMED MOLECULAR STATES

Let  $G = (V, E, x^V, x^E)$  denote a complete molecular graph, with atom and bond attributes  $x^V$  and  $x^E$ . We consider a modular chemistry  $\mathcal{R}$  with a finite set of chemically meaningful roles  $\mathcal{K}_{\mathcal{R}}$ . A design context  $c$  specifies the structural program used to instantiate the molecular state and induces a deterministic semantic map  $r_c : V \rightarrow \mathcal{K}_{\mathcal{R}} \cup \{\text{core}\}$ ; for the Ugi-3CR system studied here,  $\mathcal{K}_{\mathcal{R}} = \{\text{amine}, \text{aldehyde}, \text{isocyanide}\}$ , the isocyanide-based multicomponent condensation (Dömling & Ugi, 2000; Dömling, 2006) used to build ionizable lipids (Wang et al., 2025; Vlasova et al., 2025). The map  $r_c$  identifies the precursor-derived region associated with each generated position together

with the shared Ugi core. It is supplied by the chemistry-aware representation; it is neither inferred from the completed molecule nor modeled as a random variable.

The distinction between *semantic organization* and *generative factorization* is central. FORGE models a single conditional distribution  $G \sim p_\theta(G | c)$  over complete molecular states, rather than imposing a factorization  $p_\theta(G | c) = \prod_{k \in \mathcal{K}_R} p_{\theta,k}(G_k | c_k)$  across roles. The precursor-derived regions are therefore not independently generated reactants that are subsequently combined; a shared contextual model operates over the complete state, letting information propagate across regions while preserving their chemical interpretation. Likewise,  $c$  provides structural organization rather than molecular identity: the model receives no named amine, aldehyde, or isocyanide, no component identifier, and no stored fragment catalogue. Head topology, nitrogen environments, chain architecture, branching, unsaturation, linker chemistry, terminal groups, and constitutional identity remain generated variables. This is the sense in which FORGE uses semantic organization without generative factorization: role is determined by the design context and state layout, never predicted as an origin variable.

For Ugi-3CR lipids,  $c = (c_a, c_d, c_i)$  specifies a small set of precursor-level architectural quantities that instantiate a bounded molecular state. They define the design problem but not the lipid; within the resulting state, the flow determines the detailed chemistry of the ionizable head and hydrophobic regions. Chemistry-derived constraints thus encode what the molecular family already fixes, reserving the unresolved realization for learning.

#### 4.2 ROLE-AWARE DISCRETE FLOW MATCHING

All molecular variables in FORGE are discrete. Let  $x_u^h$  denote the clean value of variable  $u$  for prediction head  $h$ , where  $h$  ranges over the categorical variables required to specify molecular topology and chemistry. FORGE constructs a probability path from a role-specific source distribution  $p_0^{h,r}$  to the data state. For a position with semantic role  $r = r_c(u)$ ,

$$q_t(z_u^h | x_u^h, r) = t \delta_{x_u^h}(z_u^h) + (1 - t) p_0^{h,r}(z_u^h), \quad t \in [0, 1]. \quad (4)$$

Thus  $t = 0$  corresponds to the source distribution and  $t = 1$  to the observed molecular state. We use endpoint ( $x$ -) prediction: given a corrupted molecular state  $z_t$ , time  $t$ , and design context  $c$ , a contextual denoiser predicts the clean categorical state  $p_\theta(x | z_t, t, c)$ . The per-variable rates of Eq. (2) are induced by this path together with the endpoint predictor. What is specific to FORGE is the representation and the support, not the estimator: Eq. (4) differs from the generic mixture path of Eq. (3) only in that its source  $p_0^{h,r}$  is conditioned on the semantic role.

Chemically meaningful regions differ substantially in size: the hydrophobic regions of an ionizable lipid contain many more molecular variables than its head, so a naïve token-average objective would let the largest regions dominate optimization merely by carrying more categorical targets. FORGE uses a role-balanced categorical objective,

$$\mathcal{L}(\theta) = \mathbb{E}_{G,t,z_t} \left[ \underbrace{\sum_{h \in \mathcal{H}_{\text{node}}} \frac{1}{|\mathcal{K}_R|} \sum_{r \in \mathcal{K}_R} \bar{\ell}_h^r}_{\text{role-balanced}} + \underbrace{\sum_{h \in \mathcal{H}_{\text{pool}}} \bar{\ell}_h}_{\text{pooled}} \right], \quad (5)$$

where  $\bar{\ell}_h^r$  is the mean cross-entropy over the variables of head  $h$  at positions with role  $r$ , and  $\bar{\ell}_h$  the mean over all applicable variables of head  $h$ . The node-level heads  $\mathcal{H}_{\text{node}}$ , covering offspring topology, atom state and parent bond, are averaged *within* each role and then across roles, so a role contributes equally regardless of how many serialized variables it carries. The remaining heads  $\mathcal{H}_{\text{pool}}$ , covering closure bonds and the three decoration targets, are pooled, because their targets attach to specific sites rather than to every position and are not partitioned by role in the same way. Head losses enter the total unweighted. Appendix A gives the architecture.

Chemistry informs more than a role embedding: the semantic map also determines role-relative molecular coordinates and role-specific source statistics, while the shared denoiser operates contextually over the complete state and the variables inside those regions stay unconstrained by any component identity. Chemistry tells FORGE how to interpret a molecular region; the learned flow determines the structure occupying it.

### 4.3 CHEMICALLY CONSTRAINED SAMPLING AND ASSEMBLY CONSISTENCY

Not every assignment of categorical variables corresponds to a chemically admissible product. FORGE therefore restricts sampling to a declared support  $S_{\text{gen}}(c) = \{G : I_c(G) = 1\}$ , where  $I_c$  collects the structural invariants associated with the instantiated design context and generation-time masks enforce them (Appendix A). These restrictions are properties of the generative support rather than learned indicators of molecular quality: a nonlearned role-marginal sampler can satisfy the same chemical admission rules, a distinction we use explicitly in our evaluation.

The representation also provides a deterministic experimental projection. Given a terminal molecular state,

$$D_{\mathcal{R}}(G, c) = (B_a, B_d, B_i) \quad (6)$$

recovers the amine-, aldehyde-, and isocyanide-derived graphs implied by that state. A sample is chemically admitted only when the projected components satisfy their role-specific interfaces and reconstruct the generated constitutional graph under the qualified forward Ugi transform,

$$A_{\mathcal{R}}(G, c) = \mathbf{1}[\mathcal{R}(D_{\mathcal{R}}(G, c)) \cong G] \cdot \mathbf{1}[D_{\mathcal{R}}(G, c) \text{ is role-valid}]. \quad (7)$$

This ties generation to the chemistry used for experimental realization, but asserts nothing about uniqueness of the Ugi product, uniqueness of a retrosynthetic route, or achievable yield; unique assembly, precursor procurement or preparation, and route completeness are evaluated separately downstream.

The formulation extends in principle to other modular families by replacing  $r_c$ , the role-specific state interfaces, and the assembly/qualification operator for  $\mathcal{R}$ . We make no empirical cross-reaction claim: all experiments below use the Ugi-3CR lipid system.

## 5 DISCOVERY UNDER SPARSE BIOLOGICAL EVIDENCE

Here, we address the problem of how samples can be ranked or advanced experimentally. Because it is open-ended, the generator can propose molecules beyond both the chemistry represented in the biological dataset and the components directly available for purchase. This section asks how that fixed generator should be deployed under sparse biological and synthetic evidence.

We separate deployment into three decisions. First, a structure-only applicability test determines whether a generated molecule lies within the region where measured lipids support biological ranking. Second, an evidence-aware context proposal allocates more sampling effort to contexts likely to produce such molecules, without changing the trained generator or using predicted activity as guidance. Third, exact component projection and forward-verified route qualification determine whether a candidate can be advanced experimentally.

### 5.1 SEPARATING STRUCTURAL POSSIBILITY FROM BIOLOGICAL SUPPORT

Write  $S_{\text{gen}} = \bigcup_{c \in \text{supp}(\pi_0)} S_{\text{gen}}(c)$  for the structural space accessible under the broad proposal  $\pi_0(c)$  and the chemistry-constrained supports of Section 4.3. FORGE can generate throughout this space, but the biological predictor is trained on only 1,100 measured lipids. We therefore define a stricter evidence-supported subset

$$S_{\text{pred}} = \{G \in S_{\text{gen}} : E(G) = 1\}, \quad S_{\text{pred}} \subsetneq S_{\text{gen}}, \quad (8)$$

where  $E(G)$  is a frozen applicability predicate constructed independently of measured activity values and predictor residuals.

The test evaluates the complete product together with each of its three precursor-derived views, and excludes a candidate if any view falls outside the prescribed structural neighborhood (Section 6.1, Appendix B). Crucially, membership in  $S_{\text{pred}}$  is an **evidence statement**, not an activity prediction: a molecule outside this region is not declared inactive, and FORGE abstains from using the predictor to make comparative biological claims about it.

Within  $S_{\text{pred}}$ , we use an ensemble predictor to rank complete lipids by measured transfection outcome. Ranking and support remain deliberately separate: activity predictions do not define  $S_{\text{pred}}$ , modify the structural generator, or extend biological ranking into unsupported chemistry. We report

pooled residual calibration only as a descriptive characterization of predictive uncertainty rather than as a coverage guarantee or a candidate-ranking rule. This restraint is motivated by the distribution shifts induced by component- and scaffold-level novelty, for which nominal predictive coverage need not transfer automatically (Ai & Ren, 2024).

## 5.2 BROAD EXPLORATION AND EVIDENCE-AWARE DEPLOYMENT

The separation in Eq. (8) creates two distinct uses of the same trained generator. For broad exploration, precursor contexts are sampled from a reference proposal  $\pi_0(c)$ , giving  $p_{\text{explore}}(G) = \sum_c \pi_0(c) p_\theta(G | c)$ . This characterizes the molecular space learned by FORGE, including designs outside the finite reference library and outside the current biological support region.

For deployment, we seek to spend a fixed sampling budget more efficiently without turning the biological predictor into a molecular guidance function. Let

$$\rho(c) = \Pr_{G \sim p_\theta(\cdot | c)}[G \in S_{\text{pred}}] \quad (9)$$

denote the probability that context  $c$  produces an evidence-supported molecule. We estimate  $\rho(c)$  from structural generation alone and define the evidence-aware proposal

$$\pi_\gamma(c) = \frac{\pi_0(c) (\rho(c) + \epsilon)^\gamma}{\sum_{c'} \pi_0(c') (\rho(c') + \epsilon)^\gamma}, \quad \gamma \geq 0, \epsilon > 0, \quad (10)$$

with deployment distribution  $p_{\text{deploy}}(G) = \sum_c \pi_\gamma(c) p_\theta(G | c)$ . No predicted activity appears in Eqs. (9)–(10):  $p_\theta$ , its transition probabilities, and its structural support are unchanged, and only the frequency with which contexts are proposed is modified. Because  $\epsilon > 0$ , every context with nonzero mass under  $\pi_0$  retains nonzero mass under  $\pi_\gamma$ , so deployment concentrates computation without converting biological support into a hard restriction on structural discovery.

The proposal admits a simple regularized interpretation:

$$\pi_\gamma = \arg \max_q \left\{ \mathbb{E}_{c \sim q} [\log(\rho(c) + \epsilon)] - \frac{1}{\gamma} D_{\text{KL}}(q \| \pi_0) \right\}, \quad (11)$$

for  $\gamma > 0$ . Thus  $\gamma$  controls an explicit trade-off between concentrating on contexts likely to yield biologically supported candidates and remaining close to the broad exploratory proposal. Because increasing the fraction of rankable molecules can reduce chemical coverage, we report evidence-supported yield and diversity jointly rather than treating enrichment as an unconditional improvement.

## 5.3 FROM BIOLOGICAL SUPPORT TO EXPERIMENTAL ACTIONABILITY

Biological support still does not imply that a design can be taken directly to the bench. Candidates must also admit a unique qualified Ugi assembly and an explicit path to their constituent materials.

Resolution is a hierarchy rather than a single check, and it runs on generated *components* rather than on a whole product whose disconnection must first be guessed. A component that appears in the dated procurement snapshot is terminal. Otherwise it must match a documented role-specific preparation whose starting materials are themselves purchasable, and the disconnection is accepted only when the forward reaction applied to those materials reproduces the component *uniquely*. A design resolves when all three components resolve within the declared step budget, counting preparations before the final assembly and not the assembly itself, which is common to every design. Appendix F gives the full hierarchy, the proposal engines consulted where the documented set is silent, the adjudication each proposal must pass, and the procurement snapshot.

These operations do not alter the learned distribution. Instead they progressively distinguish three questions often conflated in molecular discovery: whether FORGE can generate a design ( $G \in S_{\text{gen}}$ ), whether current biology can rank it ( $G \in S_{\text{pred}}$ ), and whether it can be advanced experimentally ( $G \in S_{\text{act}}$ ). Prospective measurements answer a fourth, separately: whether the lipid is functional. This separation lets FORGE explore chemistry more broadly than current biological supervision while keeping deployment claims tied to the evidence available at selection time.

## 6 EXPERIMENTAL SETUP

We evaluate FORGE at four distinct levels: whether chemistry-informed semantics improve molecular denoising, whether the trained flow learns lipid-native structural organization, whether open generation expands the accessible design space, and whether that space can be deployed responsibly under sparse biological supervision. Unless otherwise stated, model-selection decisions, evaluation metrics, and comparison conventions were fixed before the corresponding reported measurements; complete configurations and artifact hashes are provided in Appendix E.

### 6.1 DATA AND EVALUATION PARTITIONS

The structural generator is developed from 112,386 Ugi-3CR products, using component-family-aware partitions so that molecular families can be evaluated outside the examples seen by the corresponding development checkpoint. Biological prediction uses a separate set of 1,100 experimentally measured ionizable lipids with matched transfection measurements. We additionally retain the complete 12,276-member Ugi-3CR enumeration defined by the original finite precursor inventories as a reference chemical space; membership in this enumeration is distinct from both structural-training membership and biological measurement.

For predictive support, each generated design is evaluated in four views: the complete product and its amine-, aldehyde-, and isocyanide-derived components. Each view is compared with the measured reference chemistry using a radius-2 count-Morgan fingerprint (Rogers & Hahn, 2010) and a robust physicochemical-descriptor distance. The frozen support predicate accepts candidates whose worst product/component view lies in the interpolative or boundary applicability bands and excludes extrapolative chemistry. Threshold calibration, exact-identity exclusion, and reference construction are given in Appendix B.

### 6.2 BASELINES AND MOLECULAR EVALUATION

We use three complementary classes of internal control. To isolate the contribution of chemically aligned semantics, we train matched denoisers on an identical graph-preserving serialization with identical capacity, data, corruption, optimization, and global design context. The treatment arm receives the correct node-wise chemical-role map; the ablated arm receives a constant null role; a dimension- and frequency-matched misaligned map controls for the presence of an additional embedding channel. All comparisons use paired initialization, minibatches, corruption draws, and flow times across three training seeds.

The central architectural comparison is against a learned component-factorized generator. It receives the same complete architectural context, representation, chemical support, corruption process, objective, parameter count, optimization budget, and sampling contexts as FORGE, but its recurrent state is reset at reaction-role boundaries. It therefore learns each precursor-derived region contextually while enforcing conditional independence among the regions. A deliberately generous control trains a separate full-capacity denoiser for each role. Before fitting these models, a within-context component-shuffling diagnostic tests whether the held-out structural distribution contains cross-role dependence for either model to recover. The matched comparison uses five paired training seeds; its endpoints and reporting contract are frozen before any factorized result is inspected.

To isolate what contextual learning contributes beyond chemistry constraints and one-variable statistics, we compare free-running FORGE samples against a role-marginal generator using the same support, masks, role-specific empirical marginals, and admission procedure but no contextual denoiser. Molecular fidelity is evaluated with a prespecified lipid-native panel covering ionizable-head chemistry, reactive-site multiplicity, linker and tail chemistry, branching and hydrophobic organization, together with a grouped real-versus-generated classifier. Six main-text metrics were fixed by scientific coverage rather than observed effect size; the complete 38-descriptor panel is reported in Appendix D.

### 6.3 DISCOVERY, DEPLOYMENT, AND ACTIONABILITY

Structural discovery is evaluated separately against the full structural-training corpus and the stereo-free finite enumeration; all novelty results therefore state their reference set explicitly. Proposal

[FIGURE IN PROGRESS]

Figure 2: **The FORGE evidence chain.** (a) Semantic mechanism: semantic-utility gap against  $t$  for three paired seeds, with true, no-role, and misaligned arms. (b) Joint contextual learning: conditional-dependence diagnostic, paired held-family loss, cross-role intervention, and generated dependence for FORGE versus learned factorized controls. (c) Discovery and deployment: structural reach into  $S_{\text{pred}}$ , and supported yield 3.87% to 8.64%, with the diversity concentration beside it and its distinct baseline labelled. (d) Computational actionability: exact component projection, unique Ugi assembly, and route resolution inside and outside the finite reference enumeration.

reallocation is confirmed on fresh terminal draws at matched budgets of 3,072 per arm, following a development sweep with frozen validity and diversity floors. The draws are fresh, but the design-context population matches development, so this confirms outcomes rather than generalization to unseen contexts. The headline deployment comparison uses these draws and the frozen support definition of Section 5; predicted activity does not enter the proposal.

Within  $S_{\text{pred}}$ , complete lipids are ranked by the frozen biological ensemble. Predictive calibration is reported descriptively and is not used to alter candidate ordering or expand  $S_{\text{pred}}$ . Experimental actionability is assessed separately, by a three-level cascade: exact Ugi decomposition and forward reassembly, then component-level resolution against a frozen curated evidence index extended where silent by Graph2Edits single-step proposals and AiZynthFinder multistep search, then closure of every leaf to a terminal material in a dated procurement snapshot. Route search is blind to arm and to predicted potency, and every proposal is independently adjudicated rather than accepted on a planner score. Appendix F gives the hierarchy, the adjudication criteria, the engine recovery benchmark and the snapshot date.

## 7 RESULTS

We evaluate FORGE along the sequence of claims established above: whether chemistry-informed semantics provide useful information to the generative model, whether contextual learning recovers lipid-native molecular structure, whether open generation expands the accessible chemistry, and whether that expanded space can be deployed without conflating structural novelty with biological confidence.

## 7.1 CHEMICALLY ALIGNED SEMANTICS REDUCE DENOISING AMBIGUITY

We first isolate the effect of the chemistry-informed semantic map independently of the production generator. The aligned and ablated models receive identical molecular graphs, global positional information, design context, corruption draws, optimization, and parameter count; they differ only in whether each molecular position is assigned its correct chemical role. Across three paired training seeds, providing the aligned semantic map reduces total denoising loss by  $0.248 \pm 0.058$ , with a positive improvement for all six reported heads in every seed (Fig. 2a). The effect depends strongly on corruption: the aligned-semantic advantage is 7.7–11.3 times larger near the highly corrupted end of the probability path than near the clean endpoint.

This gain requires the *correct* chemical correspondence rather than merely another categorical channel. When role labels are deliberately reassigned across molecular positions while preserving their dimensionality and label frequencies, the improvement disappears; the misaligned model is no better than the no-role model on two prediction heads and is worse on the remaining four (Fig. 2a). Chemical-role semantics therefore provide a useful representational inductive bias beyond what the same finite-capacity model extracts from the corrupted state and global context; Appendix C.5 sets out why this is a computational rather than an information-theoretic statement. Consistent with the conditional nature of this effect, a preregistered analysis found that semantic utility does *not* increase simply with role-wise marginal divergence (Appendix C).

## 7.2 JOINT CONTEXTUAL GENERATION RECOVERS CROSS-ROLE AND LIPID-NATIVE STRUCTURE

Chemical semantics organize the generative state but do not specify the molecule that occupies it. We therefore next ask whether dependencies remain among the precursor-derived regions after conditioning on the architectural context, and whether the joint denoiser recovers them more faithfully than a learned component-factorized generator.

**[RESULT: Within-context component shuffling preserves each role marginal while destroying cross-role dependence; report the grouped intact-versus-shuffled AUC, its confidence interval, and the prespecified conditional-dependence statistic.]**

**[RESULT: Across five paired seeds, report the held-component-family endpoint loss for FORGE, FACT-matched, and FACT-generous; the paired FACT-matched-minus-FORGE difference with its 95% confidence interval; and the corresponding per-role and corruption-time results.]**

**[RESULT: Report the amine-, aldehyde-, and isocyanide-target cross-role swap penalties, verify that the factorized models are invariant to the intervention, and compare the generated residual cross-role dependence matrices with the held-out real distribution.]**

Table 1: Planned reporting table for the learned joint-versus-factorized comparison. Values remain explicit placeholders until the frozen artifacts are complete. A factorized model must learn competent per-role marginals before inferior complete-product organization is attributed to factorization.

| Measurement                                | FORGE-joint | FACT-matched   | FACT-generous | Primary contrast or check                   |
|--------------------------------------------|-------------|----------------|---------------|---------------------------------------------|
| Within-context intact-versus-shuffled AUC  |             | [XXX (95% CI)] |               | [XXX above 0.5]                             |
| Held-family endpoint loss                  | [XXX]       | [XXX]          | [XXX]         | $[\Delta_{\text{joint}} \text{ (95\% CI)}]$ |
| Amine-target cross-role swap penalty       | [XXX]       | [0; verify]    | [0; verify]   | [XXX (95% CI)]                              |
| Aldehyde-target cross-role swap penalty    | [XXX]       | [0; verify]    | [0; verify]   | [XXX (95% CI)]                              |
| Isocyanide-target cross-role swap penalty  | [XXX]       | [0; verify]    | [0; verify]   | [XXX (95% CI)]                              |
| Distance to held-out cross-role dependence | [XXX]       | [XXX]          | [XXX]         | [FACT – FORGE (95% CI)]                     |
| Per-role marginal-fidelity summary         | [XXX]       | [XXX]          | [XXX]         | [competence check]                          |
| Ugi admission                              | [XXX]       | [XXX]          | [XXX]         | [noninferiority check]                      |
| Effective product diversity                | [XXX]       | [XXX]          | [XXX]         | [noncollapse check]                         |

As a complementary, deliberately weaker diagnostic, we ask whether contextual learning recovers molecular organization beyond what is supplied by the chemistry constraints and empirical one-variable statistics. A role-marginal generator using the same support and admission machinery reaches full chemical admission, yet produces none of the prespecified ionizable-head architecture in 3,072 samples. FORGE builds that architecture in 0.479 of admitted designs, overshooting the

[FIGURE IN PROGRESS]

Figure 3: **Joint versus component-factorized generation.** (a) Conditional-dependence diagnostic comparing intact held-out products with products whose components are shuffled within architectural context. (b) Paired held-component-family loss for FORGE, FACT-matched, and FACT-generous across five training seeds. (c) Increase in target-role loss after the other corrupted role blocks are shuffled within context, shown by target role and flow time. (d) Distance between the generated and held-out-real residual cross-role dependence matrices, with admission and effective diversity shown as safeguards. All values are pending completion of the frozen comparison.

0.394 rate of the real lipids, while simultaneously reducing the prevalence of multiple competing Ugi-reactive sites toward the real-lipid distribution (0.131 for FORGE versus 0.181 in the reference set). It also reproduces the dominant ester-linked tail chemistry of its training distribution (0.510 versus 0.530 in the train fold), in contrast to the marginal control (Fig. 2b).

The improvement is substantial but not perfect: a grouped classifier still separates generated from held-out real lipids with AUC 0.882. We therefore interpret FORGE as learning meaningful lipid-native molecular organization rather than matching the structural distribution exactly. The complete 38-descriptor panel, including nitrogen environments, branching, unsaturation, linker chemistry, hydrophobic balance, and nearest-neighbor statistics, is reported in Appendix D.

**Why the comparison is internal.** The natural question is what this should be compared against, and the answer is that no method shares the setting. Synthesis-constrained generators draw components from a fixed inventory, so the quantity at issue here, whether a model composes precursor chemistry the inventory does not contain, is zero for them by construction rather than by measurement. The two prior generative models targeting synthesizable ionizable lipids (Ou et al., 2024; Zhao et al., 2024) share that restriction. An unconstrained graph model would generate outside the inventory but would not admit the exact component projection that makes the question answerable at all. We therefore evaluate against internal controls that isolate specific components of the method rather than against external systems solving a different problem: the role-marginal null holds the declared support, masks, role marginals and admission fixed and removes only contextual learning, and the misaligned arm holds channel dimensionality and label frequency fixed and removes only the correspondence. Each answers what a particular part of FORGE contributes. Neither answers whether FORGE beats a system built for a different setting, and we do not claim that it does.

Because the molecular realization remains generative, this learned distribution extends well beyond the structures used to fit the model. Among distinct admitted production designs, 83.7% are absent from the full 112,386-product structural training corpus, and **86.3%** lie outside the stereo-free 12,276-product reference enumeration. These comparisons answer different questions and use dis-

tinct reference sets. Importantly, novelty survives downstream qualification: **931** route-complete designs remain outside the finite enumeration.

### 7.3 EVIDENCE-AWARE DEPLOYMENT CONCENTRATES RANKABLE GENERATION WITHOUT POTENCY GUIDANCE

Open structural generation also exposes the central asymmetry of the discovery problem: most generated chemistry lies outside the region for which the 1,100 measured lipids provide reliable functional evidence. We therefore evaluate the support-aware proposal of Eq. (10) separately from activity ranking. On a prespecified confirmation with 3,072 matched draws per arm, reallocation increases the probability of generating a molecule inside  $S_{\text{pred}}$  from 3.87% to 8.64%, a 2.23-fold increase, while leaving the molecular generator and its transitions unchanged (Fig. 2c). Neither measured activity nor predicted potency enters the proposal.

This enrichment is not diversity-neutral. In the corresponding production allocation, effective supported-molecule diversity decreases from 96.1 to 84.2, and role-resolved analysis shows that the concentration falls most strongly on amine-head diversity, which loses 60%, while aldehyde-tail diversity is essentially preserved at 1.002 times its unenriched value. We therefore treat proposal reallocation as an explicit exploration–deployment trade-off rather than an unconditional improvement.

The synthesis-grounding layer then asks whether these structurally novel, evidence-supported designs can be connected to material that can actually be ordered. Because component identity is generated rather than selected, this is a question with a real chance of a negative answer, and it is answered per component: purchasable, or reduced to purchasable material by a documented preparation whose forward reaction reproduces the component uniquely. Across designs reaching assessment, 1,472 of 1,549, or 95.0%, resolve.

Unique qualified assembly rises from 0.829 across admitted designs to 0.944 among evidence-supported designs. Route resolution remains easier inside the finite reference chemistry than outside it (1.000 versus 0.924 route completeness), showing a measurable cost to expanding the molecular search space rather than artificial parity.

Which role carries the off-registry chemistry decides how much that reach is worth, because the amine head is the dominant activity determinant and the one role for which this instantiation supplies no disconnection rule. Across admitted designs the novelty is not evenly placed: 49.7% carry an unseen amine head, 43.3% an unseen aldehyde tail and 15.2% an unseen isocyanide tail. Route assessment is not what removes them. Among designs reaching assessment, off-registry components resolve at 9/10 for amine heads, 765/800 for aldehyde tails and 49/90 for isocyanide tails. The attrition happens one gate earlier: designs carrying an unseen amine head fall from 49.7% of admitted to 0.5% of prediction-supported, whereas unseen aldehyde tails pass through and are enriched, reaching 51.6% of designs assessed. What confines the actionable set to catalogue heads is the applicability domain of the activity model, not the synthesis layer, which is  $S_{\text{pred}} \subsetneq S_{\text{gen}}$  measured per role rather than asserted. Appendix F.6 gives the full funnel.

Applying the full cascade to a 256-product route-blinded shortlist separates two quantities that are easy to conflate. Retrosynthetic proposals cover 92.6% of those products, while closure on frozen curated evidence reaches 7.4%, and the proposal engines close none of the difference. The constraint is therefore evidential rather than computational: 121 unresolved components were never expanded at all, being absent from the evidence index, against 1 unresolved after search. Open generation reaches chemistry that retrosynthetic search can address and that the curated evidence base does not yet cover, which is a statement about the evidence base rather than about the chemistry. Library 0 retains 40 fully documented computational candidates as a frozen pilot; 39 satisfy the subsequently frozen unique-assembly criterion, and no member has yet been synthesized.

### 7.4 PROSPECTIVE IONIZABLE-LIPID DISCOVERY

We finally test whether the additional molecular variation opened by FORGE translates into experimentally useful lipids. The prospective cohort is frozen before biological outcomes and contains [XXX FINAL PANEL] candidates selected from  $S_{\text{pred}}$  after unique-assembly and route qualification. Where possible, generated designs are evaluated alongside matched reference chemistry that

preserves the commercial head and broader physicochemical context while localizing differences in chain length, branching, unsaturation, or linker structure.

**[XXX PROSPECTIVE RESULT: synthesis success, formulation/LNP qualification, transfection and viability, with matched-reference comparison]**

These experiments distinguish the final question, whether a generated lipid functions experimentally, from the structural, predictive, and synthetic criteria used to select it.

## 8 DISCUSSION AND LIMITATIONS

FORGE separates three sources of information that molecular design routinely conflates: known chemistry organizes the generative state, learned discrete flows determine its molecular realization, and biological evidence determines where generated structures can responsibly be ranked.

Taking the first of those, the claim is not that flow matching can generate lipids, but that known chemistry can supply a scientifically meaningful coordinate system for generation without turning the generative model into a product of independent components. The semantic ablation makes this an empirical claim about the model rather than an appeal to interpretability: correct role alignment lowers denoising loss, a dimension- and frequency-matched misaligned map does not, and the advantage grows precisely where the corrupted state carries least information about the molecule. The marginal null supplies the complementary half. Structural constraints alone reach full chemical admission and still build no ionizable heads at all, so it is the contextual model, not the declared support, that produces lipid-native organization. Semantics and learning are doing different jobs, and each is separately necessary.

That capability is also what creates the next problem, because once component identities are generated objects rather than catalogue entries the structural generator will outrun the biological data almost by construction. We read that as the expected consequence of successful exploration rather than a defect of it. The two obvious responses both fail: restricting  $S_{\text{gen}}$  back to measured chemistry discards exactly the novelty that motivated open generation, while trusting the predictor everywhere converts extrapolation into design decisions. FORGE instead keeps the generative space broad and makes biological authority explicitly local. Membership in  $S_{\text{pred}}$  is an evidence statement about where ranking is meaningful, not a claim that unsupported molecules are inactive; and the reallocation that concentrates sampling into  $S_{\text{pred}}$  alters only how often contexts are proposed, never the generator, its transitions, or its support. This separation is what allows structural novelty and biological confidence to be reported as two quantities rather than silently merged into one.

Whether this arrangement is native to lipid discovery or merely imposed on it is a fair question, and the domain answers it: lipid chemistry suits the framework unusually well. Several modular synthetic families are in routine use, including multicomponent combinatorial synthesis, amine–aldehyde–alkyne coupling, and reductive amination (Akinc et al., 2008; Love et al., 2010; Li et al., 2023), and each opens enormous structural freedom, while functional measurement remains expensive, cell-type dependent, and small: published campaigns report hundreds to roughly a thousand measured lipids against virtual libraries orders of magnitude larger (Li et al., 2024; Xu et al., 2024). The gap between chemistry that can be generated and chemistry that can be synthesized, formulated, and assayed is therefore not incidental to lipid discovery; it is the defining shape of the problem, and it is what makes structured generative learning, open exploration, and evidence-bounded deployment fit together here rather than being imposed on the domain.

How far the arrangement travels beyond this system is a separate question, and the honest answer is bounded: FORGE is not a Ugi model, but neither do we demonstrate cross-reaction transfer. The general object is known modular construction, giving semantic organization, giving generated molecular realization. A different family supplies its own semantic role map, role-compatible state interfaces, and auditable assembly operator; the molecular identities occupying those roles remain generated. Stated precisely, FORGE is formulated for modular molecular families in which known chemistry supplies an interpretable semantic decomposition and an auditable assembly interface. It is not formulated for arbitrary chemistry, and the empirical evidence in this paper is Ugi-3CR-specific. We state that as a boundary rather than an apology. The framework therefore complements synthesis-constrained generative methods, whose reaction structure improves experimental actionability but whose reachable chemistry is fixed by predefined reactants or actions (Cretu et al., 2025;

Koziarski et al., 2024; Luo et al., 2024). Closest to this work are generative models that target synthesizable ionizable lipids directly (Ou et al., 2024; Zhao et al., 2024); both draw their components from predefined inventories, which is the restriction FORGE removes.

One consequence of generating identity rather than selecting it deserves separating out, because it inverts a convention. A generator that composes molecules from a fixed inventory inherits synthesizability from its action space: every reachable product is built from material someone can already buy, so the question never arises. That guarantee is also the constraint, because a component outside the inventory is not improbable but unreachable. FORGE gives the guarantee up deliberately, and the route layer is what buys it back. The direction of work inverts: catalogue methods compose forward from purchasable material, whereas FORGE generates the molecule and then reduces backward to material that can be ordered, accepting a transformation only when running it forward returns the intended component. We think this is the more honest arrangement for open-ended molecular design, because it makes accessibility a measured property with a rate attached rather than a property of the search space that is never put at risk. It is also where the cost shows: route resolution is harder outside the finite reference chemistry (1.000 versus 0.924), and that gap is the price of the structural reach reported in §7.2.

**One reaction family.** The formulation is stated for modular chemistries generally, but every experiment here uses Ugi-3CR, and we demonstrate no cross-reaction transfer.

**Constitutional, not stereochemical.** Generation proposes no stereocentres, so stereoisomers are neither distinguished nor generated. Every novelty comparison against a stereochemistry-bearing reference set is therefore made stereo-free on both sides, and stereochemical preference, a known determinant of LNP behaviour, is outside what we model.

**The model does not match the held-out distribution.** A grouped classifier still separates generated from real lipids at AUC 0.882, and the model overshoots rather than reproduces the real ionizable-head rate. We claim lipid-native organization in specific named respects, not distribution matching.

**The predictor is weak, and we fence it rather than fix it.** It is treated as a locally valid ranking device rather than an oracle, and its calibration is reported descriptively rather than as a coverage guarantee.  $S_{\text{pred}}$  is therefore conservative by construction, and its boundary is a modelling choice rather than a measured frontier.

**Route burden rises outside the reference chemistry.** Route completeness is 1.000 inside the finite enumeration against 0.924 outside it, so the expanded search space carries a real synthetic cost. Closure also depends on which evidence standard applies: resolution to purchasable material under forward-verified disconnection reaches 95.0%, whereas closure on frozen curated evidence alone reaches 7.4% on the route-blinded shortlist. We report both rather than the flattering one, because the gap is what identifies the curated evidence base, not search capability, as the current bottleneck.

**Nothing has been made.** Library 0 remains a frozen computational pilot with no member synthesized. Extending FORGE across lipid reaction families, and expanding  $S_{\text{pred}}$  through prospective measurement rather than through looser thresholds, are the natural next steps.

## REPRODUCIBILITY STATEMENT

We provide complete descriptions of structural-data construction, molecular representation, model architecture, training, and sampling in Appendix A; the frozen biological-support and prediction protocol, including predictor baselines, in Appendix B; all semantic-ablation contracts and the pre-declared hypotheses that were not supported in Appendix C; the full lipid-native evaluation panel in Appendix D; synthesis and route-qualification procedures in Appendix F; and derivations of the proposal and support-preservation results in Appendix G. Appendix E records frozen configurations, checkpoints, seeds, software settings, artifact hashes, comparison conventions, and analyses not performed, and indexes each main-text claim to the evidence behind it. Prospective experimental procedures and the prespecified statistical analysis are documented in Appendix H. No prospective

outcome is incorporated into model development or candidate selection after the corresponding campaign freeze. [TODO: add code availability]

## REFERENCES

- David Adams, Alejandra Gonzalez-Duarte, William D. O’Riordan, Chih-Chao Yang, Mitsuharu Ueda, Arnt V. Kristen, Ivailo Tournev, Hartmut H. Schmidt, Teresa Coelho, John L. Berk, Kon-Ping Lin, Giuseppe Vita, Shahram Attarian, Violaine Planté-Bordeneuve, Michelle M. Mezei, Josep M. Campistol, Juan Buades, Thomas H. Brannagan, Byoung J. Kim, Jeeyoung Oh, Yesim Parman, Yoshiki Sekijima, Philip N. Hawkins, Scott D. Solomon, Michael Polydefkis, Peter J. Dyck, Pritesh J. Gandhi, Sunita Goyal, Jihong Chen, Andrew L. Strahs, Saraswathy V. Nochur, Marianne T. Sweetser, Pushkal P. Garg, Akshay K. Vaishnav, Jared A. Gollob, and Ole B. Suhr. Patisiran, an RNAi therapeutic, for hereditary transthyretin amyloidosis. *New England Journal of Medicine*, 379(1):11–21, 2018. doi: 10.1056/NEJMoa1716153.
- Jiahao Ai and Zhimei Ren. Not all distributional shifts are equal: Fine-grained robust conformal inference. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 641–665. PMLR, 2024.
- Akin Akinc, Andreas Zumbuehl, Michael Goldberg, Elizaveta S. Leshchiner, Valentina Busini, Naushad Hossain, Sergio A. Bacallado, David N. Nguyen, Jason Fuller, Rene Alvarez, Anna Borodovsky, Todd Borland, Rainer Constien, Antonin de Fougères, J. Robert Dorkin, K. Narayanannair Jayaprakash, Muthusamy Jayaraman, Matthias John, Victor Koteliensky, Muthiah Manoharan, Lubomir Nechev, June Qin, Timothy Racie, Denitza Raitcheva, Kallanthottathil G. Rajeev, Dinah W. Y. Sah, Jürgen Soutschek, Ivanka Toudjarska, Hans-Peter Vornlocher, Tracy S. Zimmermann, Robert Langer, and Daniel G. Anderson. A combinatorial library of lipid-like materials for delivery of RNAi therapeutics. *Nature Biotechnology*, 26(5):561–569, 2008. doi: 10.1038/nbt1402.
- Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems*, volume 34, pp. 17981–17993, 2021.
- Lindsey R. Baden, Hana M. El Sahly, Brandon Essink, Karen Kotloff, Sharon Frey, Rick Novak, David Diemert, Stephen A. Spector, Nadine Rouphael, C. Buddy Creech, John McGettigan, Shishir Khetan, Nathan Segall, Joel Solis, Adam Brosz, Carlos Fierro, Howard Schwartz, Kathleen Neuzil, Lawrence Corey, Peter Gilbert, Holly Janes, Dean Follmann, Mary Marovich, John Mascola, Laura Polakowski, Julie Ledgerwood, Barney S. Graham, Hamilton Bennett, Rolando Pajon, Conor Knightly, Brett Leav, Weiping Deng, Honghong Zhou, Shu Han, Melanie Ivarsson, Jacqueline Miller, and Tal Zaks. Efficacy and safety of the mRNA-1273 SARS-CoV-2 vaccine. *New England Journal of Medicine*, 384(5):403–416, 2021. doi: 10.1056/NEJMoa2035389.
- John Bradshaw, Brooks Paige, Matt J. Kusner, Marwin Segler, and José Miguel Hernández-Lobato. Barking up the right tree: An approach to search over molecule synthesis DAGs. In *Advances in Neural Information Processing Systems*, volume 33, pp. 6852–6866, 2020.
- Andrew Campbell, Jason Yim, Regina Barzilay, Tom Rainforth, and Tommi Jaakkola. Generative flows on discrete state-spaces: Enabling multimodal flows with applications to protein co-design. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 5453–5512. PMLR, 2024.
- Qiang Cheng, Tuo Wei, Lukas Farbiak, Lindsay T. Johnson, Sean A. Dilliard, and Daniel J. Siegwart. Selective organ targeting (SORT) nanoparticles for tissue-specific mRNA delivery and CRISPR-Cas gene editing. *Nature Nanotechnology*, 15(4):313–320, 2020. doi: 10.1038/s41565-020-0669-6.
- Miruna Cretu, Charles Harris, Ilia Igashov, Arne Schneuing, Marwin Segler, Bruno Correia, Julien Roy, Emmanuel Bengio, and Pietro Liò. SynFlowNet: Design of diverse and novel molecules with synthesis constraints. In *International Conference on Learning Representations*, 2025.
- Pieter R. Cullis and Michael J. Hope. Lipid nanoparticle systems for enabling gene therapies. *Molecular Therapy*, 25(7):1467–1475, 2017. doi: 10.1016/j.ymthe.2017.03.013.

- Alexander Dömling. Recent developments in isocyanide based multicomponent reactions in applied chemistry. *Chemical Reviews*, 106(1):17–89, 2006. doi: 10.1021/cr0505728.
- Alexander Dömling and Ivar Ugi. Multicomponent reactions with isocyanides. *Angewandte Chemie International Edition*, 39(18):3168–3210, 2000.
- Floor Eijkelboom, Grigory Bartosh, Christian A. Naesseth, Max Welling, and Jan-Willem van de Meent. Variational flow matching for graph generation. In *Advances in Neural Information Processing Systems*, volume 37, pp. 11735–11764, 2024.
- Wenhao Gao, Rocío Mercado, and Connor W. Coley. Amortized tree generation for bottom-up synthesis planning and synthesizable molecular design. In *International Conference on Learning Representations*, 2022.
- Itai Gat, Tal Remez, Neta Shaul, Felix Kreuk, Ricky T. Q. Chen, Gabriel Synnaeve, Yossi Adi, and Yaron Lipman. Discrete flow matching. In *Advances in Neural Information Processing Systems*, volume 37, pp. 133345–133385, 2024.
- Samuel Genheden, Amol Thakkar, Veronika Chadimová, Jean-Louis Reymond, Ola Engkvist, and Esben Bjerrum. AiZynthFinder: A fast, robust and flexible open-source software for retrosynthetic planning. *Journal of Cheminformatics*, 12:70, 2020. doi: 10.1186/s13321-020-00472-1.
- Julian D. Gillmore, Ed Gane, Jorg Taubel, Justin Kao, Marianna Fontana, Michael L. Maitland, Jessica Seitzer, Daniel O’Connell, Kathryn R. Walsh, Kristy Wood, Jonathan Phillips, Yuanxin Xu, Adam Amaral, Adam P. Boyd, Jeffrey E. Cehelsky, Mark D. McKee, Andrew Schiermeier, Olivier Harari, Andrew Murphy, Christos A. Kyratsous, Brian Zambrowicz, Randy Soltys, David E. Gutstein, John Leonard, Laura Sepp-Lorenzino, and David Lebwohl. CRISPR-Cas9 in vivo gene editing for transthyretin amyloidosis. *New England Journal of Medicine*, 385(6):493–502, 2021. doi: 10.1056/NEJMoa2107454.
- Khalid A. Hajj, Rebecca L. Ball, Sarah B. Deluty, Shridhar R. Singh, Daria Strelkova, Christopher M. Knapp, and Kathryn A. Whitehead. Branched-tail lipid nanoparticles potently deliver mRNA in vivo due to enhanced ionization at endosomal pH. *Small*, 15(6):1805097, 2019. doi: 10.1002/sml.201805097.
- Xuexiang Han, Hanwen Zhang, Kamila Butowska, Kelsey L. Swingle, Mohamad-Gabriel Alameh, Drew Weissman, and Michael J. Mitchell. An ionizable lipid toolbox for RNA delivery. *Nature Communications*, 12:7233, 2021. doi: 10.1038/s41467-021-27493-0.
- Marine Z. C. Hatit, Curtis N. Dobrowolski, Melissa P. Lokugamage, David Loughrey, Huanzhen Ni, Chiara Zurla, Alejandro J. Da Silva Sanchez, Afsane Radmand, Sebastian G. Huayamares, Ryan Zenhausern, Kalina Paunovska, Hannah E. Peck, Jinwhan Kim, Manaka Sato, Jacob I. Feldman, Michael-Alexander Rivera, Ana Cristian, YongTae Kim, Philip J. Santangelo, and James E. Dahlman. Nanoparticle stereochemistry-dependent endocytic processing improves in vivo mRNA delivery. *Nature Chemistry*, 15(4):508–515, 2023. doi: 10.1038/s41557-023-01138-9.
- Xucheng Hou, Tal Zaks, Robert Langer, and Yizhou Dong. Lipid nanoparticles for mRNA delivery. *Nature Reviews Materials*, 6(12):1078–1094, 2021. doi: 10.1038/s41578-021-00358-0.
- Muthusamy Jayaraman, Steven M. Ansell, Barbara L. Mui, Ying K. Tam, Jianxin Chen, Xinyao Du, David Butler, Laxman Eltepu, Shigeo Matsuda, Jayaprakash K. Narayanannair, Kallanthottathil G. Rajeev, Ismail M. Hafez, Akin Akinc, Martin A. Maier, Mark A. Tracy, Pieter R. Cullis, Thomas D. Madden, Muthiah Manoharan, and Michael J. Hope. Maximizing the potency of siRNA lipid nanoparticles for hepatic gene silencing in vivo. *Angewandte Chemie International Edition*, 51(34):8529–8533, 2012. doi: 10.1002/anie.201203263.
- Michał Koziarski, Andrei Rekesh, Dmytro Shevchuk, Almer van der Sloot, Piotr Gaiński, Yoshua Bengio, Cheng-Hao Liu, Mike Tyers, and Robert Batey. RGFN: Synthesizable molecular generation using GFlowNets. In *Advances in Neural Information Processing Systems*, volume 37, pp. 46908–46955, 2024.

- Greg Landrum, Paolo Tosco, Brian Kelley, Ricardo Rodriguez, David Cosgrove, Riccardo Vianello, Sriniker, Peter Gedeck, Gareth Jones, Eisuke Kawashima, Nadine Schneider, Dan Nealschneider, tadhurst-cdd, Andrew Dalke, Matt Swain, Brian Cole, Samo Turk, Aleksandr Savelev, Niels Maeder, Rachel Walker, Alain Vaucher, Maciej Wójcikowski, Hussein Faara, Ichiru Take, Vincent F. Scalfani, Daniel Probst, Kazuya Ujihara, Yakov Pechersky, Jeremy Monat, and Juuso Lehtivarjo. RDKit: Open-source cheminformatics. Zenodo, 2026. Release 2025.09.6.
- Joongwon Lee, Seonghwan Kim, Seokhyun Moon, Hyunwoo Kim, and Woo Youn Kim. FragFM: Hierarchical framework for efficient molecule generation via fragment-level discrete flow matching. In *International Conference on Learning Representations*, 2026. arXiv:2502.15805.
- Sang M. Lee, Qiang Cheng, Xueliang Yu, Shuai Liu, Lindsay T. Johnson, and Daniel J. Siegwart. A systematic study of unsaturation in lipid nanoparticles leads to improved mRNA transfection in vivo. *Angewandte Chemie International Edition*, 60(11):5848–5853, 2021. doi: 10.1002/anie.202013927.
- Bowen Li, Rajith Singh Manan, Shun-Qing Liang, Akiva Gordon, Allen Jiang, Andrew Varley, Guangping Gao, Robert Langer, Wen Xue, and Daniel Anderson. Combinatorial design of nanoparticles for pulmonary mRNA delivery and genome editing. *Nature Biotechnology*, 41(10):1410–1415, 2023. doi: 10.1038/s41587-023-01679-x.
- Bowen Li, Idris O. Raji, Akiva G. R. Gordon, Lizhuang Sun, Theresa M. Raimondo, Favour A. Oladimeji, Allen Y. Jiang, Andrew Varley, Robert S. Langer, and Daniel G. Anderson. Accelerating ionizable lipid discovery for mRNA delivery using machine learning and combinatorial chemistry. *Nature Materials*, 23(7):1002–1008, 2024. doi: 10.1038/s41563-024-01867-3.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- Yaron Lipman, Marton Havasi, Peter Holderrieth, Neta Shaul, Matt Le, Brian Karrer, Ricky T. Q. Chen, David Lopez-Paz, Heli Ben-Hamu, and Itai Gat. Flow matching guide and code. *arXiv preprint arXiv:2412.06264*, 2024.
- Kevin T. Love, Kerry P. Mahon, Christopher G. Levins, Kathryn A. Whitehead, William Querbes, J. Robert Dorkin, June Qin, William Cantley, Liu Liang Qin, Timothy Racie, Maria Frank-Kamenetsky, Ka Ning Yip, Rene Alvarez, Dinah W. Y. Sah, Antonin de Fougères, Kevin Fitzgerald, Victor Kotliansky, Akin Akinc, Robert Langer, and Daniel G. Anderson. Lipid-like materials for low-dose, in vivo gene silencing. *Proceedings of the National Academy of Sciences*, 107(5):1864–1869, 2010. doi: 10.1073/pnas.0910603106.
- Shitong Luo, Wenhao Gao, Zuofan Wu, Jian Peng, Connor W. Coley, and Jianzhu Ma. Projecting molecules into synthesizable chemical spaces. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 33289–33304. PMLR, 2024.
- Yuxuan Ou, Jingyi Zhao, Austin Tripp, Morteza Rasoulboroujeni, and José Miguel Hernández-Lobato. A deep generative model for the design of synthesizable ionizable lipids. In *NeurIPS 2024 Workshop on AI for New Drug Modalities*, 2024. arXiv:2412.00928.
- Fernando P. Polack, Stephen J. Thomas, Nicholas Kitchin, Judith Absalon, Alejandra Gurtman, Stephen Lockhart, John L. Perez, Gonzalo Pérez Marc, Edson D. Moreira, Cristiano Zerbin, Ruth Bailey, Kena A. Swanson, Satrajit Roychoudhury, Kenneth Koury, Ping Li, Warren V. Kalina, David Cooper, Robert W. Frenck, Laura L. Hammitt, Özlem Türeci, Haylene Nell, Axel Schaefer, Serhat Ünal, Dina B. Tresnan, Susan Mather, Philip R. Dormitzer, Uğur Şahin, Kathrin U. Jansen, and William C. Gruber. Safety and efficacy of the BNT162b2 mRNA Covid-19 vaccine. *New England Journal of Medicine*, 383(27):2603–2615, 2020. doi: 10.1056/NEJMoa2034577.
- Yiming Qin, Manuel Madeira, Dorina Thanou, and Pascal Frossard. DeFoG: Discrete flow matching for graph generation. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 50269–50326. PMLR, 2025.
- David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.

- Staci Sabnis, E. Sathyajith Kumarasinghe, Timothy Salerno, Cosmin Mihai, Tatiana Ketova, Joseph J. Senn, Andy Lynn, Alex Bulychiev, Iain McFadyen, Joyce Chan, Örn Almarsson, Matthew G. Stanton, and Kerry E. Benenato. A novel amino lipid series for mRNA delivery: Improved endosomal escape and sustained pharmacology and safety in non-human primates. *Molecular Therapy*, 26(6):1509–1519, 2018. doi: 10.1016/j.ymthe.2018.03.010.
- Sean C. Semple, Akin Akinc, Jianxin Chen, Ammen P. Sandhu, Barbara L. Mui, Connie K. Cho, Dinah W. Y. Sah, Derrick Stebbing, Erin J. Crosley, Ed Yaworski, Ismail M. Hafez, J. Robert Dorkin, June Qin, Kieu Lam, Kallanthottathil G. Rajeev, Kim F. Wong, Lloyd B. Jeffs, Lubomir Nechev, Merete L. Eisenhardt, Muthusamy Jayaraman, Mikameh Kazem, Martin A. Maier, Masuna Srinivasulu, Michael J. Weinstein, Qingmin Chen, Rene Alvarez, Scott A. Barros, Soma De, Sandra K. Klimuk, Todd Borland, Verbena Kosovrasti, William L. Cantley, Ying K. Tam, Muthiah Manoharan, Marco A. Ciufolini, Mark A. Tracy, Antonin de Fougères, Ian MacLachlan, Pieter R. Cullis, Thomas D. Madden, and Michael J. Hope. Rational design of cationic lipids for siRNA delivery. *Nature Biotechnology*, 28(2):172–176, 2010. doi: 10.1038/nbt.1602.
- Clément Vignac, Igor Krawczuk, Antoine Siraudin, Bohan Wang, Volkan Cevher, and Pascal Frossard. DiGress: Discrete denoising diffusion for graph generation. In *International Conference on Learning Representations*, 2023.
- K. Yu. Vlasova, A. Kerr, N. D. Pennock, A. Jozic, D. K. Sahel, M. Gautam, N. T. V. Murthy, A. Roberts, M. W. Ali, K. D. MacDonald, J. M. Walker, R. Luxenhofer, and G. Sahay. Synthesis of ionizable lipopolymers using split-Ugi reaction for pulmonary delivery of various size RNAs and gene editing. *Nature Communications*, 16:4021, 2025. doi: 10.1038/s41467-025-59136-z.
- Yashuai Wang, Xiaoyin Liu, Haiyin Yang, Shuxiang Wang, Shutao Guo, Yuhua Weng, Jinchao Zhang, Luwei Li, Kun Ge, and Xing-Jie Liang. Ugi-reaction-derived ionizable lipids with cyclic tertiary amine heads enable spleen-targeted mRNA delivery. *Biomaterials Science*, 13(16):4494–4501, 2025. doi: 10.1039/d5bm00738k.
- Kathryn A. Whitehead, J. Robert Dorkin, Arturo J. Vegas, Philip H. Chang, Omid Veischi, Jonathan Matthews, Owen S. Fenton, Yunlong Zhang, Karsten T. Olejnik, Volkan Yesilyurt, Delai Chen, Scott Barros, Boris Klebanov, Tatiana Novobrantseva, Robert Langer, and Daniel G. Anderson. Degradable lipid nanoparticles with predictable in vivo siRNA delivery activity. *Nature Communications*, 5:4277, 2014. doi: 10.1038/ncomms5277.
- Yue Xu, Shihao Ma, Haotian Cui, Jingan Chen, Shufen Xu, Fanglin Gong, Alex Golubovic, Muye Zhou, Kevin Chang Wang, Andrew Varley, Rick Xing Ze Lu, Bo Wang, and Bowen Li. AGILE platform: A deep learning powered approach to accelerate LNP development for mRNA delivery. *Nature Communications*, 15:6305, 2024. doi: 10.1038/s41467-024-50619-z.
- Kevin Yang, Kyle Swanson, Wengong Jin, Connor Coley, Philipp Eiden, Hua Gao, Angel Guzman-Perez, Timothy Hopper, Brian Kelley, Miriam Mathea, Andrew Palmer, Volker Settels, Tommi Jaakkola, Klavs Jensen, and Regina Barzilay. Analyzing learned molecular representations for property prediction. *Journal of Chemical Information and Modeling*, 59(8):3370–3388, 2019. doi: 10.1021/acs.jcim.9b00237. Correction: *J. Chem. Inf. Model.* 59(12):5304–5305, 2019, doi:10.1021/acs.jcim.9b01076.
- Jingyi Zhao, Yuxuan Ou, Austin Tripp, Morteza Rasoulboroujeni, and José Miguel Hernández-Lobato. Generative model for synthesizing ionizable lipids: A Monte Carlo tree search approach. arXiv:2412.00807, 2024.
- Weihe Zhong, Ziduo Yang, and Calvin Yu-Chian Chen. Retrosynthesis prediction using an end-to-end graph generative architecture for molecular graph editing. *Nature Communications*, 14:3009, 2023. doi: 10.1038/s41467-023-38851-5.

Table 2: Reference sets used throughout the paper, and what each comparison means.

| Set                | Size    | Labels? | Used for                        | Meaning of absence                                           |
|--------------------|---------|---------|---------------------------------|--------------------------------------------------------------|
| Structural corpus  | 112,386 | no      | training the flow               | the design was not seen during structural training           |
| Development fold   | 66,464  | no      | development only                | <i>not</i> a valid novelty denominator for production        |
| Measured lipids    | 1,100   | yes     | predictor and $S_{\text{pred}}$ | no functional evidence, <i>not</i> predicted inactive        |
| Finite enumeration | 12,276  | no      | reference chemical space        | outside the precursor inventories the library was built from |

## APPENDIX

### A MODEL, REPRESENTATION, DATA, AND TRAINING

#### A.1 NOTATION, CORPORA, AND REFERENCE SETS

**Notation.** Table 3 fixes the symbols used throughout. Symbols belonging to the stratified-conformal lower-bound rule used for the pilot are absent; that rule is documented in §F.9.

Table 3: Notation.

| Symbol                                            | Meaning                                                                     |
|---------------------------------------------------|-----------------------------------------------------------------------------|
| $G = (V, E, x^V, x^E)$                            | complete molecular graph, with atom and bond attributes                     |
| $\mathcal{R}$                                     | the modular chemistry; here the Ugi three-component reaction                |
| $\mathcal{KR}$                                    | precursor roles; here amine, aldehyde, isocyanide                           |
| $c = (c_a, c_d, c_i)$                             | design context: per-role architectural counts, carrying no identity         |
| $r_c$                                             | role map, serialized position $\mapsto$ role; deterministic given $c$       |
| $z_t, t$                                          | corrupted state and flow time; $t = 0$ source, $t = 1$ data                 |
| $p_0^{h,r}$                                       | role-specific empirical source marginal for head $h$                        |
| $\lambda_h, w_r$                                  | head weight and role weight in the objective                                |
| $D_{\mathcal{R}}$                                 | component projection, $(G, c) \mapsto (B_a, B_d, B_i)$                      |
| $A_{\mathcal{R}}(G, c)$                           | forward-assembly admission predicate                                        |
| $I_c$                                             | declared structural invariants of the instantiated context                  |
| $S_{\text{gen}}, S_{\text{pred}}, S_{\text{act}}$ | generative, evidence-supported and actionable supports                      |
| $E(G)$                                            | frozen applicability predicate defining $S_{\text{pred}}$                   |
| $\rho(c)$                                         | probability that context $c$ terminates in $S_{\text{pred}}$                |
| $\pi_0, \pi_\gamma$                               | broad and evidence-aware context proposals                                  |
| $\gamma, \epsilon$                                | tilt exponent and floor in $\pi_\gamma$                                     |
| $\Delta_{\text{sem}}$                             | empirical semantic-utility gap, $L_{\text{no-role}} - L_{\text{true-role}}$ |

**Corpora.** The broad ionizable-lipid corpus contains 15,229 unique connected molecular graphs after standardization. Measured AGILE records were reconciled against the primary dataset and the independently curated LANTERN release: of 1,200 nominal experimental records, 1,100 corresponded to exact single-compound structures and were retained for molecular supervision, while 100 mixture-derived executions were retained as experimental provenance and excluded from single-graph supervision. The associated virtual dataset contributed 12,276 transform-consistent products, and after constitutional deduplication the model union contained 12,386 unique product-component records.

To widen the available chemistry without introducing arbitrary structures, components were assembled into an expanded registry admitted only after standardization, validation of the required functional handle, and exact forward assembly. That registry holds 424 components: 264 amines, 107 aldehydes and 53 isocyanides. Exhaustive forward enumeration over it yields 1,486,415 valid products. The 112,386-product production set is the complete model union plus 100,000 deterministically selected expanded products.

Sampling of expanded products was hierarchical rather than uniform over the Cartesian product. Products were stratified by component provenance, component family, branching, unsaturation and linker-related chemistry, so that heavily represented component classes could not dominate the training distribution by combinatorial weight alone. Development partitions contain 66,464 training, 15,800 calibration and 30,122 component-family-held-out products. Table 2 records which set each comparison in the paper is made against.

**Standardization and identity.** The corpus is standardized at the level of constitutional molecular identity. Salts, disconnected structures, mixtures and duplicate source records were reconciled under a prespecified procedure, and stereochemical annotation was retained separately from the identity used for modelling. Unless stated otherwise, molecular identity is canonical non-isomeric SMILES after preprocessing. Generation is constitutional, so FORGE proposes no stereochemistry; every comparison against a reference set that carries stereochemistry is therefore made stereo-free on *both* sides. The enumerated library retains stereochemistry on 3,652 of its members, which is why comparing raw canonical SMILES against it inflates novelty. All structure processing and graph comparison used RDKit release 2025.09.6 (Landrum et al., 2026). That release produced every frozen artifact behind a number reported in this paper; three ancillary diagnostic artifacts that back no reported number were produced under a later release, and are identified in Appendix E.2. Parsing uses RDKit’s default sanitization through `MolFromSmiles`, and identity is `MolToSmiles` with `canonical=True` and `isomericSmiles=False`, which is what makes the identity constitutional. Parser logging is suppressed during bulk reads so that a warning stream cannot be mistaken for a failure, and an unparseable SMILES raises rather than being silently dropped, so a malformed record cannot quietly shrink a denominator. Descriptor and fingerprint computations run on the same sanitized molecule object, so no comparison in this paper mixes a sanitized structure with an unsanitized one.

## A.2 REACTION-DEFINED STRUCTURAL PROGRAM AND SEMANTIC MAP

The design context supplies only coarse per-role architecture: the number of exterior atoms, the junction budget, the cycle rank and the core-attachment count for each precursor-derived region. It contains no component identity, no stored molecular fragment and no lookup key. Those counts are nonetheless what fix which serialized positions belong to which role, which is the sense in which  $r_c$  is deterministic given  $c$  rather than predicted. Appendix C.5 takes up the consequence of that determinism for how the semantic ablation must be interpreted.

Component-family partitions were defined before products were assigned to data splits, and **splitting was performed before component extraction**, so information from a held-out product cannot re-enter training through a reused component. This matters more here than in a typical molecular split: the corpus is a combinatorial enumeration, and a product-level random split leaks badly because a held-out product shares two of its three components with training products. Family-level folds make the held-out set genuinely held out.

## A.3 MOLECULAR SERIALIZATION AND CATEGORICAL STATE SPACE

Each accepted product is encoded as a deterministic spanning tree together with its sparse residual closure edges. Serialization is anchored to the fixed five-atom Ugi core and divides the remainder into three role regions, concatenated in role-blocked order. Positions therefore carry both a global index and a role-relative coordinate; the ablation arms of Appendix C.2 differ in whether the role-relative coordinate is available, never in whether a global coordinate is.

Table S15 gives the seven categorical vocabularies the model emits and the eight declared support bounds. Generation is non-enumerative but bounded: at most 80 exterior atoms per product and 32 per precursor-derived component, at most three children per node, seven junction units, two cycles per component and two core attachments. Structures outside those bounds are *not produced*, rather than produced and filtered afterwards, which is what makes  $S_{\text{gen}}$  a declared support rather than a post-hoc screen.

#### A.4 PREDICTION HEADS

The production model emits seven categorical outputs through one shared role-aware backbone rather than through separate topology and chemistry models: offspring branching state, atom state, parent-bond state, closure-bond state, decoration atom state, decoration bond state, and the decoration anchor.

Two analyses in this paper report over six heads, and *they are not the same six*. Stating this here is not pedantry: substituting one set for the other converts a true claim into a false one.

The semantic pilot of §7.1 reports the aggregate total together with atom state, parent bond, offspring topology, closure bond and decoration anchor. All six improve under correct alignment in every seed. This is what “six heads” means in the main text.

The role-divergence analysis of §C.6 instead requires a role-conditional target *distribution* per head, so it drops the decoration anchor, whose target is a position rather than a categorical chemical state, and adds the decoration atom and decoration bond heads. Over that set the semantic gap is not uniformly positive, and no claim is made over it.

#### A.5 TRAINING OBJECTIVE

All molecular variables are discrete, and topology and chemical state are interpolated independently at each sampled time between the role-specific source marginal and the target, along the path of Eq. (4). Training times are sampled uniformly on  $[0.02, 0.98]$ ; the interval is truncated because at  $t$  exactly 0 or 1 the path endpoints carry no gradient signal about the interior. The denoiser is trained by endpoint prediction rather than by predicting a transition kernel, so a single forward pass supplies a distribution over clean states at every position.

The objective is Eq. (5). Role-balanced weighting exists because the roles differ systematically in size: an ionizable lipid’s hydrophobic regions carry many more serialized variables than its head, so an unweighted token average would let tail chemistry dominate the gradient purely by count, and the ionizable head is the region whose chemistry most determines delivery. Targets that are inapplicable at a given position, such as a decoration target where no decoration exists, are excluded from the sum rather than supervised toward a null class; the aggregate is rebuilt from the defined terms so that an empty group contributes nothing instead of poisoning the total. Two properties of Eq. (5) are worth stating explicitly because they are easy to assume otherwise. There are no per-head weights: the seven head losses enter the total with coefficient one, and the balance between topology and chemistry comes from the averaging structure rather than from tuned coefficients. And role balancing is not a per-position weight. It is realized as a mean of role-conditional cross-entropies, which is not expressible as fixed constants  $w_r$ , because the implied per-position weight  $1/(|\mathcal{K}_{\mathcal{R}}| n_r)$  depends on how many positions  $n_r$  that role happens to carry in the molecule at hand.

Two further terms, cycle rank and attachment count, are added to the total when the corresponding global-state outputs are supervised, so the objective carries nine terms in that configuration rather than seven. Where a head has no applicable target in a batch, for instance a closure loss in a batch containing no closure bonds, the term is contributed as an explicit zero rather than skipped, so an empty group cannot make the total undefined.

The flat arms of Appendix C.2 replace the role-balanced average with a pooled one over the whole molecule. This is deliberate and is part of the parity contract: a role-balanced objective is itself role structure, so leaving it in an arm whose treatment is the absence of the role channel would confound the comparison. The pooled objective is applied identically to every flat arm, including the true-role one, so those arms differ in the role channel and not in the objective.

**Role-specific source marginals.** The source distributions  $p_0^{h,r}$  are empirical frequencies stored with each checkpoint: offspring branching, atom state and parent bond as  $3 \times |\mathcal{V}_h|$  tables indexed by role, and closure bonds, decoration presence, decoration atoms and decoration bonds as single pooled vectors, since those targets are not partitioned by role.

The three role rows differ substantially, which is the point of conditioning on role at all. Table 4 gives the atom-state and offspring rows for the production generator.

Table 4: Production source marginals by role, leading entries. The amine role is markedly more heteroatom-rich than either tail, and the two tails are close to pure carbon, so a position’s role alone already narrows what it is likely to be at high corruption.

| Role                      | atom state 0 | atom state 1 | atom state 3 | atom state 8 |
|---------------------------|--------------|--------------|--------------|--------------|
| amine                     | 0.7294       | 0.1187       | 0.1120       | 0.0206       |
| aldehyde                  | 0.9586       | 0.0000       | 0.0000       | 0.0414       |
| isocyanide                | 0.9740       | 0.0125       | 0.0000       | 0.0115       |
| offspring branching state |              |              |              |              |
| amine                     | 0.3095       | 0.5325       | 0.1520       | 0.0059       |
| aldehyde                  | 0.0839       | 0.8964       | 0.0197       | 0.0000       |
| isocyanide                | 0.1261       | 0.8371       | 0.0284       | 0.0085       |

This is the concrete content behind the claim in §C.5 that role identifies which marginal a position was drawn from: at small  $t$  a position resampled from the amine marginal looks different from one resampled from either tail, and knowing the role is what tells the denoiser which it is.

The pilot arms of Appendix C.2 instead use a single pooled source marginal, identical across roles, in *every* arm including the true-role one. This is the same parity reasoning that governs the objective: a role-specific source is itself role structure, so leaving it in place would mean the arms differed in two things rather than one.

## A.6 ARCHITECTURE

Table S14 gives the architecture and the parameter budget in full. In summary: a four-layer bidirectional recurrent backbone at hidden dimension 192 with dropout 0.15, totalling 1,159,916 trainable parameters, of which the backbone carries 57.6%, the output heads 23.1%, the shared projection and normalization 9.7% and the input embeddings 9.6%. Inputs combine the current categorical state with role, within-role position, flow time and the coarse per-role architecture variables.

The architectural point that matters for the paper’s claim is negative: the model holds no component identifiers and no stored molecular fragments. There is no table from which a head group or a tail can be retrieved, so every component is constructed atom by atom under the declared support. A generated component that happens to coincide with a catalogue material does so because the flow built it, not because it was selected.

## A.7 SAMPLING AND SUPPORT MASKS

Sampling initializes every position from the same role-specific source marginals used in training and takes eight discrete reverse-star transitions. At each transition the denoiser predicts a clean categorical state per position, and the reverse kernel induced by the path of Eq. (4) is formed from that endpoint prediction; masks are applied to the resulting logits and the survivors are renormalized, so a forbidden transition receives exactly zero probability rather than a small one.

Four mask families enforce the declared support: valence masks forbid an atom exceeding its allowed degree; duplicate-edge and self-edge masks forbid a multigraph or a loop; the adapter-boundary mask forbids an edge crossing a role boundary anywhere other than at the core; and counting masks forbid exceeding the per-component atom, child, junction, cycle and core-attachment budgets. Because the masks are predicates on candidate successors evaluated against those bounds, and because a masked state always retains a completion within them, sampling cannot paint itself into a corner. That is Proposition 3.

At termination the state has no pending frontier and is decoded to a molecular graph, projected to its three components by  $D_{\mathcal{R}}$ , and admitted only if  $A_{\mathcal{R}}(G, c) = 1$ . Samples failing admission are discarded, never repaired and never resampled toward a target; the source-probability floor is  $10^{-5}$ .

---

**Algorithm 1** Training FORGE. One step; role-balanced endpoint prediction.

---

**Require:** corpus  $\mathcal{D}$ , role map  $r_c$ , source marginals  $p_0^{h,r}$ , weights  $\lambda_h, w_r$

- 1: sample a minibatch  $\{(G, c)\} \sim \mathcal{D}$
- 2: sample  $t \sim \text{Uniform}[0.02, 0.98]$
- 3: **for** each position  $u$  and head  $h$  **do**
- 4:    $r \leftarrow r_c(u)$
- 5:   draw  $z_u^h \sim q_t(\cdot \mid x_u^h, r)$  {Eq. (4): mix  $\delta_x$  with  $p_0^{h,r}$ }
- 6: **end for**
- 7: predict  $p_\theta(x \mid z_t, t, c)$  with the shared role-aware backbone
- 8: accumulate  $\mathcal{L}$  over defined targets only, weighting head  $h$  by  $\lambda_h$  and position  $u$  by  $w_{r_c(u)}$
- 9: take an AdamW step on  $\nabla_\theta \mathcal{L}$ , clipping the global norm at 1.0

---

**Algorithm 2** Sampling from FORGE. Masked reverse transitions, then admission.

---

**Require:** design context  $c$ , transition count  $N = 8$ , invariants  $I_c$

- 1:  $r_c \leftarrow$  role layout induced by the per-role counts in  $c$
- 2: initialize  $z^{(0)}$  from  $p_0^{h,r_c(u)}$  at every position  $u$  and head  $h$
- 3: **for**  $n = 0$  **to**  $N - 1$  **do**
- 4:    $t \leftarrow n/N$
- 5:    $\hat{p} \leftarrow p_\theta(x \mid z^{(n)}, t, c)$  {endpoint prediction}
- 6:   form the reverse kernel induced by  $q_t$  and  $\hat{p}$
- 7:   zero every transition violating valence, duplicate or self edges, adapter boundaries, or a declared budget
- 8:   renormalize and sample  $z^{(n+1)}$
- 9: **end for**
- 10:  $G \leftarrow$  decode  $z^{(N)}$ ;  $(B_a, B_d, B_i) \leftarrow D_{\mathcal{R}}(G, c)$
- 11: **if**  $A_{\mathcal{R}}(G, c) = 1$  **then**
- 12:   **return**  $G$  with its components
- 13: **else**
- 14:   discard {never repaired, never resampled toward a target}
- 15: **end if**

---

## A.8 TRAINING AND CHECKPOINT PROVENANCE

Table S16 gives the optimizer, schedule and hardware. Two features of the protocol bear on how the results should be read.

First, architecture selection did not use likelihood. Candidate configurations had to satisfy prespecified hard gates on validity, exact forward and inverse reconstruction, component-handle qualification, uniqueness and declared-support compliance, and among those the criterion was usable *de novo* generation: the fraction of attempts that produced a valid product, recovered all three precursor graphs, reconstructed exactly through the Ugi transform, satisfied handle requirements and contained at least one precursor absent from the 424-component registry. Selection ran on a separate paired draw of 1,024 programs. Likelihood would have chosen differently: the best development calibration-loss checkpoint occurred at step 500 and was not the configuration shipped.

Second, the production model was refit once from random initialization on all 112,386 records for a fixed 5,100 optimization steps, with no early stopping and no post-refit checkpoint selection. No held-out quantity chose the shipped weights. This is also why the novelty denominator in Table 2 is the full corpus rather than the development fold: the production model saw every structural fold, so measuring novelty against the 66,464-product fold would count calibration and held-out products as unseen and overstate novelty at 89.0% instead of 83.7%.

**Checkpoint provenance, which must not be conflated.** Two distinct checkpoints back different claims in this paper. The *production refit* at step 5,100 backs the novelty, supported-yield, route-completeness and unique-assembly results. The *step-3000* checkpoints back only the semantic pilot and the held-family structural panel. Any figure or table stating a parameter count, a training length or a sample population must name which of the two it refers to.

## B APPLICABILITY SUPPORT AND BIOLOGICAL PREDICTION

### B.1 BIOLOGICAL DATASET AND PREDICTOR

Prediction is trained on the 1,100 exact single-compound measurements described in §A.1. HeLa and RAW 264.7 transfection were treated as separate biological endpoints, and the campaign reported here was selected on *HeLa activity only*; RAW measurements carried zero weight in ranking. The selected predictor is a role-aware directed message-passing ensemble (Yang et al., 2019), evaluated under prespecified random, structural and held-component splits.

**The predictor is modest, and that is load-bearing.** The ensemble reaches an aggregate HeLa calibration  $R^2$  of 0.504 and a descriptive post-selection test  $R^2$  of 0.253 with  $\rho = 0.612$ . Performance varies substantially across component holdouts. More pointedly, a component-additive ridge matches the production message-passing ensemble on held-aldehyde-family Spearman correlation, so most of the transferable signal at that level of holdout is additive in the components rather than requiring a learned interaction model.

We report this rather than burying it, because it is the reason the architecture of §5 is what it is. A predictor a linear component model can match is not an oracle, and promoting it to a global generative guidance signal would extend an unreliable extrapolation into every molecular transition. Restricting it to a ranking device inside  $S_{\text{pred}}$  is the response the evidence supports. Each ensemble member is a sparse directed message-passing network with hidden dimension 128, depth 3, dropout 0.1 and a concatenated sum-and-mean readout. Members differ only by initialization seed, with three seeds fit per evaluation fold, and the ensemble prediction is the mean over members with the across-member standard deviation retained alongside it. Ranking within  $S_{\text{pred}}$  uses that ensemble mean on the measured transfection scale, ordered so that higher is better; the across-member standard deviation is recorded but does not enter the ordering. Two points of bookkeeping. Ties in the frozen selection break deterministically on canonical product SMILES; the current formulation has not yet been used to select a panel, so its tie-break is inherited rather than exercised. And the comparator predictors, used to establish how much of the signal a simpler model captures, are a ridge at  $\alpha = 1.0$  on standardized features, a random forest with 300 trees at  $\sqrt{d}$  features and a minimum leaf of 2, a gradient-boosted ensemble with 400 trees at depth 4, learning rate 0.03 and 0.8 subsampling, and a three-layer perceptron of widths 256, 128 and 64 with early stopping on a 15% validation split.

The two  $R^2$  values quoted above come from the development record rather than from a ledger entry, and §E.4 flags them for re-derivation before submission.

### B.2 THE FOUR-VIEW APPLICABILITY TEST

Support is assessed in four views: the complete product and its amine-, aldehyde- and isocyanide-derived components. Each view is compared against the corresponding measured reference chemistry under two criteria, a fingerprint distance and a physicochemical-descriptor distance, and a view passes only when *both* fall under threshold.

Fingerprints are radius-2 Morgan at 2048 bits (Rogers & Hahn, 2010), taken as *count* rather than binary fingerprints. Count fingerprints retain multiplicity information for repeated local environments that binary folding discards; they do not guarantee that homologues remain distinguishable after hashing, and the paper makes no such claim. The motivation is concrete: alkyl chains of different length share the same local environments, so under binary fingerprints a long-chain homologue can score as identical to a short one, and chain length is a real potency determinant for ionizable lipids.

The descriptor distance is computed over 15 physicochemical descriptors covering heavy-atom, carbon and heteroatom counts, molecular weight, logP, TPSA, rotatable bonds, ring count, absolute formal charge, double and triple bonds, branch excess, and ester, amide and ether counts. Standardization is robust rather than Gaussian: the center is the per-descriptor median of the reference set and the scale is the interquartile range, falling back to the standard deviation where the range is degenerate. That choice matters because the reference sets are small and contain long-chain outliers that would otherwise dominate a mean-and-variance scaling.

### B.3 THRESHOLD CALIBRATION

Thresholds are estimated per fold. For each view and each distance type, the interpolative maximum is the lower calibration quantile and the boundary maximum is the upper one, so the three bands follow directly from the calibration distance distribution rather than from a chosen radius. A view is interpolative when both its distances fall under the interpolative maxima, boundary when both fall under the boundary maxima, and extrapolative otherwise; the overall bin is the *worst* of the four views.

The support predicate is therefore

$$E(G) = 1 \iff \max_{v \in \{\text{product, amine, aldehyde, isocyanide}\}} \text{bin}(v) \leq \text{boundary},$$

that is, interpolative or boundary in every view, extrapolative in none. The frozen predicate is recorded as `valid_exact_ll_and_fixed_multiview_radius_at_most_one`, the radius naming exactly this: bin index at most one on the ordering interpolative, boundary, extrapolative. Thresholds are computed from structures and split stage only, never from measured activity values or predictor residuals, which is what allows  $E$  to be described as an evidence statement rather than a quality prediction.

Table 5: Per-view calibration thresholds. A view is interpolative when *both* distances fall under the interpolative maxima and boundary when both fall under the boundary maxima.

| View       | fingerprint |          | descriptor |          | calib. records |
|------------|-------------|----------|------------|----------|----------------|
|            | interp.     | boundary | interp.    | boundary |                |
| product    | 0.0368      | 0.0833   | 0.0369     | 0.1671   | 1,648          |
| amine      | 0.5366      | 0.8464   | 0.3635     | 1.0898   | 80             |
| aldehyde   | 0.1860      | 0.3301   | 0.2625     | 0.5410   | 44             |
| isocyanide | 0.1229      | 0.2667   | 0.4705     | 0.6949   | 20             |

The calibration counts differ by an order of magnitude across views, and that asymmetry is worth noting rather than smoothing over: the product view rests on 1,648 calibration products while the isocyanide view rests on 20 unique identity-excluded components, so the isocyanide thresholds are the least precisely estimated of the four. The product unit is the calibration product itself; the three component units are unique exact-identity-excluded components per fold.

### B.4 EXCLUSION CONTRACT AND ITS ASYMMETRY

For each unique calibration component, that component’s exact identity is removed from the component reference set before its distance is measured; the calibration unit is recorded as `unique_exact_identity_excluded_component_per_fold`. This exists because without it the comparison degenerates: a calibration product’s component commonly occurs in other training-fold products, which would make its component distance exactly zero and the threshold meaningless.

The exclusion is asymmetric and the asymmetry should be stated plainly. The component identity is removed from the *component* reference. Training products that contain that component remain in the *product* reference. The component-level comparison is therefore leave-one-identity-out, and it does not constitute an equally strong leave-component-out evaluation at the product level. An earlier draft of this work described the exclusion as also removing products sharing the component; that describes a stronger control than the one implemented, and the claim is withdrawn here rather than repeated.

### B.5 DESCRIPTIVE CALIBRATION

Pooled record-weighted empirical coverage of the nominal 90% residual interval is 88.4%, over 1,218 scheme-level rows drawn from 1,100 measured lipids across four evaluation schemes.

Three cautions travel with that number and none is optional. The rows exceed the lipids, because a lipid appearing in more than one scheme contributes more than one row, so the rows are not

independent observations and we report no independence-based interval. The figure is descriptive: it is not a validated 90% guarantee, and it is not used to alter candidate ordering or to widen  $S_{\text{pred}}$ . And the pooling hides how unevenly the calibration rows sit: in the frozen pilot the quantile applying to most candidates came from a single fold of twenty rows. A figure pooled over rows that concentrate that way describes the aggregate and does not transfer to any individual candidate, which is why any bound built on it is reported as descriptive. Support, ranking and uncertainty are three separate objects in the current formulation, and this quantity informs only the third.

## B.6 THE LOWER-BOUND RANKING STATISTIC

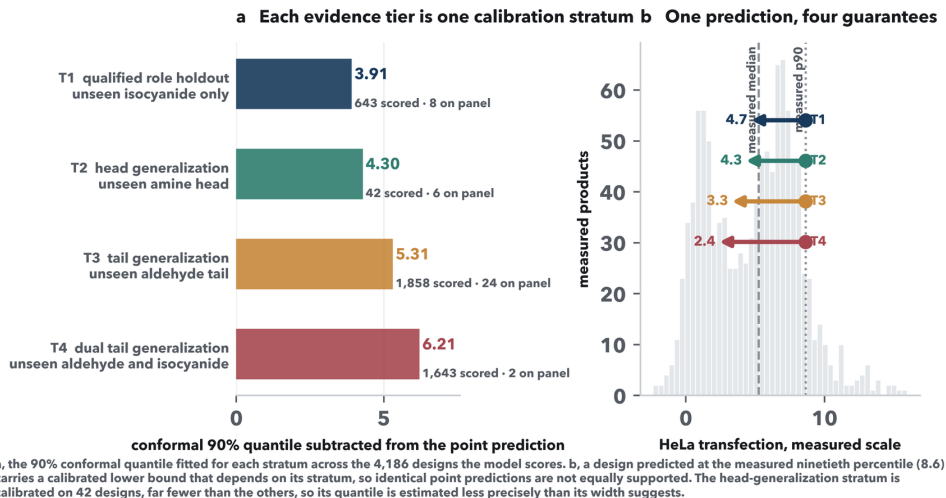


Figure 4: **Conformal calibration by evidence tier, as used for the pilot.** Reproduced for Library 0 only. The stratum quantile takes four discrete values across the pool, so within any one stratum this ranking is identical to ranking by the point prediction, and the head-generalization stratum rests on 42 calibration designs.

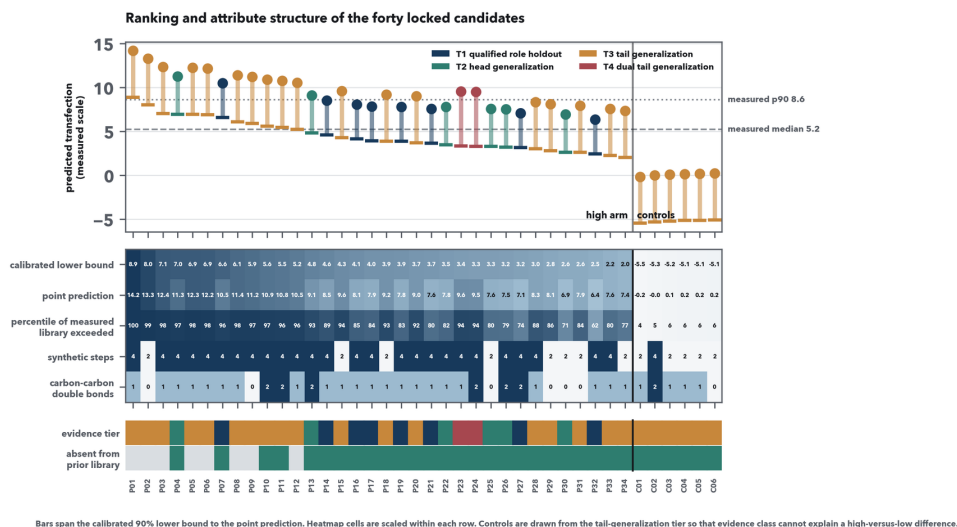


Figure 5: **The frozen ranking of the Library 0 pilot.** The ordering under the lower-bound statistic. It does not define the prospective campaign.

The frozen pilot of §F.9 was ranked by a calibrated lower bound,  $\text{lcb}_{90} = \hat{y} - q_{0.90}$ , with  $q_{0.90}$  a per-stratum conformal quantile. The current formulation ranks on the ensemble mean. The difference is

worth stating precisely, because it is a change in where applicability sits rather than the correction of an error.

**What the lower bound was doing.** The stratum quantile takes only four values across the eligible pool, so within any one stratum ranking by the lower bound is *identical* to ranking by the point prediction (Figure 4). Its whole effect is between strata, where it penalises membership in a weakly calibrated region. That is an applicability signal embedded in the score, and it was chosen deliberately for that reason, not overlooked.

**Where applicability sits instead.** Applicability now enters as the admission predicate  $E(G)$  of §B.3 rather than as a penalty on the score. One statistic was doing two jobs, deciding both whether a molecule can be ranked and where it ranks; separating them lets each be stated and checked on its own. The ranking statistic differs between the two formulations because that second job moved.

**Neither ranking can be shown superior.** Re-running the frozen selector with only the sort key changed, and with eligibility, the physicochemical envelope, route completeness, the Tanimoto ceiling, the head cap and both tie-breaks held fixed, the two rules share 22 of 34 candidates and differ on 12. On axes that neither statistic optimises the panels are close to equivalent: 13 against 14 distinct amine heads, 22 against 24 aldehyde tails, and 26 against 25 candidates absent from the enumerated library. Comparisons on median predicted potency or on the lower-bound floor are circular, each rule winning on the quantity it sorts by, and no prospective measurement exists that could break the tie. We therefore report the frozen panel under the rule it was selected with, and treat the alternative as a sensitivity analysis rather than a correction.

**What does not survive, and is not a matter of preference.** The quantity must not be described as a validated 90% bound. Its coverage was never established prospectively, and for most of the panel the offset came from a fold carrying 20 calibration rows in a scheme that missed its own coverage gate at 0.159 against a 0.100 maximum. Reporting it as a calibrated interval would be a claim the calibration does not support, independently of which statistic is used to rank.

## C SEMANTIC-UTILITY ANALYSES

### C.1 ESTIMAND AND IDENTIFIABILITY

The estimand is the paired difference in denoising loss between two arms that are identical except for the local role channel,

$$\Delta_{\text{sem}}(t) = L_{\text{no-role}}(t) - L_{\text{true-role}}(t),$$

reported per prediction head and aggregated over the six eligible heads. Positive values mean the aligned arm reconstructs the clean state better.

The first version of this experiment was designed differently and was withdrawn, and the reason determines the shape of the final design. That version sought to remove semantics entirely and compare against a semantics-free generator. It is not identifiable. Role is recoverable from the structural organization the model needs anyway: the per-role counts in  $c$  determine where each role block begins and ends, so an arm denied the explicit role tag still receives everything required to derive role. Removing role information would mean removing  $c$ , and removing  $c$  removes the design problem rather than the semantics.

What is identifiable is narrower and, we think, more interesting: the value of presenting an explicitly derivable quantity to a network of finite capacity, holding the global state and context fixed. That is what  $\Delta_{\text{sem}}$  measures. Appendix C.5 sets out the consequence, which is that  $\Delta_{\text{sem}}$  is not an information quantity and must not be reported as one.

### C.2 MATCHED EXPERIMENTAL CONTRACT

Two features of the contract deserve emphasis because they are what make the comparison a comparison rather than a confound.

*Capacity parity is exact, not approximate.* The no-role arm does not simply omit the role embedding, which would leave it with fewer parameters and confound semantics with capacity. It receives a

Table 6: Parity across the three arms. Only the local role channel differs.

| Quantity                  | no-role   | true-role | misaligned |
|---------------------------|-----------|-----------|------------|
| Graph serialization       | same      | same      | same       |
| Parameter count           | 1,150,892 | 1,150,892 | 1,150,892  |
| Initialization            | paired    | paired    | paired     |
| Minibatches               | paired    | paired    | paired     |
| Flow times $t$            | paired    | paired    | paired     |
| Corruption draws          | paired    | paired    | paired     |
| Design program $c$        | same      | same      | same       |
| Global position embedding | present   | present   | present    |
| Local role channel        | null tag  | aligned   | permuted   |

constant null tag through the same embedding of the same dimension, so every arm has an identical parameter count and an identical computation graph. The misaligned arm receives a permuted role assignment matched in dimensionality and in label frequency.

*The ablation removes experiment-specific semantics, never generic modelling capacity.* Every arm retains a global position embedding sized to the maximum total atom count. An earlier version of the specification would have denied the ablated arm any positional signal at all, which would have crippled it and made the comparison meaningless; what is removed is the role-relative coordinate and the role tag, not the ability to know where in the molecule a position sits.

Randomness is paired at every level that could otherwise dominate a paired difference: initialization, minibatch composition, corruption draws and the flow-time grid are shared across arms within a seed, so  $\Delta_{\text{sem}}$  is a within-seed contrast rather than a between-run one. Two invariants were checked rather than assumed: that the flat and structured serializations induce the same graph up to permutation, and that the design program is unchanged across arms. All seven pilot checkpoints carry exactly 1,150,892 parameters at step 3000, which is the count reported above and is distinct from the production generator’s 1,159,916.

### C.3 MAIN RESULT

Across three paired seeds the aligned map reduces total denoising loss by  $0.248 \pm 0.058$ , and the improvement is positive for all six reported heads in every seed. Sign consistency across every reported head and every seed is the load-bearing observation here rather than the magnitude: with three seeds a mean difference alone would be weak evidence, whereas eighteen of eighteen head-by-seed contrasts sharing a sign is not plausibly noise. Table 14 gives every cell.

The effect is strongly corruption-dependent. Taking the ratio of  $\Delta_{\text{sem}}$  near the corrupted end of the path to  $\Delta_{\text{sem}}$  near the clean endpoint gives 7.7, 9.89 and 11.3 across the three seeds (Table 15). We report the three values rather than a range because the spread is itself informative: the direction is identical in every seed while the magnitude varies by roughly 50%, so the qualitative dependence replicates even though its size is not tightly estimated at  $n = 3$ .

**Two different six-head sets exist in this project; they are not interchangeable.** This report’s six are the semantic pilot’s reported heads: the aggregate total together with atom state, parent bond, offspring topology, closure bond and decoration anchor. All six improve in all three seeds. A *different* six is used by the role-divergence analysis of §C.6, which drops the decoration anchor because its target is positional rather than categorical and adds the decoration atom and decoration bond heads. Those two added heads are *not* uniformly positive: decoration atom is  $-0.00219$  and decoration bond is  $-0.05537$  in seed s2, and decoration bond is marginally negative averaged over seeds. Any statement of the form “all six heads” must therefore name which six it means, and no result stated over one set may be restated over the other.

The dependence is monotone in all three seeds and spans roughly an order of magnitude between the ends of the grid, which is the pattern §C.5 accounts for: as  $t$  falls, the corrupted state carries less about the molecule while the arithmetic the ablated arm must perform on  $c$  is unchanged, so explicit role becomes progressively more load-bearing.

Table 7: The semantic-utility gap along the probability path, per paired seed. Low  $t$  is the strongly corrupted end. The gap falls monotonically toward the clean endpoint in every seed.

| $t$  | s1      | s2      | s3      | mean    |
|------|---------|---------|---------|---------|
| 0.10 | +0.5126 | +0.3206 | +0.4617 | +0.4316 |
| 0.25 | +0.4449 | +0.2801 | +0.4138 | +0.3796 |
| 0.50 | +0.2909 | +0.1827 | +0.2637 | +0.2458 |
| 0.75 | +0.1630 | +0.1017 | +0.1427 | +0.1358 |
| 0.90 | +0.0666 | +0.0324 | +0.0410 | +0.0467 |

#### C.4 MISALIGNMENT CONTROL

Misalignment recovers -1.9% of the aligned arm’s benefit. The sign is negative: the misaligned arm is not merely worse than the aligned arm, it is on aggregate slightly worse than receiving no role at all, and it is worse than the no-role arm on four of six eligible heads while being approximately tied on the remaining two.

That is the result the control was built to obtain. A channel matched in dimensionality and in label frequency, differing only in whether the correspondence to chemistry is correct, produces none of the gain. The benefit therefore cannot be attributed to the presence of an extra categorical input, to added embedding parameters, or to the label distribution; it requires the correspondence itself. Reading it alongside §C.5: a wrong-but-well-formed semantic channel gives a finite network something to fit that does not organize the molecular state, which is worse than leaving it to derive the organization from  $c$ .

#### C.5 INFORMATION-THEORETIC INTERPRETATION

Write  $X$  for the clean categorical state,  $Z_t$  for the corrupted state at flow time  $t$ ,  $C$  for the design context, and  $R$  for the per-position role assignment. Under log loss the Bayes-optimal risk of a predictor given a conditioning set is the conditional entropy of  $X$  given that set, so for the two arms

$$L_{\emptyset}^*(t) = H(X \mid Z_t, C), \quad L_R^*(t) = H(X \mid Z_t, C, R),$$

and therefore

$$L_{\emptyset}^*(t) - L_R^*(t) = I(X; R \mid Z_t, C). \quad (12)$$

Equation (12) is the identity that makes a role channel look like an information channel, and it is why the first version of this experiment was designed as an information-removal ablation. That design was abandoned, and the reason is worth stating plainly rather than leaving implicit.

**Why the measured gap is not a mutual information.** In FORGE the role map is a deterministic function of the design context:  $c$  carries the per-role atom counts, junction budgets, cycle ranks and attachment counts, and those counts fix which serialized positions belong to which role. Hence  $R$  is  $\sigma(C)$ -measurable, and the right-hand side of Eq. (12) is identically zero. At the Bayes optimum, with unbounded capacity, the two arms would achieve the same risk and  $\Delta_{\text{sem}}$  would vanish.

The gap we measure is nonetheless large and reproducible. It follows that  $\Delta_{\text{sem}}$  does not estimate an information quantity at all. It measures the *computational* value of presenting a derivable quantity explicitly to a capacity-limited network: the ablated arm must recover block boundaries from the counts in  $c$  by arithmetic over positions before it can use role at all, and at finite width and depth it does not fully do so. We therefore report  $\Delta_{\text{sem}}$  as an empirical semantic-utility gap throughout, never as a conditional mutual information, and we make no claim that it bounds one. No inequality direction is available in either sense, because neither arm attains the Bayes optimum and the ablated arm’s shortfall need not be smaller than the treated arm’s.

The capacity at which the gap would close is not reachable in this setting, which is what makes the limit a mathematical statement rather than a practical one. Capacity here is bounded by data and not by compute: the corpus holds 112,386 products, and the pilot resolves the gap at 1,150,892 parameters. Scaling toward the regime where the two arms converge would overfit this corpus long before it reached it. The claim we make is correspondingly scoped, and it is the one that matters for anyone working under comparable evidence: presenting role explicitly is worth a measurable amount at every capacity this dataset supports.

This is also what forced the design of the third arm. Since role information cannot be removed from  $C$  without removing the structural program the model needs, a genuine control cannot be an information-removal arm. The misaligned arm instead holds the channel, its dimensionality and its label frequencies fixed while destroying the correspondence, which isolates explicit correct alignment rather than channel presence. Appendix C.2 records the parity that makes the three arms comparable.

**Why utility grows as  $t \rightarrow 0$ .** As  $t \rightarrow 0$  the path in Eq. (4) approaches the role-specific source marginals, so  $Z_t$  becomes progressively less informative about  $X$  while the arithmetic the ablated arm must perform on  $c$  is unchanged. Two effects compound. The corrupted state supplies less positional disambiguation, so role must carry more of the conditioning; and the role-specific source distributions  $p_0^{h,r}$  differ across roles, so at small  $t$  the correct role is precisely what identifies which marginal a position was drawn from. Both make explicit role more load-bearing exactly where the molecular state is least informative, which is the dependence reported in §7.1.

#### C.6 PREDECLARED HYPOTHESES THAT WERE NOT SUPPORTED

Three hypotheses were declared before the corresponding quantity was computed and were not supported. All three are reported here because each constrains what the paper claims, and none is re-tested under an alternative measure.

**Semantic utility does not track role-wise marginal divergence.** The natural mechanistic story is that role helps most where the roles differ most, so we declared a measure in advance: the Jensen–Shannon divergence among the three role-conditional empirical target distributions per head, uniform mixture reference, base-2 logarithms. The association with per-head semantic utility is *negative*, Spearman  $\rho = -0.543$  over six heads. The power limitation was also stated in advance and stands: with six points only a near-perfect ordering could reach conventional significance, so this analysis is descriptive and secondary whatever it shows, and it is not featured as a primary result. No alternative divergence measure was computed after seeing this one. What it rules out is the simplest mechanistic account; it does not rule out the effect, which is established by the paired design and the misalignment control rather than by this correlation.

**Semantics do not act as a data-efficiency advantage.** We declared that if role functions as an inductive bias, its benefit should be larger under sparser supervision. Five of six heads moved in the wrong direction. The advantage is therefore not a sample-efficiency effect in the sense that hypothesis proposed, and we make no data-efficiency claim.

**Semantics do not confer held-family extrapolation.** We declared strata of increasing structural distance from the training families and predicted that semantic utility would grow with distance. The two most distant strata are negative, at  $-0.10$  and  $-0.39$ . We therefore make no claim that chemistry-informed semantics improve generalization to unseen component families, and §7.1 confines itself to denoising ambiguity under corruption, which is what the evidence supports.

Reported together these bound the claim tightly. The effect is real, replicated, and not explained by channel presence or label frequency; and it is not a data-efficiency effect, not an extrapolation effect, and not predicted by how much the role-conditional distributions differ.

Table 8: Role differentiation against semantic utility, ordered by divergence. If the simple mechanistic account held, these columns would rise together. They do not.

| Head            | JSD (bits) | mean $\Delta_{\text{sem}}$ | $\Delta_{\text{sem}}$ at $t = 0.10$ |
|-----------------|------------|----------------------------|-------------------------------------|
| decoration bond | 0.6638     | $-0.00047$                 | $-0.00463$                          |
| closure bond    | 0.2180     | $+0.01713$                 | $+0.06127$                          |
| decoration atom | 0.2081     | $+0.00280$                 | $+0.00143$                          |
| atom state      | 0.1510     | $+0.02656$                 | $+0.04720$                          |
| offspring       | 0.0953     | $+0.14092$                 | $+0.20645$                          |
| parent bond     | 0.0818     | $+0.01703$                 | $+0.03236$                          |

Spearman  $\rho = -0.543$  with a two-sided permutation  $p = 0.297$  over  $10^5$  label permutations for the seed-averaged gap, and  $\rho = -0.429$  with  $p = 0.420$  at  $t = 0.10$ . The ordering is close to inverted: the head with the largest role differentiation, decoration bond at 0.664 bits, has the only negative mean gap, while offspring topology has the smallest divergence at 0.095 bits and much the largest benefit.

## D LIPID-NATIVE MOLECULAR EVALUATION

### D.1 WHY THESE DESCRIPTORS

The panel was assembled to cover the axes along which ionizable-lipid design is known to vary, not to maximize the difference between FORGE and its baselines. That distinction is procedural rather than rhetorical, and it was enforced by a prior commitment.

An earlier version of this analysis proposed selecting the main-text metrics as “the ones the panel already discriminates on”. That is post-hoc metric selection: with 38 descriptors and a free choice of which to feature, a difference can be manufactured from noise. The commitment adopted in response fixes the Six main-text metrics by *scientific coverage* rather than by observed effect size, one per design axis, chosen before the per-descriptor comparison was read. Every remaining descriptor is reported here regardless of what it shows, which is the other half of the same commitment: a panel is only a guard against cherry-picking if the unselected members are also published.

### D.2 DESCRIPTOR DICTIONARY

Table 9: The complete lipid-native descriptor panel. Counts are over the region named in the second column; SMARTS patterns are given where a descriptor is a substructure match.

| Descriptor           | Region          | Definition                                               |
|----------------------|-----------------|----------------------------------------------------------|
| aldalkenes           | aldehyde tail   | SMARTS [CX3]=[CX3]                                       |
| aldbranches          | aldehyde tail   | SMARTS [CX4;\$ (C(-[#6])(-[#6])-[#6])]                   |
| aldcarbons           | aldehyde tail   | carbon count                                             |
| aldesters            | aldehyde tail   | SMARTS [CX3](=[OX1])[OX2][CX4]                           |
| aldethers            | aldehyde tail   | SMARTS [CX4][OX2][CX4]                                   |
| aldheavyatoms        | aldehyde tail   | heavy-atom count                                         |
| aldketones           | aldehyde tail   | SMARTS [CX4][CX3](=[OX1])[CX4]                           |
| aldlongestchain      | aldehyde tail   | longest contiguous carbon chain                          |
| clogp                | whole lipid     | Crippen logP                                             |
| degradable.linkages  | whole lipid     | ald.esters + iso.esters                                  |
| hba                  | whole lipid     | H-bond acceptors                                         |
| hbd                  | whole lipid     | H-bond donors                                            |
| head.hba             | ionizable head  | H-bond acceptors                                         |
| head.hbd             | ionizable head  | H-bond donors                                            |
| head.heavyatoms      | ionizable head  | heavy-atom count                                         |
| head.max.n.spacing   | ionizable head  | largest topological distance between two head nitrogens  |
| head.min.n.spacing   | ionizable head  | smallest topological distance between two head nitrogens |
| head.nitrogens       | ionizable head  | nitrogen count                                           |
| head.primary.amine   | ionizable head  | SMARTS [NX3;H2;!\$(NC=O)]                                |
| head.ring.nitrogen   | ionizable head  | SMARTS [nX2,NX3;R]                                       |
| head.rings           | ionizable head  | ring count                                               |
| head.secondary.amine | ionizable head  | SMARTS [NX3;H1;!\$(NC=O)]                                |
| head.tertiary.amine  | ionizable head  | SMARTS [NX3;H0;!\$(NC=O);!\$(N=*)]                       |
| head.to.tail.ratio   | ionizable head  | head.heavyatoms / total.tail.carbons                     |
| isoalkenes           | isocyanide tail | SMARTS [CX3]=[CX3]                                       |
| isobranches          | isocyanide tail | SMARTS [CX4;\$ (C(-[#6])(-[#6])-[#6])]                   |
| isocarbons           | isocyanide tail | carbon count                                             |
| isoesters            | isocyanide tail | SMARTS [CX3](=[OX1])[OX2][CX4]                           |
| isoethers            | isocyanide tail | SMARTS [CX4][OX2][CX4]                                   |
| isoheavyatoms        | isocyanide tail | heavy-atom count                                         |

| Descriptor         | Region          | Definition                                          |
|--------------------|-----------------|-----------------------------------------------------|
| iso_ketones        | isocyanide tail | SMARTS [CX4] [CX3] (= [OX1]) [CX4]                  |
| iso_longest_chain  | isocyanide tail | longest contiguous carbon chain                     |
| molecular_weight   | whole lipid     | molecular weight                                    |
| rotatable_bonds    | whole lipid     | rotatable bonds                                     |
| tail_asymmetry     | whole lipid     | $ \text{ald\_carbons} - \text{iso\_carbons} $       |
| tail_balance       | whole lipid     | $\text{ald\_carbons} / \text{total\_tail\_carbons}$ |
| total_tail_carbons | whole lipid     | $\text{ald\_carbons} + \text{iso\_carbons}$         |
| tpsa               | whole lipid     | topological polar surface area                      |

Counts are integers over the named region; `tail_balance` and `headto_tail_ratio` are dimensionless ratios, and `clogp`, `tpsa` and `molecular_weight` carry their usual units. The three regional prefixes follow the Ugi decomposition, so a descriptor prefixed `head_` is computed on the amine-derived component graph rather than on the whole lipid, which is what allows head chemistry and tail chemistry to be compared separately in §D.3.

### D.3 FULL RESULTS

The comparison is three-way: held-out real lipids, FORGE samples, and the role-marginal null. The null is the informative comparator rather than a straw man, because it shares FORGE’s declared support, masks, role-specific empirical marginals and admission procedure, and differs only in having no contextual denoiser. Any descriptor on which the null already matches the real lipids is a descriptor explained by one-variable statistics plus chemistry constraints, and FORGE deserves no credit for it.

Two results anchor the reading. The null reaches full chemical admission while building the pre-specified ionizable-head architecture in 0 of 3,072 samples, so declared support plus marginals is demonstrably not sufficient for lipid-native head chemistry. And FORGE overshoots rather than matching: 0.479 against 0.394 in the real set, a roughly 22% overshoot. We report the overshoot as an overshoot. A model that built the architecture more often than real lipids do has not reproduced the target distribution, and the grouped classifier of §D.4 says the same thing in aggregate. Table 17 in §I.2 reports all 38 descriptors.

### D.4 GROUPED REAL-VERSUS-GENERATED CLASSIFIER

A classifier trained to separate generated from held-out real lipids reaches AUC 0.882. Grouping is what makes this number meaningful: products sharing a component are not independent observations, so the split is grouped rather than random, and the reported AUC is aggregated across groups rather than pooled over products.

We interpret 0.882 as *residual separability* and nothing stronger. It is well above chance, so FORGE has not matched the real distribution and we make no distribution-matching claim; it is also well below one, so the generated designs are not trivially separable either. Read together with the per-descriptor panel, the picture is a model that has acquired lipid-native organization in the specific respects the panel names while remaining distinguishable in aggregate. Table 18 gives the per-fold values and the grouping unit. The classifier is a histogram gradient-boosted tree with 200 iterations at a fixed seed, taking the full 38-descriptor vector as its feature set, so it sees exactly the panel and no additional structural information. Folds are `GroupKFold` over the grouping unit, and the reported figure is the mean of the per-fold areas rather than a pooled curve.

### D.5 ESTER-LINKAGE DECOMPOSITION

Ester-linked tail chemistry is the clearest case of a miss that was diagnosed rather than absorbed. Early configurations produced ester linkages in 0.029 of admitted designs against a training-fold prevalence of 0.530, a nearly twenty-fold shortfall on the single most characteristic linker motif in the corpus.

The shortfall decomposed into two separable causes. Changing the architecture moved the rate from 0.029 to 0.190. Changing the training budget moved it from 0.190 to 0.510, against the 0.530

training prevalence. Each stage changed one factor, and the stages must be read that way: this is not an architecture-only causal progression, and the second step is a budget effect that would have been misattributed to representation had both been changed together. The ester-enriched held-family reference sits at 0.708, which is the relevant comparator for a matched-family evaluation rather than the train-fold prevalence.

## E REPRODUCIBILITY, CONFIGURATIONS, AND ARTIFACT REGISTRY

### E.1 WHERE THE CLAIM-TO-EVIDENCE INDEX LIVES

The claim-to-evidence index is the Evidence Map at the head of this appendix rather than a table here, so that a reviewer meets it before the technical material rather than after it.

### E.2 ARTIFACT REGISTRY

Every reported quantity traces to a frozen artifact carrying its own SHA-256, and every artifact records the configuration, input hashes, seeds and runtime that produced it. Rather than reproduce those strings in prose, where they are unreadable and cannot be checked, this section states the conventions and points at the artifacts; the Evidence Map at the head of this appendix maps each main-text claim to its source.

Four conventions govern how the artifacts should be read.

*Numbers are single-sourced.* Every quantity in the main text and in this appendix is a macro resolved from one ledger entry, never retyped at the point of use. Tables S1–S16 are emitted by a generator directly from artifacts and are marked do-not-edit-by-hand; Table S17 is generated the same way. A number that appears twice in this document appears twice because one macro was used twice.

*Checkpoints are named, not assumed.* The production refit at 5,100 steps and the step-3000 pilot checkpoints back disjoint sets of claims, per §A.8.

*Seeds are recorded per analysis rather than globally.* Bootstrap, permutation and blinding seeds differ across analyses and are stored with the artifact that used them, because a single global seed would make independent analyses spuriously correlated.

*Software is pinned to the artifact, not the environment.* The reported version is the one that produced the frozen artifacts, which is not necessarily the one currently installed; see §A.1.

Table 10 lists the 26 ledger entries with their artifacts. SHA-256 values, input hashes and per-analysis seeds are recorded inside each artifact rather than reproduced here, where they would be unreadable and unverifiable. Three ancillary diagnostic artifacts, covering distribution overlap, route-saturation sampling and an early semantic-annotation probe, were produced under a later RD-Kit release than the one reported in §A.1; none of them backs a number reported in this paper. Each artifact additionally records its own runtime block: elapsed seconds, platform string, Python version and, where a model was trained or evaluated, the PyTorch version. Structural analyses ran on macOS arm64 under Python 3.14.2 with PyTorch 2.11.0; generator training and the semantic pilot ran in deterministic float32 on a single NVIDIA L4, per Table S16. Because the runtime block travels inside each artifact rather than being asserted centrally, a reader checking one number does not have to trust a global claim about the environment.

### E.3 ANALYSES NOT PERFORMED AND CLAIMS NOT MADE

The following are *not* claimed anywhere in this paper. Each is listed because the surrounding material could otherwise be read as implying it.

- **No cross-reaction generalization.** The formulation is stated for modular families generally; every experiment uses Ugi-3CR. No model was trained or evaluated on a second chemistry.
- **No stereochemical generation.** Generation is constitutional. FORGE proposes no stereo-centres and makes no claim about stereochemical preference, which is a known determinant of LNP behaviour (Hatit et al., 2023).

Table 10: Artifact registry. Every reported quantity traces to one frozen artifact; each artifact records its own configuration, input hashes, seeds and runtime.

| Claim                                                      | Artifact                                            | Checkpoint                                                |
|------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------|
| semantic-utility gap, total head                           | results/phase1/forged.semantics.pilot.v1/closure    | flat.true.role / step 3000                                |
| corruption amplification, $t=0.10$ over $t=0.90$           | results/phase1/forged.semantics.pilot.v1/closure    | step 3000                                                 |
| misaligned control, total head                             | results/phase1/forged.semantics.pilot.v1/closure    | step 3000                                                 |
| role-divergence association (Amendment 14)                 | results/phase1/forged.role.divergence.association   | in, not traininglt. json                                  |
| wanted head architecture (1 handle + tertiary ionizable N) | results/phase1/forged.held.family.stage2.v1/amp     | ugidecoratocoupling.y3/challenger/checkpoint_step_3000.pt |
| two or more Ugi-reactive sites                             | results/phase1/forged.held.family.stage2.v1/amp     | ugidecoratocoupling.y3/challenger/checkpoint_step_3000.pt |
| ester-linked aldehyde tail                                 | results/phase1/forged.held.family.stage2.v1/amp     | ugidecoratocoupling.y3/challenger/checkpoint_step_3000.pt |
| grouped two-sample AUC against real lipids                 | results/phase1/forged.held.family.stage2.v1/amp     | challenger_step3000/3. json                               |
| role-marginal null builds an ionizable head                | results/phase1/forged.held.family.stage2.v1/null    | denoiser, not trainedson weights                          |
| absent from the production training corpus                 | results/phase1/uginovalty.audit.v1/result. js       | decoration-coupling production refit                      |
| outside the finite enumeration                             | docs/FORGE_PENDING_MANUSCRIPT_CHANGES.md            | production refit                                          |
|                                                            | 3g; tests/test.enumeration.membership.convention.py | pins the convention                                       |
| distinct aldehyde components generated                     | results/phase1/uginovalty.audit.v1/result. js       | production refit                                          |
| distinct amine head components generated                   | results/phase1/uginovalty.audit.v1/result. js       | production refit                                          |
| distinct isocyanide components generated                   | results/phase1/uginovalty.audit.v1/result. js       | production refit                                          |
| route-complete out-of-enumeration designs                  | results/phase1/forged.open.world.actionability      | productionrefit result. json                              |
| fresh prediction-supported yield                           | results/phase1/ugimorphology.proposal.challenger    | productionrefit result. json                              |
| exploration cost of enrichment                             | results/phase1/ugimorphology.proposal.challenger    | productionrefit result. json                              |
| pooled descriptive calibration coverage                    | results/phase1/uginterpolative.conformal.v1         | frozenl predictive ensemble                               |
| unique forward assembly, all admitted                      | results/phase1/forged.forward.outcome.census.v1     | production refit                                          |
| unique forward assembly, prediction-supported              | results/phase1/forged.forward.outcome.census.v1     | production refit                                          |
| unique forward assembly, frozen 40 / Library 0             | results/phase1/forged.forward.outcome.census.v1     | production refit                                          |
| route completeness by enumeration membership               | results/phase1/forged.open.world.actionability      | productionrefit result. json                              |
| designs resolving to purchasable starting material, pooled | results/phase1/forged.open.world.actionability      | productionrefit result. json                              |
| single-step route recovery, top 5                          | results/phase1/graph2edits.single.step.recovery     | in, external engines / score. json                        |
| bounded hybrid route cascade on the 256-product shortlist  | results/phase1/ugibounded.hybrid.route.cascade      | productionrefit. json                                     |
| Library 0 status                                           | results/phase1/ugiprospective.panel.v6/prospective  | productionrefit1. json1.gz                                |

- **No predictive coverage guarantee.** Calibration is descriptive. No conformal or other finite-sample coverage claim is made for any candidate, inside or outside  $S_{\text{pred}}$ .
- **No distribution matching.** A grouped classifier separates generated from real lipids at AUC 0.882, and the model overshoots the real ionizable-head rate.
- **No semantic data-efficiency advantage and no semantic held-family generalization,** both predeclared and both unsupported; see §C.6.
- **No mechanistic LNP model.** Nothing here models particle packing, apparent  $\text{pK}_a$ , endosomal escape, or any physical mechanism linking structure to delivery.
- **No synthesis, formulation or biological result.** No member of any panel has been made. Every prospective quantity is a marked placeholder.
- **No route proposal treated as evidence, and no unresolved chemistry called unsynthesizable;** see §F.7.
- **No claim that either ranking statistic is superior.** The two cannot be ordered without prospective outcomes; see §B.6.
- **No comparison against external generative models.** This is a scope choice rather than an omission. The claim under test is that known chemistry can organize open-ended generation for a molecular family, and the quantity that would distinguish methods, composition of precursor chemistry outside a fixed inventory, is identically zero for inventory-constrained systems. A comparison would therefore measure a definitional difference

rather than a performance one. The evidence here is internal: the semantic ablation with its misaligned control, and the role-marginal null. See §7.2.

- **The declared support is not itself ablated.** Section 4.3 and Proposition 3 state that the support constrains sampling, but no arm removes it: the role-marginal null operates *inside* the same support and so cannot measure its contribution. What the support does for sample quality, as opposed to what it guarantees, is therefore unmeasured here.

#### E.4 COMPLETION CHECKLIST

What remains open before submission falls into three kinds, and they are not equally tractable.

*Values only the campaign can supply.* The candidate-lock date and hash, the procurement snapshot refresh, the identity and affiliation of the reviewing chemist, which candidates were selected for synthesis, and every prospective outcome.

*Values recoverable from the record.* The tie-breaking rule applied by the current ranking, and the comparator-predictor hyperparameters.

*One thing that is not a missing value but a missing experiment.* An earlier plan specified matched-budget comparisons against other generative models. Those runs do not exist, and the placeholders that stood in for them in a previous draft should not be read as work awaiting write-up. The controls reported here, the semantic ablation and the role-marginal null, answer a different question, and §E.3 states the absence rather than implying coverage.

Two numbers also carry a weaker provenance than the rest and are flagged where they appear: the predictor’s  $R^2$  values of 0.504 in calibration and 0.253 in post-selection test are quoted from the development record rather than resolved from a ledger entry, and should be re-derived or pinned before submission.

## F SYNTHESIS QUALIFICATION AND ACTIONABILITY

### F.1 ADMISSION IS NOT SYNTHESIS

Four properties are distinct and are routinely conflated. *Structural admission* says the generated graph is chemically well formed and reassembles from its projected components. *Unique assembly* says those components produce that product and no other under the qualified forward transform. *Route completeness* says every component is purchasable or reducible to purchasable material by a documented preparation. *Successful synthesis* says someone made it.

Each is necessary for the next and none implies it. A design can be admitted and not uniquely assembling, uniquely assembling and not route-complete, and route-complete and still fail at the bench. The whole of this appendix is organized by that ladder, and no result in it should be read as evidence about a rung above the one it measures.

### F.2 THE ADMISSION OPERATOR

Admission is Eq. (7): the projected components must satisfy their role-specific interfaces, and the qualified forward transform applied to them must reproduce the generated constitutional graph up to isomorphism. Failures are rejected rather than repaired, and no sample is resampled toward a target after failing.

Forward enumeration is capped at 100 products per component assignment. The cap was never reached: the frozen census records `enumeration_cap` 100 with `designs.hitting.enumeration_cap` 0, so uniqueness is decided by exhaustive enumeration rather than truncated search for every design in the population. The value 128 appears elsewhere in the route code for a different parameter and does not apply here.

### F.3 UNIQUE FORWARD ASSEMBLY

Uniqueness asks whether the target is the *only* distinct product enumerable from its own components. Across admitted designs the rate is 0.829; among evidence-supported designs it is 0.944. Within the frozen Library 0 pilot it is 39 of 40.

The mechanism behind non-uniqueness is fully determined and worth stating because it is narrower than one might guess. Every non-unique design carries at least two reactive amine sites, and no other cause was identified: with two such sites the Ugi condensation can proceed at either, giving a second constitutional product from the same three components. An earlier reading attributed part of the effect to competing ketone chemistry; that reading was withdrawn once the rate was compared against its base rate in the population. Admission verifies that the target is produced; uniqueness verifies that it is produced alone. These are different properties and the gap between them is chemistry, not model error.

### F.4 COMPONENT REALIZATION

Routing runs after generation and after the exact forward check, and never modifies a design. Its pseudocode is Table S13; the per-candidate assignments are Table S11, the per-component records with supplier counts are Table S10, and the consolidated purchase list is Table S12.

The three roles are asymmetric by design. *Amine heads* carry no disconnection rule in this instantiation, so an unseen head must be purchasable outright. *Ester-linked aldehyde tails* disconnect by esterification of a carboxylic acid with an  $\alpha,\omega$ -diol followed by oxidation of the remaining primary alcohol, two steps, resolved only if both starting materials have supplier records. *Isocyanide tails* disconnect to the corresponding primary amine by formylation and dehydration, two steps, resolved only if the amine has a supplier record. Both tail routes are published chemistry applied to generated structures, not model proposals. Admission additionally requires total steps at most four, counting across components and excluding the final Ugi assembly, which is common to every design.

For the frozen pilot this resolves to 49 distinct catalogue materials, with 53 components purchased directly and 67 prepared by a forward-verified route; 13 candidates require two synthetic steps and 27 require four. Figure 6 shows one design’s full resolution. Every branch terminates in a catalogue material, with no unresolved branch. Supplier records are dated snapshots and are not guarantees of stock, purity or lead time.

### F.5 ROUTE COMPLETENESS OUTSIDE THE FINITE CHEMISTRY

Route completeness is 1.000 inside the enumerated reference chemistry, 541 of 541, against 0.924 outside it, 931 of 1,008. The difference is -0.076 with a 95% interval of [-0.103, -0.055], over 1,549 designs reaching route assessment.

The interval is a cluster bootstrap over 162 aldehyde components, not a product-level i.i.d. bootstrap. That choice is substantive rather than conservative bookkeeping: products sharing an aldehyde are not independent evidence about whether that aldehyde can be prepared, since a single successful disconnection resolves every design using it. Resampling products independently would count one preparation many times and shrink the interval to something the data do not support.

Two limits bound the claim. Route completeness is *conditional on reaching route assessment*, so designs removed earlier in the funnel have unmeasured route completeness and the outside-enumeration figure is not a marginal rate over all generated chemistry. And route completeness is computational: it predicts that a documented disconnection exists with purchasable starting materials, and it is not evidence that any specific reaction will succeed at scale, in yield or in purity. The measured gap is the honest form of the cost of leaving the finite library. It is a real cost, not parity, and the interval marginally clears the prespecified  $\pm 10$  point band.

### F.6 WHERE THE OFF-REGISTRY CHEMISTRY SITS, AND WHICH GATE REMOVES IT

The headline novelty figure counts designs carrying at least one component outside the 424-component training registry. That phrasing hides the role, and the roles are not interchangeable: the amine head drives activity, and §F.4 gives it no disconnection rule, so an unseen head has to be

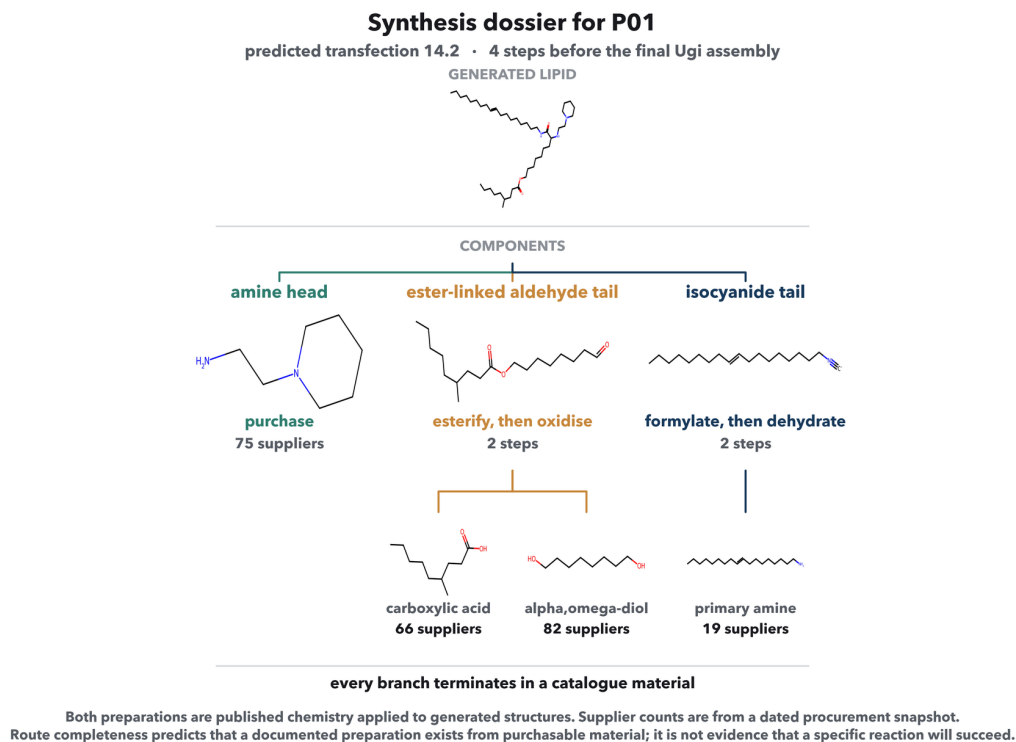


Figure 6: **Component routing for a single generated design.** The generated lipid is projected to its three components; the amine head terminates directly in catalogue material, while both tails are reduced to purchasable starting materials by published preparations. Supplier counts come from a dated procurement snapshot. The predicted transfection value shown in the header is provenance from the lower-bound ranking rule of §F.9 and is not a current ranking quantity.

purchasable outright. If novelty concentrated in the tails, which are a homologous series where an unseen component often means a different chain length, the reach would be worth much less than the number suggests.

Table 11 splits the funnel by role. Two things follow. Novelty is heaviest on the amine head at generation, not on the tails. And route assessment is not the filter: off-registry components that reach it resolve at rates close to their on-registry counterparts for both the amine head and the aldehyde tail. The isocyanide tail is the weakest role at 49/90, which is the opposite of what the disconnection asymmetry predicts.

The filter is the prediction-support gate. Designs with an unseen amine head are 49.7% of admitted and 0.5% of prediction-supported, a collapse of roughly two orders of magnitude in one step, while unseen aldehyde tails survive and are enriched. This is the intended behavior of §4.3, not a defect: the activity model refuses to score head chemistry it has no evidence for, and the head is exactly where it has least. It does mean the actionable set is confined to catalogue heads by the predictor rather than by the chemistry, and that the way to widen it is prospective measurement on new heads rather than a head disconnection rule.

Two cautions. The off-registry amine cell at route assessment holds ten designs, so 9/10 is an observation and not a rate worth an interval. And all resolution figures are conditional on reaching assessment, so designs removed by earlier gates have unmeasured route status.

Resolution also degrades with how much novelty a single design carries, which the per-role view hides: designs with no off-registry component resolve at 664/664, with one at 802/870, and with two at 6/15. Nothing in the assessed population carries three.

Table 11: **Off-registry share by role through the funnel, and resolution at route assessment.** Population is the union of both rescoring ledgers, 29,850 distinct admitted products; the abstract’s pooled figure is over the main ledger alone and is not comparable row for row. Resolution uses the accessibility standard of §F.4, purchasable or reducible to purchasable material under a forward-verified disconnection within the step budget, not adjudicated closure.

| Stage                                 | off-registry share of designs |               |                 | <i>n</i> |
|---------------------------------------|-------------------------------|---------------|-----------------|----------|
|                                       | amine head                    | aldehyde tail | isocyanide tail |          |
| Admitted                              | 49.7%                         | 43.3%         | 15.2%           | 29,850   |
| Prediction-supported                  | 0.5%                          | 31.4%         | 3.4%            | 2,629    |
| Absent from train fold                | 0.7%                          | 51.2%         | 5.6%            | 1,612    |
| Reaching route assessment             | 0.6%                          | 51.6%         | 5.8%            | 1,549    |
| <i>Resolution at route assessment</i> |                               |               |                 |          |
| On-registry components                | 1,539/1,539                   | 749/749       | 1,459/1,459     |          |
| Off-registry components               | 9/10                          | 765/800       | 49/90           |          |

## F.7 TOOLING

**Route levels.** Resolution is organized into three levels. **L1** is exact Ugi decomposition and forward reassembly of the product itself. **L2** is synthesis of each generated amine, aldehyde and isocyanide component, or a declared direct commercial path for it. **L3** is closure of every route leaf to a terminal material under the declared procurement and evidence snapshot. Routing is post hoc to graph generation but *not* post hoc to panel selection: the cascade is applied identically to an arm-balanced, route-blinded shortlist before any panel is locked, so route attrition is part of the reported outcome rather than an invisible cleanup step.

**The resolution hierarchy.** Within L2 the order is fixed and each stage is consulted only where the previous one is silent. Exact evidence comes first: a frozen curated index of qualified component routes terminating in a current terminal-material ledger. Where the index is silent, **Graph2Edits** (Zhong et al., 2023) acts as the primary source-neutral single-step proposal engine. Where that leaves a residual, **AiZynthFinder 4.4.1** (Genheden et al., 2020) acts as the multistep and public-stock challenger. Both engines are unmodified and neither is ours; what this work contributes is the object they run on, since the recursion is over generated *components* rather than over a product whose disconnection would first have to be guessed.

**Proposals are adjudicated, not counted.** Every hypothesis, whether it originates in the curated index or in an engine, must pass the same independent adjudication: reaction direction, reactive site, precursor and product consistency, substrate compatibility, evidence level, and terminal-material availability. A planner score is never sufficient. A route found by an engine is not penalized merely because the hand-built index lacked it, and a route is not accepted merely because a general model proposed it. Correspondingly, an AiZynthFinder public-stock solution is not current procurement evidence, and an unresolved component is unresolved under the frozen bounded cascade and dated snapshot rather than chemically unsynthesizable.

**Engine recovery, measured separately.** The engines were scored on a hidden exact single-step recovery benchmark of 36 targets, against a scoring truth frozen before the proposal ledgers. Graph2Edits recovers 29/36 within the top five and AiZynthFinder 19/36; their union did not improve on Graph2Edits alone. Stratified, Graph2Edits reaches 16 of 18 on targets with known exact routes but 13 of 18 on held reaction families. Graph2Edits recovered oxidation, esterification and formamide dehydration; neither engine recovered amine formylation. AiZynthFinder nonetheless earns its place in the cascade for a different reason: on a residual checkpoint it closed 44 of 64 Graph2Edits-missed components to public stock and resolved 29 of 97 previously unresolved leaves. Both benchmark artifacts record `exact_recovery_is_route_evidence` as false: recovery measures proposal quality and does not create route evidence.

**What the cascade returned, and the number that matters.** Applied to the 256-product route-blinded shortlist, with arm and potency blinded from route search, the cascade produced engine proposals covering 92.6% of products, from 909 Graph2Edits proposals across 120 unresolved com-

ponent targets and 713 AiZynthFinder single-step proposals across 72 residual targets, 25 of which full search solved to public stock. Adjudicated closure on the frozen curated exact evidence was 7.4%. The engines closed **zero** routes.

That gap is the result rather than a disappointment, and it locates the binding constraint. Search capability is not what limits closure here; the evidence standard is. Of the unresolved components, 121 were never expanded at all because they are absent from the evidence index and returned at depth zero having attempted no disconnection, while only 1 was unresolved after search. Reporting closure without that split would present an index-miss rate as a search outcome. The procurement snapshot is dated 3 August 2026 and is explicitly not live availability; the artifact requires a refresh before any panel lock.

**Two senses of “proposal” in this paper.** The route *proposal engines* above generate retrosynthetic hypotheses about how to make a component. The evidence-aware *proposal*  $\pi_\gamma$  of §5.2 is a distribution over design contexts. They share a word and nothing else: no route information enters  $\pi_\gamma$ , and no biological quantity enters route search.

## F.8 BLINDED CHEMISTRY REVIEW PROTOCOL

Table 12: What the reviewing chemist saw, and what was withheld. The withheld column is why the chemistry selection cannot have reproduced the model’s ranking.

| Provided                                  | Withheld                                 |
|-------------------------------------------|------------------------------------------|
| Product structure and SMILES              | Predicted activity                       |
| All three component structures and SMILES | Calibrated lower bound                   |
| Which components are purchasable          | Conformal quantile                       |
| Suggested route per prepared component    | Evidence stratum                         |
| Intermediates and starting materials      | Arm assignment                           |
| Steps before final assembly               | Pool identifiers, which are rank-ordered |
| Molecular weight                          | The six low-ranked controls entirely     |
| How many candidates share each component  |                                          |

Synthetic capacity rather than computational throughput limits a prospective campaign, so a subset of the frozen pool was selected for synthesis by a reviewing chemist. The review took place after the candidate lock and before any synthesis was attempted, so no experimental outcome informed it.

Table 12 lists what was disclosed and what was withheld, and Figures 7–9 reproduce representative pages of the packet as delivered. The blinding is not a formality. Predicted activity had already determined which designs entered the pool, so withholding it during the review is what makes the chemistry selection something other than a second application of the model’s own ranking. The pool identifiers were withheld for the same reason: they are ordered by rank and would have conveyed the ranking on their own. Codes were assigned in shuffled order under a recorded seed, and the map from code to identifier was held separately and not supplied with the review materials. The six low-ranked controls were withheld entirely, because presenting them would have disclosed that a ranking existed.

No structural-diversity target, quota or coverage requirement was imposed. Any such constraint would be a constraint the model did not choose, and it would have to be described as part of the selection rule; leaving the selection unconstrained means the structures and their chemistry were the only inputs to the decision. **[TODO: F.8: the reviewing chemist’s identity and affiliation, which candidates were selected, any adjustment made before the freeze together with its stated reason, and the disposition of the controls]**

## F.9 HISTORICAL LIBRARY 0 PRIORITIZATION PROTOCOL

*This protocol generated the frozen Library 0 computational pilot. It is documented here because a frozen selection whose rule is unpublished cannot be audited. It is not the ranking used in the current formulation, and §B.6 sets out how the two differ and why neither can be shown superior without prospective outcomes.*

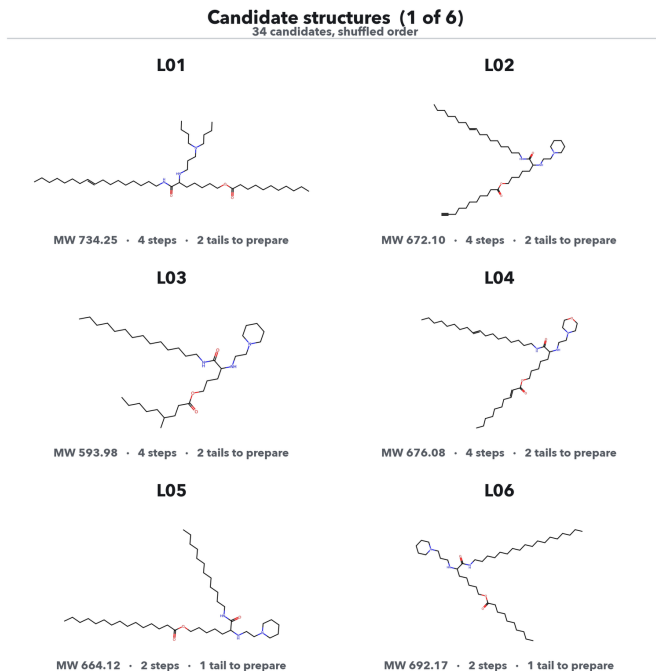


Figure 7: **Blinded dossier, candidate structures.** One of six structure pages supplied to the reviewing chemist. Designs carry shuffled codes rather than the rank-ordered pool identifiers, and no predicted activity, calibrated bound, evidence stratum or arm appears anywhere in the packet. Native aspect is square.

The rule that produced the pilot is recorded here in full, because a frozen selection whose rule is not published cannot be audited. Eligibility required valid molecular and Ugi reconstruction, a calibrated activity estimate, route closure within four upstream steps, membership of the frozen physicochemical envelope, and absence from the generative training fold. Ranking used the 90% conformational lower bound. The twelve highest-ranked eligible designs were taken directly; the remaining twenty-two of the high arm preferred designs absent from the enumerated library, applied only where that preference did not lower the ranking; six low-ranked controls were drawn from the lowest-ranked members of the tail-generalization stratum so that the controls match the bulk of the high arm in evidence class. Diversity was enforced at a maximum pairwise product Tanimoto of 0.95 and at most four candidates per amine head, with ties broken deterministically on canonical product SMILES. Tables S2, S6 and S8 give the resulting panel and rule; Table S17 gives the product structures.

The ranking statistic is where this rule and the current formulation part company, and §B.6 treats that comparison in full: the two rules share 22 of 34 candidates, and neither can be shown superior without prospective outcomes. Support, ranking and uncertainty are separate objects in §5 so that no single statistic has to express all three at once. The pilot stands as provenance: no member has been synthesized, all forty are route-complete, and 39 of 40 uniquely assemblable.

## G DERIVATIONS

### G.1 EXACT COMPONENT PROJECTION

Decomposition is not a learned inverse and not a retrosynthetic search. It is a partition of the terminal state induced by a map the design context already fixes.

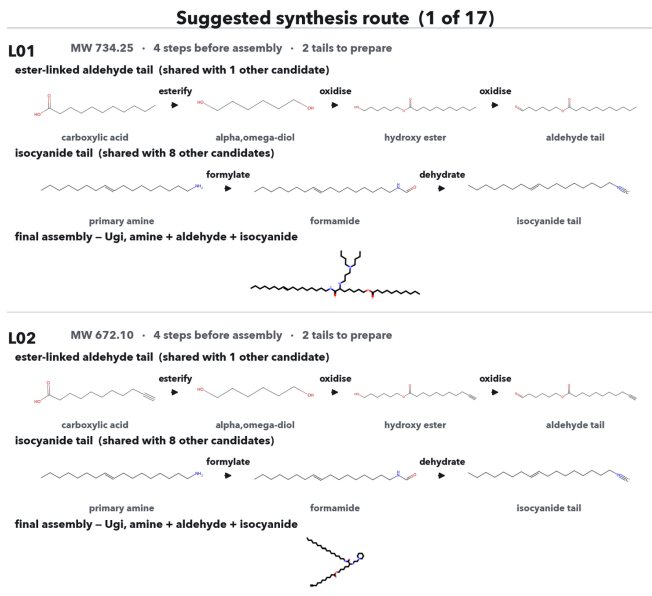


Figure 8: **Blinded dossier, suggested route.** One route page of seventeen. Intermediates are derived per component and retained only where carrying them forward returns the target structure exactly. Components shared with other candidates are marked, since one tail preparation can serve several designs.

**Proposition 1** (Exact component projection). *Let  $G$  be a terminal state instantiated under design context  $c$ , with role map  $r_c : V \rightarrow \mathcal{K}_{\mathcal{R}} \cup \{\text{core}\}$ . Then  $D_{\mathcal{R}}(G, c)$  is computable in  $O(|V| + |E|)$  time consulting no external resource, and if  $A_{\mathcal{R}}(G, c) = 1$  then  $\mathcal{R}(D_{\mathcal{R}}(G, c)) \cong G$ .*

*Proof.* The map  $r_c$  partitions  $V$  into the three role blocks and the shared core. Each edge of  $E$  is classified by the roles of its endpoints in one pass, giving intra-role edges, intra-core edges, and the cut edges crossing a role boundary at the core. For role  $k$ , take the subgraph induced on  $r_c^{-1}(k)$  together with its core attachment atoms, and close each cut edge with the valence  $\mathcal{R}$  assigns to that attachment. Both the classification and the closure visit each vertex and edge once, so the projection is linear and involves no catalogue, no template library and no search. The second claim is immediate:  $A_{\mathcal{R}}(G, c) = 1$  requires by definition that  $\mathcal{R}(D_{\mathcal{R}}(G, c)) \cong G$ , so certification costs one canonicalization and one graph isomorphism test rather than a route search.  $\square$

Two things are worth separating here. The projection is exact by construction, because  $r_c$  is supplied rather than inferred; this is what makes the decomposition available before generation rather than recovered after it. What the proposition does not give is procurement: it says the three components are determined and reassemble the product, not that any of them can be obtained. That is Appendix F.

## G.2 THE EVIDENCE-AWARE PROPOSAL IS A KL-REGULARIZED TILT

Write  $f(c) = \log(\rho(c) + \epsilon)$  and consider

$$\mathcal{J}(q) = \mathbb{E}_{c \sim q}[f(c)] - \frac{1}{\gamma} D_{\text{KL}}(q \parallel \pi_0), \quad \gamma > 0,$$

| Starting materials                                                                        |                          |                                |
|-------------------------------------------------------------------------------------------|--------------------------|--------------------------------|
| 25 catalogue items cover all 34 candidates. Uses counts how many candidates draw on each. |                          |                                |
| 17                                                                                        | OC(C)CCCC                | alpha,omega-diol, 93 suppliers |
| 14                                                                                        | CCCCCCCCC=CCCCCCCCCN     | primary amine, 19 suppliers    |
| 10                                                                                        | OC(C)CCO                 | alpha,omega-diol, 68 suppliers |
| 9                                                                                         | CCCCCCCCC=CCCCCCCCCN     | primary amine, 3 suppliers     |
| 7                                                                                         | OC(C)CCCCCO              | alpha,omega-diol, 82 suppliers |
| 6                                                                                         | CCCCCCCCC(=O)O           | carboxylic acid, 107 suppliers |
| 5                                                                                         | CCCCCCC(C)CCC(=O)O       | carboxylic acid, 66 suppliers  |
| 3                                                                                         | CCCCCCCCC(=O)O           | carboxylic acid, 101 suppliers |
| 3                                                                                         | C=CCCCCCCCC(=O)O         | carboxylic acid, 115 suppliers |
| 2                                                                                         | C#CCCCCCCCC(=O)O         | carboxylic acid, 74 suppliers  |
| 2                                                                                         | CCCCCCCCC(C)CCN          | primary amine, 68 suppliers    |
| 2                                                                                         | CCCCCCCCC=CC(=O)O        | carboxylic acid, 39 suppliers  |
| 2                                                                                         | CCCCCCCCC(=O)O           | carboxylic acid, 113 suppliers |
| 1                                                                                         | CCCCCCCCC(C)CC(=O)O      | carboxylic acid, 92 suppliers  |
| 1                                                                                         | CCCCCCCCC=CCCCCCCCC(=O)O | carboxylic acid, 20 suppliers  |
| 1                                                                                         | CCCCCCCCC=CCCC(=O)O      | carboxylic acid, 8 suppliers   |
| 1                                                                                         | CCCCCCCCC(C)CC(=O)O      | carboxylic acid, 88 suppliers  |
| 1                                                                                         | CCCCCCCCC=CC(=O)O        | carboxylic acid, 18 suppliers  |
| 1                                                                                         | CCCCCCCCC=CCCCCCCCC(=O)O | carboxylic acid, 27 suppliers  |
| 1                                                                                         | CCCCCCCCC(C)CC(=O)O      | carboxylic acid, 158 suppliers |
| 1                                                                                         | CCCCCCCCC=CCCCCCCCC(=O)O | carboxylic acid, 8 suppliers   |
| 1                                                                                         | CCCCCCCCC(C)CC(C)CCN     | primary amine, 38 suppliers    |
| 1                                                                                         | C#CCCCCCCCC(=O)O         | carboxylic acid, 32 suppliers  |
| 1                                                                                         | CCCCCCCCC(C)CC(=O)O      | carboxylic acid, 103 suppliers |
| 1                                                                                         | CCCCCCCCC=CCCCCCCCC(=O)O | carboxylic acid, 2 suppliers   |

Figure 9: **Blinded dossier, consolidated starting materials.** The reviewed designs draw on a small shared pool of catalogue items, so preparation effort is not proportional to candidate count. Native aspect is portrait, 1739x2240.

maximized over distributions  $q$  on the countable context set. Introducing a multiplier  $\mu$  for  $\sum_c q(c) = 1$  and differentiating the Lagrangian with respect to  $q(c)$  gives

$$f(c) - \frac{1}{\gamma} \left( \log \frac{q(c)}{\pi_0(c)} + 1 \right) - \mu = 0,$$

so  $\log q(c) = \log \pi_0(c) + \gamma f(c) + \text{const}$ , and therefore

$$q^*(c) \propto \pi_0(c) e^{\gamma f(c)} = \pi_0(c) (\rho(c) + \epsilon)^\gamma.$$

Normalizing recovers  $\pi_\gamma$  exactly as defined in Eq. (10). Since  $\mathcal{J}$  is strictly concave in  $q$  and the constraint set is convex, this stationary point is the unique maximizer. The exponent  $\gamma$  is therefore not a free tuning knob layered on top of the proposal: it is the inverse temperature of an explicit trade-off between supported yield and divergence from the broad exploratory proposal.

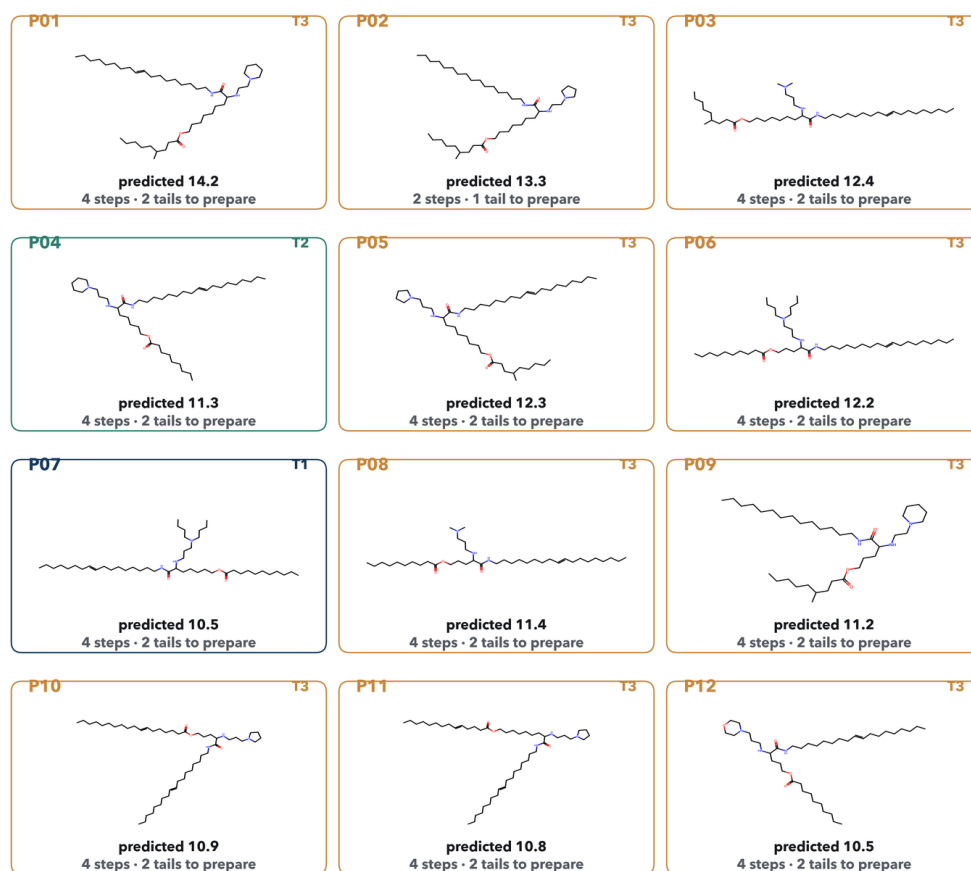
### G.3 SUPPORT PRESERVATION, AND WHY IT IS NOT DIVERSITY PRESERVATION

**Proposition 2.** For any  $\epsilon > 0$  and finite  $\gamma \geq 0$ ,  $\text{supp}(\pi_\gamma) = \text{supp}(\pi_0)$ .

*Proof.* For every context  $c$  we have  $\rho(c) + \epsilon \geq \epsilon > 0$ , so  $(\rho(c) + \epsilon)^\gamma > 0$  for finite  $\gamma$ . The normalizing sum in Eq. (10) is finite and strictly positive, hence  $\pi_\gamma(c) > 0$  if and only if  $\pi_0(c) > 0$ .  $\square$

The proposition is what prevents evidence-aware deployment from silently becoming a support restriction: no architecture is made impossible. It says nothing at all about how probability mass

**The forty prospectively locked designs (1 of 4)**  
**Selected on predicted activity alone**  
 the twelve highest-ranked designs in the eligible pool



Predicted values are on the measured transfection scale. Steps count the preparations required before the final Ugi assembly, common to every design. T1 unseen isocyanide only, T2 unseen amine head, T3 unseen aldehyde tail, T4 both tails unseen.

**Figure 10: The frozen Library 0 pilot, structures 1 of 4.** Predicted values and the T1–T4 tier badges come from the lower-bound prioritization rule described in this section and are not current ranking quantities.

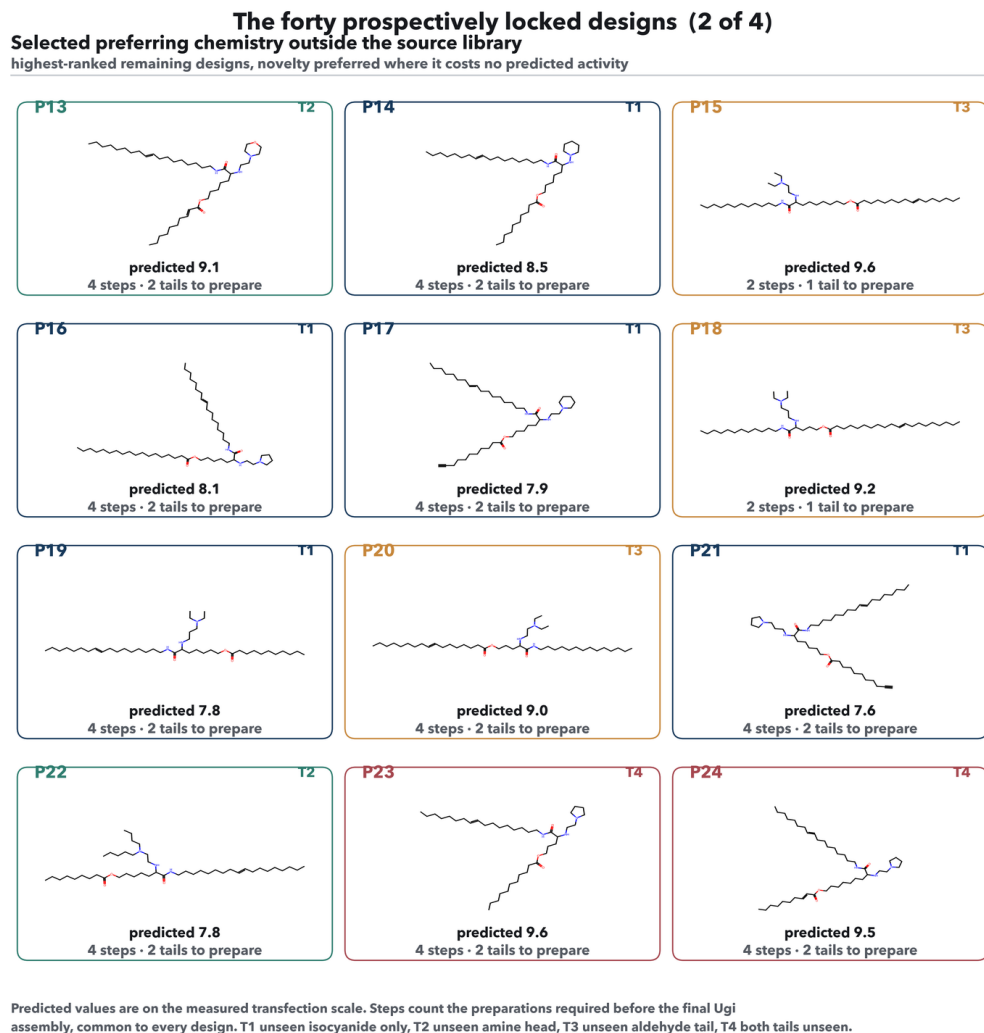


Figure 11: The frozen Library 0 pilot, structures 2 of 4.

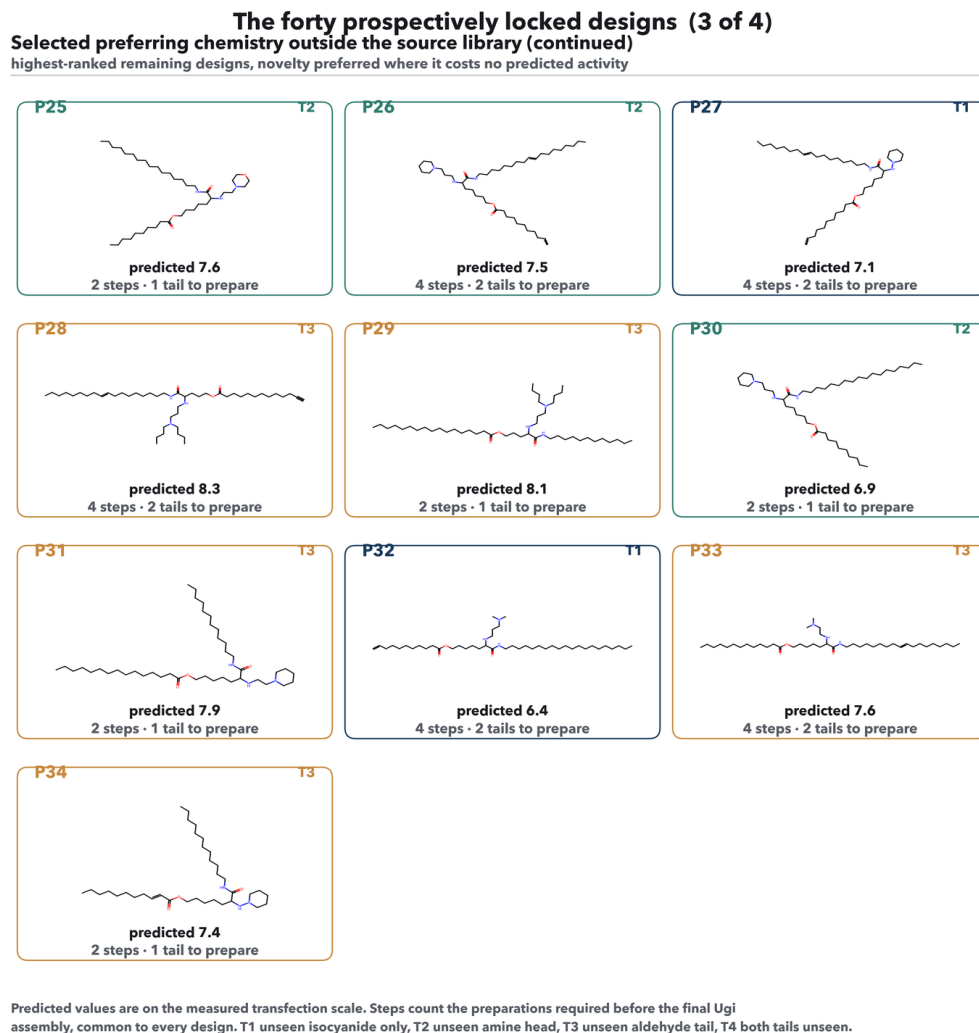


Figure 12: The frozen Library 0 pilot, structures 3 of 4.

is distributed within that support, and the empirical concentration is substantial. Effective amine-head diversity falls by 60% under the enriched proposal, while effective aldehyde-tail diversity is essentially unchanged at 1.002 times its unenriched value.

The mechanism behind that asymmetry is worth naming, because it is a property of the evidence rather than of the tilt. Membership of  $S_{\text{pred}}$  requires *all four* views to pass, and the measured library is far denser at the amine head than at either tail: 20 measured heads against 264 in the corpus registry, compared with 11 measured aldehydes against 107. Tilting toward support therefore pulls hardest on the role where the reference chemistry is most concentrated, which is why head diversity absorbs almost all of the loss while the aldehyde tail is left essentially untouched. A tilt that concentrated on the tails instead would require a measured library with the opposite shape.

#### G.4 SAMPLER SUPPORT INVARIANT

**Proposition 3** (Sampler support invariant). *If the initial partial state is admissible and every mask assigns zero probability to any successor violating a declared bound, then every state reached during sampling, including the terminal state, lies in  $S_{\text{gen}}(c)$ .*

*Proof.* By induction on the transition index  $n$ . At  $n = 0$  the initial partial state is admissible by assumption. Suppose the state after  $n$  transitions is admissible. Each mask is a predicate on candidate successors evaluated against the declared bounds: valence masks forbid an atom exceeding its allowed degree, duplicate-edge and self-edge masks forbid a multigraph or a loop, the adapter-boundary mask forbids an edge crossing a role boundary anywhere but the core, and the counting masks forbid exceeding the per-component atom, child, junction, cycle and core-attachment budgets. Every successor assigned nonzero probability therefore satisfies the bounds and retains a completion within them, so the state after  $n + 1$  transitions is admissible. Sampling terminates after finitely many transitions at a state with no pending frontier, and that state is admissible, hence in  $S_{\text{gen}}(c)$ .  $\square$

This is a property of the sampler, not a learned capability of the model. A nonlearned role-marginal sampler operating under the same masks satisfies it identically, which is exactly why declared support cannot be offered as evidence that the model has learned lipid chemistry, and why the marginal-null contrast of §7.2 is needed.

## H PROSPECTIVE EXPERIMENTAL PROTOCOL

### [PROSPECTIVE PROTOCOL PENDING EXECUTION]

#### H.1 PRESPECIFIED REPORTING TABLE

Rows are the designs selected by the chemistry review, in the order fixed before any experiment. No row is added, removed or reordered once synthesis begins, and a design that fails at any stage keeps its row with the failure recorded against it. The table ships blank: a prespecified reporting frame is what makes the campaign legible as prospective rather than narrated after the fact.

Table 13: Prespecified reporting frame. Row order is fixed before synthesis begins.

| Design | Precursors prepared | Final assembly | Identity and purity | LNP formed | Size, PDI | Encapsulation | HeLa / viability |
|--------|---------------------|----------------|---------------------|------------|-----------|---------------|------------------|
| 1      |                     |                |                     |            |           |               |                  |
| 2      |                     |                |                     |            |           |               |                  |
| 3      |                     |                |                     |            |           |               |                  |
| 4      |                     |                |                     |            |           |               |                  |
| 5      |                     |                |                     |            |           |               |                  |
| 6      |                     |                |                     |            |           |               |                  |
| 7      |                     |                |                     |            |           |               |                  |
| 8      |                     |                |                     |            |           |               |                  |
| 9      |                     |                |                     |            |           |               |                  |
| 10     |                     |                |                     |            |           |               |                  |
| 11     |                     |                |                     |            |           |               |                  |
| 12     |                     |                |                     |            |           |               |                  |

Matched-reference comparisons are recorded alongside each row where a matched reference exists.

## H.2 PROTOCOL

[TODO: H.3: candidate-freeze date and hash; selection protocol; matched-reference construction; synthesis and purification; structural characterization; LNP formulation composition and procedure; encapsulation measurement; particle size and PDI; cell line and culture conditions; mRNA dose; treatment duration; reporter assay; viability assay; the definition of technical versus biological replicate; positive and negative controls; randomization and blinding if used; the primary endpoint; the prespecified statistical test; the handling of failed synthesis or formulation; *in vivo* methods and ethics or animal approvals if applicable.]

## I GENERATED DATA TABLES

### I.1 PER-HEAD AND PER-SEED SEMANTIC RESULTS

Table 14: **Semantic-utility gap per head and per paired seed.** Each entry is  $L_{\text{no-role}} - L_{\text{true-role}}$  for that head, within the seed’s paired run, so positive means the aligned map reconstructs better. The six rows above the rule are the semantic pilot’s reported set, matching PRIMARY\_HEADS in the closure script. Below it are the two heads belonging only to the role-divergence analysis, a different six; those two are not uniformly positive and no claim is made over them.

| head                                                                                               | s1      | s2      | s3      | mean    | s.d.   | all positive |
|----------------------------------------------------------------------------------------------------|---------|---------|---------|---------|--------|--------------|
| atom state                                                                                         | +0.0282 | +0.0261 | +0.0254 | +0.0266 | 0.0015 | yes          |
| parent bond                                                                                        | +0.0189 | +0.0148 | +0.0175 | +0.0170 | 0.0021 | yes          |
| offspring topology                                                                                 | +0.1402 | +0.1439 | +0.1386 | +0.1409 | 0.0027 | yes          |
| closure bond                                                                                       | +0.0172 | +0.0146 | +0.0195 | +0.0171 | 0.0025 | yes          |
| decoration anchor                                                                                  | +0.0501 | +0.0417 | +0.0400 | +0.0439 | 0.0054 | yes          |
| <b>total</b>                                                                                       | +0.2956 | +0.1835 | +0.2646 | +0.2479 | 0.0579 | yes          |
| <i>Heads used only by the role-divergence analysis of Amendment 14, not part of the six above:</i> |         |         |         |         |        |              |
| decoration atom                                                                                    | +0.0054 | -0.0022 | +0.0052 | +0.0028 | 0.0043 | 2/3          |
| decoration bond                                                                                    | +0.0355 | -0.0554 | +0.0185 | -0.0005 | 0.0483 | 2/3          |

Table 15: **Corruption dependence of the semantic-utility gap.** The gap near the corrupted end of the probability path, against the gap near the clean endpoint, per seed. The direction replicates in every seed; the magnitude does not, which is why the three values are reported rather than a range.

| seed | gap, corrupted end | gap, clean end | ratio  |
|------|--------------------|----------------|--------|
| s1   | +0.5126            | +0.0666        | 7.70×  |
| s2   | +0.3206            | +0.0324        | 9.89×  |
| s3   | +0.4617            | +0.0410        | 11.25× |

Table 16: **The misalignment control.** Denoising loss under the three arms; lower is better. A negative *none* – *misaligned* means the misaligned arm is *worse* than receiving no role at all, which is what rules out attributing the benefit to the presence of an extra categorical channel.

| head               | true   | none   | misaligned | none – mis. | true – mis. |
|--------------------|--------|--------|------------|-------------|-------------|
| atom state         | 0.1180 | 0.1463 | 0.1459     | +0.0003     | -0.0279     |
| parent bond        | 0.0747 | 0.0935 | 0.0945     | -0.0010     | -0.0198     |
| offspring topology | 0.1648 | 0.3050 | 0.3108     | -0.0057     | -0.1460     |
| closure bond       | 0.1235 | 0.1407 | 0.1463     | -0.0056     | -0.0228     |
| decoration anchor  | 0.1273 | 0.1775 | 0.1727     | +0.0047     | -0.0454     |
| <b>total</b>       | 0.8138 | 1.1094 | 1.1149     | -0.0056     | -0.3012     |

## I.2 THE COMPLETE DESCRIPTOR PANEL

Table 17: **The complete lipid-native descriptor panel.** Median value per descriptor for held-out real lipids, FORGE samples, and the role-marginal null. Every descriptor is reported, not only those on which FORGE compares well: the panel is a guard against post-hoc metric selection only if its unselected members are published. Populations: real  $n = 3,072$ , FORGE  $n = 2,985$ , null  $n = 3,072$ .

| descriptor          | region          | real    | FORGE   | null    |
|---------------------|-----------------|---------|---------|---------|
| aldalkenes          | aldehyde tail   | 1.000   | 0.000   | 0.000   |
| aldbranches         | aldehyde tail   | 0.000   | 0.000   | 0.000   |
| aldcarbons          | aldehyde tail   | 15.000  | 15.000  | 16.000  |
| aldesters           | aldehyde tail   | 1.000   | 1.000   | 0.000   |
| aldeethers          | aldehyde tail   | 0.000   | 0.000   | 0.000   |
| aldheavyatoms       | aldehyde tail   | 18.000  | 18.000  | 18.000  |
| aldketones          | aldehyde tail   | 0.000   | 0.000   | 1.000   |
| aldlongestchain     | aldehyde tail   | 11.000  | 11.000  | 16.000  |
| clogp               | whole lipid     | 7.694   | 8.175   | 6.762   |
| degradable_linkages | whole lipid     | 1.000   | 1.000   | 0.000   |
| hba                 | whole lipid     | 5.000   | 5.000   | 6.000   |
| hbd                 | whole lipid     | 2.000   | 2.000   | 3.000   |
| headhba             | ionizable head  | 2.000   | 2.000   | 2.000   |
| headhbd             | ionizable head  | 1.000   | 1.000   | 2.000   |
| headheavyatoms      | ionizable head  | 9.000   | 9.000   | 10.000  |
| headmax_nspacing    | ionizable head  | 3.000   | 3.000   | 0.000   |
| headmin_nspacing    | ionizable head  | 3.000   | 2.000   | 0.000   |
| headnitrogens       | ionizable head  | 2.000   | 2.000   | 1.000   |
| headprimary_amine   | ionizable head  | 1.000   | 1.000   | 0.000   |
| headring_nitrogen   | ionizable head  | 0.000   | 0.000   | 0.000   |
| headrings           | ionizable head  | 1.000   | 1.000   | 1.000   |
| headsecondary_amine | ionizable head  | 0.000   | 0.000   | 0.000   |
| headtertiary_amine  | ionizable head  | 0.000   | 1.000   | 0.000   |
| headto_tail_ratio   | ionizable head  | 0.312   | 0.310   | 0.364   |
| isoalkenes          | isocyanide tail | 0.000   | 0.000   | 0.000   |
| isobranches         | isocyanide tail | 0.000   | 0.000   | 0.000   |
| isocarbons          | isocyanide tail | 11.000  | 11.000  | 11.000  |
| isoesters           | isocyanide tail | 0.000   | 0.000   | 0.000   |
| isoethers           | isocyanide tail | 0.000   | 0.000   | 0.000   |
| isoheavyatoms       | isocyanide tail | 13.000  | 12.000  | 14.000  |
| isoketones          | isocyanide tail | 0.000   | 0.000   | 1.000   |
| isolongestchain     | isocyanide tail | 8.000   | 8.000   | 8.000   |
| molecular_weight    | whole lipid     | 565.821 | 563.956 | 616.840 |
| rotatable_bonds     | whole lipid     | 26.000  | 27.000  | 28.000  |
| tail_asymmetry      | whole lipid     | 6.000   | 6.000   | 6.000   |
| tail_balance        | whole lipid     | 0.578   | 0.571   | 0.579   |
| total_tail_carbons  | whole lipid     | 27.000  | 27.000  | 28.000  |
| tpsa                | whole lipid     | 70.670  | 70.670  | 129.640 |

Table 18: **Grouped real-versus-generated separability.** Grouping is by aldehyde component family, so repeated component combinations cannot leak across folds. Per-fold values are given because the mean alone hides the spread.

| arm                | groups | mean AUC | per-fold AUC                      |
|--------------------|--------|----------|-----------------------------------|
| FORGE              | 11     | 0.8820   | 0.852, 0.969, 0.869, 0.883, 0.838 |
| role-marginal null | 11     | 1.0000   | 1.000, 1.000, 1.000, 1.000, 1.000 |
| earlier checkpoint | 11     | 0.9581   | 0.982, 0.992, 0.939, 0.949, 0.928 |

## J SUPPLEMENTARY TABLES

All tables are generated from the frozen selection artifact and its pinned inputs. Regenerating this file after any change to the selection is required; the numbers here are not transcribed.

- Selection configuration: configs/bio/phase1-ugi-prospective-panel-v6.json, sha256 0e04c17daa7e68a7
- Panel artifact: results/phase1/ugi-prospective-panel-v6/prospective-panel.jsonl.gz, sha256 d9b7c0a70c8d8528
- Derived numbers: results/phase1/ugi-manuscript-numbers-v1/result.json, sha256 32e1e00eb3e79256

### J.1 TABLE S1 | EVIDENCE TIERS AND WHAT EACH ONE ASSUMES

The activity model issues a calibrated interval for every candidate in the panel, but the evidence behind those intervals is not uniform. Tiers are assigned by which component identities were absent from the measured training set, and are recorded per candidate. The panel is not powered for per-tier claims; the tier is reported so that an outcome can be read against the evidence supporting it.

| Tier                        | Component absent from the measured set | Panel     | High arm  | Low arm  | Eligible pool |
|-----------------------------|----------------------------------------|-----------|-----------|----------|---------------|
| T1 qualified role holdout   | isocyanide only                        | 8         | 8         | 0        | 119           |
| T2 head generalization      | amine head                             | 6         | 6         | 0        | 24            |
| T3 tail generalization      | aldehyde tail                          | 24        | 18        | 6        | 671           |
| T4 dual tail generalization | aldehyde and isocyanide                | 2         | 2         | 0        | 553           |
| <b>Total</b>                |                                        | <b>40</b> | <b>34</b> | <b>6</b> | <b>1367</b>   |

**T1 qualified role holdout.** Every component except the isocyanide appears in the measured training set, and the isocyanide identity was held out during model fitting. The activity model was validated on exactly this kind of substitution, so its calibrated interval carries the most direct evidence available in the panel.

**T2 head generalization.** The amine head group is absent from the measured set. The head sets the ionizable amine's pKa and is the dominant activity determinant in the model's own saliency analysis, so this is the furthest reach the panel makes. The model still issues a calibrated interval, but no measured product carries this head.

**T3 tail generalization.** The ester-linked aldehyde tail is absent from the measured set while the head and isocyanide are measured. Aldehyde tails form a homologous series in which chain length and unsaturation vary continuously, so interpolation within this role is better supported than in the other two.

**T4 dual tail generalization.** Both the aldehyde and the isocyanide are absent from the measured set. Two simultaneous substitutions place these designs furthest from the measured factorial, and any outcome here is attributable to the combination rather than to either component alone.

The low arm is drawn entirely from the tail-generalization tier. No design with in-domain component chemistry occupies the low end of the predicted range, so drawing controls from the same tier as the bulk of the high arm removes evidence class as an alternative explanation for any difference between the arms.

### J.2 TABLE S2 | THE FORTY LOCKED CANDIDATES

Candidates appear in their prespecified selection order. LCB90 is the calibrated conformal lower bound used for ranking; mean is the ensemble point prediction on the measured transfection scale; percentile is the fraction of the 1,100 measured products that the point prediction exceeds.

| ID  | Arm  | LCB90 | Mean  | Percentile | Tier | Novel vs library | Steps | C=C | Branched |
|-----|------|-------|-------|------------|------|------------------|-------|-----|----------|
| P01 | HIGH | 8.89  | 14.20 | 100%       | T3   | no               | 4     | 1   | yes      |

| ID  | Arm  | LCB90 | Mean  | Percentile | Tier | Novel<br>vs<br>library | Steps | C=C | Branched |
|-----|------|-------|-------|------------|------|------------------------|-------|-----|----------|
| P02 | HIGH | 8.01  | 13.32 | 99%        | T3   | no                     | 2     | 0   | yes      |
| P03 | HIGH | 7.05  | 12.36 | 98%        | T3   | no                     | 4     | 1   | yes      |
| P04 | HIGH | 6.96  | 11.25 | 97%        | T2   | yes                    | 4     | 1   | no       |
| P05 | HIGH | 6.94  | 12.25 | 98%        | T3   | no                     | 4     | 1   | yes      |
| P06 | HIGH | 6.89  | 12.20 | 98%        | T3   | no                     | 4     | 1   | no       |
| P07 | HIGH | 6.59  | 10.51 | 96%        | T1   | yes                    | 4     | 1   | no       |
| P08 | HIGH | 6.10  | 11.40 | 98%        | T3   | no                     | 4     | 1   | no       |
| P09 | HIGH | 5.93  | 11.24 | 97%        | T3   | no                     | 4     | 0   | yes      |
| P10 | HIGH | 5.60  | 10.91 | 97%        | T3   | yes                    | 4     | 2   | no       |
| P11 | HIGH | 5.48  | 10.78 | 96%        | T3   | yes                    | 4     | 2   | no       |
| P12 | HIGH | 5.23  | 10.54 | 96%        | T3   | no                     | 4     | 1   | no       |
| P13 | HIGH | 4.83  | 9.13  | 93%        | T2   | yes                    | 4     | 2   | no       |
| P14 | HIGH | 4.62  | 8.53  | 89%        | T1   | yes                    | 4     | 1   | no       |
| P15 | HIGH | 4.31  | 9.62  | 94%        | T3   | yes                    | 2     | 1   | no       |
| P16 | HIGH | 4.14  | 8.06  | 85%        | T1   | yes                    | 4     | 1   | no       |
| P17 | HIGH | 3.95  | 7.87  | 84%        | T1   | yes                    | 4     | 1   | no       |
| P18 | HIGH | 3.91  | 9.22  | 93%        | T3   | yes                    | 2     | 1   | no       |
| P19 | HIGH | 3.90  | 7.81  | 83%        | T1   | yes                    | 4     | 1   | no       |
| P20 | HIGH | 3.71  | 9.02  | 92%        | T3   | yes                    | 4     | 1   | no       |
| P21 | HIGH | 3.66  | 7.57  | 80%        | T1   | yes                    | 4     | 1   | no       |
| P22 | HIGH | 3.49  | 7.78  | 82%        | T2   | yes                    | 4     | 1   | no       |
| P23 | HIGH | 3.36  | 9.58  | 94%        | T4   | yes                    | 4     | 1   | no       |
| P24 | HIGH | 3.31  | 9.53  | 94%        | T4   | yes                    | 4     | 2   | no       |
| P25 | HIGH | 3.29  | 7.59  | 80%        | T2   | yes                    | 2     | 0   | no       |
| P26 | HIGH | 3.23  | 7.53  | 79%        | T2   | yes                    | 4     | 2   | no       |
| P27 | HIGH | 3.16  | 7.08  | 74%        | T1   | yes                    | 4     | 2   | no       |
| P28 | HIGH | 3.03  | 8.34  | 88%        | T3   | yes                    | 4     | 1   | no       |
| P29 | HIGH | 2.82  | 8.13  | 86%        | T3   | yes                    | 2     | 0   | no       |
| P30 | HIGH | 2.63  | 6.93  | 71%        | T2   | yes                    | 2     | 0   | no       |
| P31 | HIGH | 2.62  | 7.93  | 84%        | T3   | yes                    | 2     | 0   | no       |
| P32 | HIGH | 2.45  | 6.37  | 62%        | T1   | yes                    | 4     | 1   | no       |
| P33 | HIGH | 2.25  | 7.56  | 80%        | T3   | yes                    | 4     | 1   | no       |
| P34 | HIGH | 2.05  | 7.36  | 77%        | T3   | yes                    | 2     | 1   | no       |
| C01 | LOW  | -5.46 | -0.15 | 4%         | T3   | yes                    | 2     | 1   | no       |
| C02 | LOW  | -5.31 | -0.00 | 5%         | T3   | yes                    | 4     | 2   | no       |
| C03 | LOW  | -5.21 | 0.10  | 6%         | T3   | yes                    | 2     | 1   | no       |
| C04 | LOW  | -5.15 | 0.16  | 6%         | T3   | yes                    | 2     | 1   | no       |
| C05 | LOW  | -5.13 | 0.18  | 6%         | T3   | yes                    | 2     | 1   | no       |
| C06 | LOW  | -5.08 | 0.23  | 6%         | T3   | yes                    | 2     | 0   | no       |

### J.3 TABLE S3 | FROM GENERATED DESIGNS TO THE LOCKED PANEL

| Stage                                                                                                                                         | Count  |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Attempted draws                                                                                                                               | 32,768 |
| Chemically valid and exactly reconstructed through the Ugi reaction                                                                           | 30,979 |
| Passing terminal chemical screening                                                                                                           | 30,180 |
| Distinct admitted products                                                                                                                    | 26,235 |
| Carrying a calibrated activity estimate, route-complete within four steps, inside the physicochemical envelope, absent from the training fold | 1,367  |
| Selected into the panel                                                                                                                       | 40     |

No design was repaired or resampled after failing a check, and no candidate was replaced after selection.

## J.4 TABLE S4 | NOVELTY AGAINST EACH REFERENCE SET

Novelty takes different values against different reference sets, and the sets answer different questions. The production generator was refit on every structural fold, so the reference set for “never seen during fitting” is the full corpus, not the development training fold. Absence from the enumerated candidate library means the design uses a building block that library never contained, since that library is the complete cross-product of its block set. Comparisons against the enumerated library are made stereochemistry-free on both sides, because the generative pipeline is constitutional.

| Reference set                                 | What absence means                                               | Generated designs                                      | Panel |
|-----------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------|-------|
| Full structural corpus, 112,386 products      | the production model never saw this molecule during fitting      | 21,960 (83.7%)                                         | 40/40 |
| Enumerated candidate library, 12,276 products | the design uses a building block the library never contained     | 22,645 (86.3%)                                         | 32/40 |
| Measured products, 1,100                      | no measurement exists for any component combination of this kind | 25,982 (99.0%) carry at least one unmeasured component | 40/40 |

Every candidate on the panel is unmeasured, whether or not it appears in a previously published enumeration. The enumerated library contains 12,276 designs of which 1,100 were synthesized, so membership in it does not imply that a molecule has been made or tested.

## J.5 TABLE S5 | COMPONENT SPACE, MEASURED AGAINST GENERATED

| Component role             | Measured products (1,100) | Enumerated library | Training corpus | Designs the model can score |
|----------------------------|---------------------------|--------------------|-----------------|-----------------------------|
| Amine head                 | 20                        | 22                 | 264             | 23                          |
| Ester-linked aldehyde tail | 11                        | 62                 | 107             | 163                         |
| Isocyanide tail            | 5                         | 9                  | 53              | 15                          |

The aldehyde role expands 14.8-fold over the measured set. The measured set and the enumerated library are distinct references and are not interchangeable: the library spans  $22 \times 62 \times 9 = 12,276$  products, of which the 1,100 measured products are a subset.

## J.6 TABLE S6 | PANEL COMPOSITION

| Property                           | High arm     | Low arm        | Panel |
|------------------------------------|--------------|----------------|-------|
| Candidates                         | 34           | 6              | 40    |
| LCB90 range                        | 2.05 to 8.89 | -5.46 to -5.08 |       |
| Predicted mean, median             | 9.07         | 0.13           |       |
| Above the measured median          | 34           | 0              |       |
| Above the measured 90th percentile | 18           | 0              |       |
| Distinct amine heads               | 13           | 4              | 17    |
| Distinct aldehyde tails            | 22           | 6              | 26    |
| Distinct isocyanide tails          | 7            | 4              | 7     |
| Unsaturated tail                   | 28           | 5              |       |
| Branched aldehyde                  | 5            | 0              |       |
| Absent from the enumerated library | 26           | 6              | 32    |

No two candidates exceed 0.9483 pairwise Tanimoto similarity, and at most 4 candidates share an amine head.

## J.7 TABLE S7 | SYNTHETIC BURDEN

| Quantity                                        | Value |
|-------------------------------------------------|-------|
| Distinct catalogue materials to purchase        | 49    |
| Components purchased directly                   | 53    |
| Components prepared by a forward-verified route | 67    |
| Candidates requiring 2 synthetic steps          | 13    |
| Candidates requiring 4 synthetic steps          | 27    |

Every component is either purchasable or reduced to purchasable material by a forward-verified route, with no unresolved branch. Supplier records are dated snapshots and are not guarantees of stock or purity.

## J.8 TABLE S8 | THE SELECTION RULE, AS FROZEN

| Step                           | Rule                                                                                                                                                                                                |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Population                     | designs carrying a calibrated conformal lower bound, chemically admitted, route-complete within four synthetic steps, inside the frozen physicochemical envelope, and absent from the training fold |
| Ranking statistic              | calibrated conformal lower bound, mean minus conformal quantile                                                                                                                                     |
| High arm, first twelve         | highest ranked, on predicted activity alone                                                                                                                                                         |
| High arm, remaining twenty-two | preferring designs absent from the enumerated library, applied only where it does not lower predicted activity                                                                                      |
| Low arm                        | lowest ranked within the tail-generalization tier                                                                                                                                                   |
| Diversity                      | pairwise product Tanimoto at most 0.95; at most four candidates per amine head                                                                                                                      |
| Tie-breaking                   | canonical product SMILES, deterministic                                                                                                                                                             |

The configuration was frozen before the selection ran. The ranking statistic penalises membership in a weakly calibrated region rather than per-candidate ensemble variance: the conformal quantile takes four discrete values across the pool, so within a calibration stratum this ranking is identical to ranking by the point prediction.

## J.9 TABLE S9 | CONFORMAL CALIBRATION AND HOW THE EVIDENCE TIERS WERE DERIVED

The evidence tiers are not labels applied after the fact. Each is a distinct conformal calibration stratum, and the activity model fits one 90% quantile per stratum. The calibrated lower bound used for ranking is the ensemble mean minus that quantile, so a design in a weakly supported stratum is penalised by exactly the width its own calibration data justify.

| Tier                        | Conformal q90 | Interval width relative to T1 | Panel | Eligible pool |
|-----------------------------|---------------|-------------------------------|-------|---------------|
| T1 qualified role holdout   | 3.9147        | 1.00x                         | 8     | 119           |
| T2 head generalization      | 4.2959        | 1.10x                         | 6     | 24            |
| T3 tail generalization      | 5.3094        | 1.36x                         | 24    | 671           |
| T4 dual tail generalization | 6.2145        | 1.59x                         | 2     | 553           |

The quantile widens monotonically as the design moves away from the measured factorial: an unseen isocyanide costs least, an unseen head next, an unseen tail next, and two simultaneous unseen tails most. One caveat is worth stating. The head-generalization stratum is calibrated on only 42 designs, far fewer than the other three, so its interval is estimated less precisely than its width alone suggests.

Two consequences follow for how the panel should be read. Ranking by the calibrated lower bound is identical to ranking by the point prediction within any single tier, and differs between tiers only by the fixed offset above. And because the offset is a stratum constant rather than a per-molecule uncertainty, the ensemble standard deviation, whose median across the eligible pool is 0.36, never enters the ranking.

#### J.10 TABLE S10 | EVERY COMPONENT USED BY THE FORTY CANDIDATES

Components are listed by role with the candidates that use them. A component is marked purchasable where a supplier record exists in the procurement snapshot, and otherwise carries the route by which it is prepared.

##### J.10.1 AMINE HEAD GROUPS (17 DISTINCT)

| SMILES                       | Source   | Suppliers | Used by            |
|------------------------------|----------|-----------|--------------------|
| <chem>NCCN1CCCC1</chem>      | purchase | 75        | P01, P09, P17, P31 |
| <chem>NCCN1CCCC1</chem>      | purchase | 82        | P02, P16, P23, P24 |
| <chem>NCCCN1CCCC1</chem>     | purchase | 75        | P05, P10, P11, P21 |
| <chem>CCCCN(CCCC)CCCN</chem> | purchase | 62        | P06, P07, P28, P29 |
| <chem>CN(C)CCCN</chem>       | purchase | 72        | P03, P08, P32      |
| <chem>NCCCN1CCCC1</chem>     | purchase | 64        | P04, P26, P30      |
| <chem>NN1CCCC1</chem>        | purchase | 67        | P14, P27, P34      |
| <chem>NCCN1CCOCC1</chem>     | purchase | 95        | P13, P25           |
| <chem>CCN(CC)CCN</chem>      | purchase | 64        | P15, P20           |
| <chem>CCN(CC)CCCN</chem>     | purchase | 46        | P18, P19           |
| <chem>CCN1CCCC1CN</chem>     | purchase | 126       | C01, C05           |
| <chem>CN1CCC(N)CC1</chem>    | purchase | 84        | C02, C03           |
| <chem>NCCCN1CCOCC1</chem>    | purchase | 78        | P12                |
| <chem>CCCCCN(CCN)CCCC</chem> | purchase | 1         | P22                |
| <chem>CN(C)CCN</chem>        | purchase | 84        | P33                |
| <chem>NN1CCOCC1</chem>       | purchase | 75        | C04                |
| <chem>NCCN1CCNCC1</chem>     | purchase | 64        | C06                |

##### J.10.2 ESTER-LINKED ALDEHYDE TAILS (26 DISTINCT)

| SMILES                                      | Source                | Suppliers      | Used by            |
|---------------------------------------------|-----------------------|----------------|--------------------|
| <chem>CCCCC(C)CCC(=O)</chem>                | prepare: AGILE Tail A | see precursors | P01, P02, P03, P05 |
| <chem>OCCCCCCCC=O</chem>                    |                       |                |                    |
| <chem>CCCCCCCCC(=O)OC</chem>                | prepare: AGILE Tail A | see precursors | P06, P08, P12      |
| <chem>CCC=O</chem>                          |                       |                |                    |
| <chem>CCCCCCCCC(=O)OC</chem>                | prepare: AGILE Tail A | see precursors | P14, P25, P30      |
| <chem>CCCCC=O</chem>                        |                       |                |                    |
| <chem>C=CCCCCCCCC(=O)</chem>                | prepare: AGILE Tail A | see precursors | P26, P27, P32      |
| <chem>OCCCCC=O</chem>                       |                       |                |                    |
| <chem>CCCCCCCCC(=O)OCC</chem>               | prepare: AGILE Tail A | see precursors | P04, P22           |
| <chem>CCCC=O</chem>                         |                       |                |                    |
| <chem>CCCCCCCCCCCC(=O)O</chem>              | prepare: AGILE Tail A | see precursors | P07, P19           |
| <chem>CCCCC=O</chem>                        |                       |                |                    |
| <chem>CCCCCCCCC=CCCCC(=O)OCCCCCCCC=O</chem> | prepare: AGILE Tail A | see precursors | P11, C04           |
| <chem>C#CCCCCCCCC(=O)</chem>                | prepare: AGILE Tail A | see precursors | P17, P21           |
| <chem>OCCCCC=O</chem>                       |                       |                |                    |
| <chem>CCCCCCCCCCCCCCCC(=O)OCCCCC=O</chem>   | prepare: AGILE Tail A | see precursors | P31, C06           |
| <chem>CCCCC(C)CCC(=O)</chem>                | prepare: AGILE Tail A | see precursors | P09                |
| <chem>OCCCC=O</chem>                        |                       |                |                    |
| <chem>CCCCCCCCCCCC=CCCC</chem>              | prepare: AGILE Tail A | see precursors | P10                |
| <chem>CCC(=O)OCCCC=O</chem>                 |                       |                |                    |
| <chem>CCCCCCCC=CC(=O)O</chem>               | prepare: AGILE Tail A | see precursors | P13                |
| <chem>CCCCC=O</chem>                        |                       |                |                    |
| <chem>CCCCCCC=CCCCCCCC</chem>               | prepare: AGILE Tail A | see precursors | P15                |
| <chem>C(=O)OCCCCCCCC=O</chem>               |                       |                |                    |

| SMILES                                     | Source                | Suppliers      | Used by |
|--------------------------------------------|-----------------------|----------------|---------|
| <chem>CCCCCCCCCCCCCCCC</chem>              | prepare: AGILE Tail A | see precursors | P16     |
| <chem>CC (=O) OCCCCC=O</chem>              | prepare: AGILE Tail A | see precursors | P18     |
| <chem>CCCCCCCC=CCCCC</chem>                | prepare: AGILE Tail A | see precursors | P20     |
| <chem>CCCCC (=O) OCCCC=O</chem>            | prepare: AGILE Tail A | see precursors | P23     |
| <chem>CCCCCCCC=CCCCC</chem>                | prepare: AGILE Tail A | see precursors | P24     |
| <chem>CC (=O) OCCCC=O</chem>               | prepare: AGILE Tail A | see precursors | P28     |
| <chem>CCCCCCCCCCC (=O) O</chem>            | prepare: AGILE Tail A | see precursors | P29     |
| <chem>CCCC=O</chem>                        | prepare: AGILE Tail A | see precursors | P33     |
| <chem>CCCCCCCC=CC (=O) O</chem>            | prepare: AGILE Tail A | see precursors | P34     |
| <chem>CCCCCCCC=O</chem>                    | prepare: AGILE Tail A | see precursors | C01     |
| <chem>C#CCCCCCCCCCCC (=O) OCCCC=O</chem>   | prepare: AGILE Tail A | see precursors | C02     |
| <chem>CCCCCCCCCCCCCCCC</chem>              | prepare: AGILE Tail A | see precursors | C03     |
| <chem>C (=O) OCCCC=O</chem>                | prepare: AGILE Tail A | see precursors | C05     |
| <chem>CCCCCCCCCCCCCCC (=O) OCCCCC=O</chem> | prepare: AGILE Tail A | see precursors |         |
| <chem>CCCCCCCC=CC (=O) OCCCCC=O</chem>     | prepare: AGILE Tail A | see precursors |         |
| <chem>CCCCCCCC=CCCCCCC</chem>              | prepare: AGILE Tail A | see precursors |         |
| <chem>CCC (=O) OCCCC=O</chem>              | prepare: AGILE Tail A | see precursors |         |
| <chem>CCCCCCCC=CCCCC</chem>                | prepare: AGILE Tail A | see precursors |         |
| <chem>C (=O) OCCCCC=O</chem>               | prepare: AGILE Tail A | see precursors |         |
| <chem>CCCCCCC=CCCCCCCC</chem>              | prepare: AGILE Tail A | see precursors |         |
| <chem>CCC (=O) OCCCC=O</chem>              | prepare: AGILE Tail A | see precursors |         |
| <chem>CCCC=CCCCCCCCCCC</chem>              | prepare: AGILE Tail A | see precursors |         |
| <chem>CCC (=O) OCCCCCCCC</chem>            | prepare: AGILE Tail A | see precursors |         |
| <chem>=O</chem>                            |                       |                |         |

### J.10.3 ISOCYANIDE TAILS (7 DISTINCT)

| SMILES                          | Source                                 | Suppliers      | Used by                                                                   |
|---------------------------------|----------------------------------------|----------------|---------------------------------------------------------------------------|
| <chem>[C-] # [N+] CCCCCC</chem> | prepare: isocyanide from primary amine | see precursors | P01, P03, P04, P05, P06, P08, P10, P11, P12, P13, P22, P26, P28, P33, C02 |
| <chem>CC=CCCCCCCC</chem>        | prepare: isocyanide from primary amine | see precursors | P07, P14, P16, P17, P19, P21, P23, P24, P27                               |
| <chem>[C-] # [N+] CCCCCC</chem> | purchase                               | 3              | P15, P18, P29, P31, P34, C01, C05, C06                                    |
| <chem>CCCCC</chem>              | purchase                               | 1              | P02, P25, C04                                                             |
| <chem>[C-] # [N+] CCCCCC</chem> | prepare: isocyanide from primary amine | see precursors | P09, P20                                                                  |
| <chem>CCCCCCC</chem>            | purchase                               | 1              | P30, C03                                                                  |
| <chem>[C-] # [N+] CCCCCC</chem> | prepare: isocyanide from primary amine | see precursors | P32                                                                       |
| <chem>CCCCCCCCCCCCC</chem>      |                                        |                |                                                                           |

### J.11 TABLE S11 | SYNTHESIS ROUTE FOR EVERY CANDIDATE

Each candidate is assembled by a single Ugi three-component reaction from the three components below. Components marked prepare require the upstream step shown, from the starting materials listed. Step counts are the sum across all three components; the final Ugi assembly is not counted, since it is common to every candidate.

| ID  | Role  | Component   | Action   | Route     | Starting materials (suppliers) |
|-----|-------|-------------|----------|-----------|--------------------------------|
| P01 | amine | NCCN1CCCCC1 | purchase | catalogue | n/a                            |

| ID  | Role       | Component                                | Action     | Route                               | Starting materials (suppliers)                          |
|-----|------------|------------------------------------------|------------|-------------------------------------|---------------------------------------------------------|
| P02 | aldehyde   | CCCCC (C) C<br>CC (=O) OCCC<br>CCCC=O    | synthesise | AGILE Tail A                        | CCCCC (C) C<br>CC (=O) O<br>(66),<br>OCCCCCCCCO<br>(82) |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCC=C<br>CCCCCCCC  | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=<br>CCCCCCCCCN<br>(19)                        |
|     | amine      | NCCN1CCCC1                               | purchase   | catalogue                           | n/a                                                     |
|     | aldehyde   | CCCCC (C) C<br>CC (=O) OCCC<br>CCCC=O    | synthesise | AGILE Tail A                        | CCCCC (C) C<br>CC (=O) O<br>(66),<br>OCCCCCCCCO<br>(82) |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC<br>CCCC       | purchase   | catalogue                           | n/a                                                     |
| P03 | amine      | CN (C) CCCN                              | purchase   | catalogue                           | n/a                                                     |
|     | aldehyde   | CCCCC (C) C<br>CC (=O) OCCC<br>CCCC=O    | synthesise | AGILE Tail A                        | CCCCC (C) C<br>CC (=O) O<br>(66),<br>OCCCCCCCCO<br>(82) |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCC | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=<br>CCCCCCCCCN<br>(19)                        |
| P04 | amine      | NCCCN1CCCC1                              | purchase   | catalogue                           | n/a                                                     |
|     | aldehyde   | CCCCCCCC (=<br>O) OCCCCC<br>=O           | synthesise | AGILE Tail A                        | CCCCCCCC (=<br>O) O (113),<br>OCCCCCO<br>(93)           |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCC | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=<br>CCCCCCCCCN<br>(19)                        |
| P05 | amine      | NCCCN1CCCC1                              | purchase   | catalogue                           | n/a                                                     |
|     | aldehyde   | CCCCC (C) C<br>CC (=O) OCCC<br>CCCC=O    | synthesise | AGILE Tail A                        | CCCCC (C) C<br>CC (=O) O<br>(66),<br>OCCCCCCCCO<br>(82) |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCC | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=<br>CCCCCCCCCN<br>(19)                        |
| P06 | amine      | CCCCN (CCCC<br>) CCN                     | purchase   | catalogue                           | n/a                                                     |
|     | aldehyde   | CCCCCCCCC<br>(=O) OCCCC=<br>O            | synthesise | AGILE Tail A                        | CCCCCCCCC<br>(=O) O (107),<br>OCCCCO (68)               |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCC | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=<br>CCCCCCCCCN<br>(19)                        |
| P07 | amine      | CCCCN (CCCC<br>) CCN                     | purchase   | catalogue                           | n/a                                                     |
|     | aldehyde   | CCCCCCCCC<br>C (=O) OCCCC<br>CC=O        | synthesise | AGILE Tail A                        | CCCCCCCCC<br>C (=O) O (101),<br>OCCCCCO<br>(93)         |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCC | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=C<br>CCCCCCCCCN<br>(3)                        |
| P08 | amine      | CN (C) CCCN                              | purchase   | catalogue                           | n/a                                                     |

| ID  | Role       | Component                                        | Action     | Route                               | Starting materials (suppliers)                                 |
|-----|------------|--------------------------------------------------|------------|-------------------------------------|----------------------------------------------------------------|
| P09 | aldehyde   | CCCCCCCCC<br>(=O) OCCCC=O                        | synthesise | AGILE Tail A                        | CCCCCCCCC<br>(=O) O (107),<br>OCCCCO (68)                      |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC        | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=C<br>CCCCCCCCCN<br>(19)                              |
|     | amine      | NCCN1CCCCC1                                      | purchase   | catalogue                           | n/a                                                            |
|     | aldehyde   | CCCCC (C) C<br>CC (=O) OCCC<br>C=O               | synthesise | AGILE Tail A                        | CCCCC (C) C<br>CC (=O) O<br>(66), OCCCCO<br>(68)               |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC<br>CCC                | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC<br>CCCN (68)                                         |
| P10 | amine      | NCCCN1CCCC1                                      | purchase   | catalogue                           | n/a                                                            |
|     | aldehyde   | CCCCCCCCC<br>C=CCCCC (=<br>=O) OCCCC=O           | synthesise | AGILE Tail A                        | CCCCCCCCC<br>C=CCCCC (=<br>=O) O (8),<br>OCCCCO (68)           |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC        | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=C<br>CCCCCCCCCN<br>(19)                              |
| P11 | amine      | NCCCN1CCCC1                                      | purchase   | catalogue                           | n/a                                                            |
|     | aldehyde   | CCCCCCCCC=C<br>CCCCC (=O) O<br>CCCCCCCCC=O       | synthesise | AGILE Tail A                        | CCCCCCCCC=C<br>CCCCC (=O) O<br>(8),<br>OCCCCCCCCO<br>(82)      |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC        | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=C<br>CCCCCCCCCN<br>(19)                              |
| P12 | amine      | NCCCN1CCOCC1                                     | purchase   | catalogue                           | n/a                                                            |
|     | aldehyde   | CCCCCCCCC<br>(=O) OCCCC=O                        | synthesise | AGILE Tail A                        | CCCCCCCCC<br>(=O) O (107),<br>OCCCCO (68)                      |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC        | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=C<br>CCCCCCCCCN<br>(19)                              |
| P13 | amine      | NCCN1CCOCC1                                      | purchase   | catalogue                           | n/a                                                            |
|     | aldehyde   | CCCCCCCCC=C<br>C (=O) OCCCC<br>CC=O              | synthesise | AGILE Tail A                        | CCCCCCCCC=C<br>C (=O) O (39),<br>OCCCCCO<br>(93)               |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC        | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=C<br>CCCCCCCCCN<br>(19)                              |
| P14 | amine      | NN1CCCCC1                                        | purchase   | catalogue                           | n/a                                                            |
|     | aldehyde   | CCCCCCCCC<br>(=O) OCCCC<br>C=O                   | synthesise | AGILE Tail A                        | CCCCCCCCC<br>(=O) O (107),<br>OCCCCCO<br>(93)                  |
|     | isocyanide | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC        | synthesise | isocyanide<br>from primary<br>amine | CCCCCCCCC=C<br>CCCCCCCCCN<br>(3)                               |
| P15 | amine      | CCN (CC) CCN                                     | purchase   | catalogue                           | n/a                                                            |
|     | aldehyde   | CCCCCCCC=CC<br>CCCCCCC (=O)<br>) OCCCCCCCC<br>=O | synthesise | AGILE Tail A                        | CCCCCCCC=CC<br>CCCCCCC (=O)<br>) O (27),<br>OCCCCCCCCO<br>(82) |

| ID  | Role              | Component                                                     | Action                 | Route                               | Starting materials (suppliers)                                 |
|-----|-------------------|---------------------------------------------------------------|------------------------|-------------------------------------|----------------------------------------------------------------|
| P16 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>C                               | purchase               | catalogue                           | n/a                                                            |
| P16 | amine<br>aldehyde | NCCN1CCCC1<br>CCCCCCCCC<br>CCCCCCCC (=O)<br>OCCCCC=O          | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC<br>CCCCCCCC (=O) O (158),<br>OCCCCCO (93)     |
| P17 | isocyanide        | [C-] # [N+] C<br>CCCCCCCC=C<br>CCCCCCC                        | synthesise             | isocyanide<br>from primary<br>amine | CCCCCCCC=C<br>CCCCCCCCN (3)                                    |
| P17 | amine<br>aldehyde | NCCN1CCCC1<br>C#CCCCCCCC<br>CC (=O) OCCC<br>CCC=O             | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>C#CCCCCCCC<br>CC (=O) O (74),<br>OCCCCCO (93)           |
| P18 | isocyanide        | [C-] # [N+] C<br>CCCCCCCC=C<br>CCCCCCC                        | synthesise             | isocyanide<br>from primary<br>amine | CCCCCCCC=C<br>CCCCCCCCN (3)                                    |
| P18 | amine<br>aldehyde | CCN (CC) CCCN<br>CCCCCCCCC<br>CCCCCCCCC<br>C (=O) OCCCC<br>=O | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC<br>CCCCCCCCC<br>C (=O) O (20),<br>OCCCCO (68) |
| P19 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>C                               | purchase               | catalogue                           | n/a                                                            |
| P19 | amine<br>aldehyde | CCN (CC) CCCN<br>CCCCCCCCC<br>C (=O) OCCCC<br>CC=O            | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC<br>C (=O) O (101),<br>OCCCCCO (93)            |
| P20 | isocyanide        | [C-] # [N+] C<br>CCCCCCCC=C<br>CCCCCCC                        | synthesise             | isocyanide<br>from primary<br>amine | CCCCCCCC=C<br>CCCCCCCCN (3)                                    |
| P20 | amine<br>aldehyde | CCN (CC) CCN<br>CCCCCCCCC<br>CCCCCCCC (=O)<br>OCCCC=O         | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC<br>CCCCCCCC (=O) O (2),<br>OCCCCO (68)        |
| P21 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>CCC                             | synthesise             | isocyanide<br>from primary<br>amine | CCCCCCCCC<br>CCCN (68)                                         |
| P21 | amine<br>aldehyde | NCCN1CCCC1<br>C#CCCCCCCC<br>CC (=O) OCCC<br>CCC=O             | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>C#CCCCCCCC<br>CC (=O) O (74),<br>OCCCCCO (93)           |
| P22 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>CCCCCCC                         | synthesise             | isocyanide<br>from primary<br>amine | CCCCCCCCC<br>CCCCCCCCN (3)                                     |
| P22 | amine<br>aldehyde | CCCCCN (CCN)<br>) CCCC<br>CCCCCCCC (=O)<br>OCCCCC=O           | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCC (=O) O (113),<br>OCCCCCO (93)                  |

| ID  | Role              | Component                                                         | Action                 | Route                                                            | Starting materials (suppliers)                               |
|-----|-------------------|-------------------------------------------------------------------|------------------------|------------------------------------------------------------------|--------------------------------------------------------------|
| P23 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC                         | synthesise             | isocyanide<br>from primary<br>amine<br>catalogue<br>AGILE Tail A | CCCCCCCCC=C<br>CCCCCCCCCN<br>(19)<br>n/a                     |
|     | amine<br>aldehyde | NCCN1CCCC1<br>CCCCCCCCC=C<br>C (=O) OCCCC<br>=O                   | purchase<br>synthesise |                                                                  | CCCCCCCCC=C<br>C (=O) O (101),<br>OCCCCO (68)                |
| P24 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC                         | synthesise             | isocyanide<br>from primary<br>amine<br>catalogue<br>AGILE Tail A | CCCCCCCCC=C<br>CCCCCCCCCN<br>(3)<br>n/a                      |
|     | amine<br>aldehyde | NCCN1CCCC1<br>CCCCCCCCC=C<br>C (=O) OCCCC<br>CCCC=O               | purchase<br>synthesise |                                                                  | CCCCCCCCC=C<br>C (=O) O (39),<br>OCCCCCCCCO<br>(82)          |
| P25 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC                         | synthesise             | isocyanide<br>from primary<br>amine<br>catalogue<br>AGILE Tail A | CCCCCCCCC=C<br>CCCCCCCCCN<br>(3)<br>n/a                      |
|     | amine<br>aldehyde | NCCN1CCOCC1<br>CCCCCCCCC=C<br>C (=O) OCCCCC<br>C=O                | purchase<br>synthesise |                                                                  | CCCCCCCCC=C<br>C (=O) O (107),<br>OCCCCCCO<br>(93)<br>n/a    |
| P26 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC                         | purchase               | catalogue                                                        | n/a                                                          |
|     | amine<br>aldehyde | NCCN1CCCC1<br>C=CCCCCCCC<br>CC (=O) OCCC<br>CCC=O                 | purchase<br>synthesise | catalogue<br>AGILE Tail A                                        | n/a<br>C=CCCCCCCC<br>CC (=O) O<br>(115),<br>OCCCCCCO<br>(93) |
| P27 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC                         | synthesise             | isocyanide<br>from primary<br>amine<br>catalogue<br>AGILE Tail A | CCCCCCCCC=C<br>CCCCCCCCCN<br>(19)<br>n/a                     |
|     | amine<br>aldehyde | NN1CCCC1<br>C=CCCCCCCC<br>CC (=O) OCCC<br>CCC=O                   | purchase<br>synthesise |                                                                  | C=CCCCCCCC<br>CC (=O) O<br>(115),<br>OCCCCCCO<br>(93)        |
| P28 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC                         | synthesise             | isocyanide<br>from primary<br>amine<br>catalogue                 | CCCCCCCCC=C<br>CCCCCCCCCN<br>(3)<br>n/a                      |
|     | amine<br>aldehyde | CCCCN (CCCC<br>) CCCN<br>C#CCCCCCCC<br>CCCC (=O) OC<br>CCC=O      | purchase<br>synthesise | AGILE Tail A                                                     | C#CCCCCCCC<br>CCCC (=O) O<br>(32), OCCCCO<br>(68)            |
| P29 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC                         | synthesise             | isocyanide<br>from primary<br>amine<br>catalogue                 | CCCCCCCCC=C<br>CCCCCCCCCN<br>(19)<br>n/a                     |
|     | amine<br>aldehyde | CCCCN (CCCC<br>) CCCN<br>CCCCCCCCC=C<br>CCCCCCCC (=O<br>) OCCCC=O | purchase<br>synthesise | AGILE Tail A                                                     | CCCCCCCCC=C<br>CCCCCCCC (=O<br>) O (103),<br>OCCCCO (68)     |

| ID  | Role              | Component                                                   | Action                 | Route                               | Starting materials (suppliers)                                    |
|-----|-------------------|-------------------------------------------------------------|------------------------|-------------------------------------|-------------------------------------------------------------------|
| P30 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>C                             | purchase               | catalogue                           | n/a                                                               |
| P30 | amine<br>aldehyde | NCCCN1CCCCC1<br>CCCCCCCCC<br>(=O) OCCCC<br>C=O              | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC<br>(=O) O (107),<br>OCCCCCO<br>(93)              |
| P31 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>CCCCCCC                       | purchase               | catalogue                           | n/a                                                               |
| P31 | amine<br>aldehyde | NCCN1CCCCC1<br>CCCCCCCCC<br>CCCCC (=O) O<br>CCCCC=O         | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC<br>CCCCC (=O) O<br>(92),<br>OCCCCCO<br>(93)      |
| P32 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>C                             | purchase               | catalogue                           | n/a                                                               |
| P32 | amine<br>aldehyde | CN (C) CCCN<br>C=CCCCCCCC<br>CC (=O) OCCC<br>CCC=O          | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>C=CCCCCCCC<br>CC (=O) O<br>(115),<br>OCCCCCO<br>(93)       |
| P33 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>CCCCCCCCC                     | synthesise             | isocyanide<br>from primary<br>amine | CCCCCCCCC<br>CCCCCCCCC<br>N (38)                                  |
| P33 | amine<br>aldehyde | CN (C) CCN<br>CCCCCCCCC<br>CCC (=O) OCC<br>CCCC=O           | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC<br>CCC (=O) O<br>(88),<br>OCCCCCO<br>(93)        |
| P34 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCCC                   | synthesise             | isocyanide<br>from primary<br>amine | CCCCCCCCC=<br>CCCCCCCCCN<br>(19)                                  |
| P34 | amine<br>aldehyde | NN1CCCCC1<br>CCCCCCCCC=<br>CC (=O) OCCC<br>CCC=O            | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC=<br>CC (=O) O<br>(18),<br>OCCCCCO<br>(93)        |
| C01 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>C                             | purchase               | catalogue                           | n/a                                                               |
| C01 | amine<br>aldehyde | CCN1CCCCC1CN<br>CCCCCCCCC=C<br>CCCCCCCCC (=O) OCCCC=O       | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC=C<br>CCCCCCCCC (=O) O (6),<br>OCCCCO (68)        |
| C02 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>C                             | purchase               | catalogue                           | n/a                                                               |
| C02 | amine<br>aldehyde | CN1CCC (N) CC1<br>CCCCCCCCC=<br>CCCCCCCC (=O)<br>) OCCCCC=O | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC=<br>CCCCCCCC (=O)<br>) O (8),<br>OCCCCCO<br>(93) |

| ID  | Role              | Component                                                       | Action                 | Route                               | Starting materials (suppliers)                                         |
|-----|-------------------|-----------------------------------------------------------------|------------------------|-------------------------------------|------------------------------------------------------------------------|
| C03 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC=C<br>CCCCCCCC                        | synthesise             | isocyanide<br>from primary<br>amine | CCCCCCCCC=CC<br>CCCCCCCCCN<br>(19)                                     |
|     | amine<br>aldehyde | CN1CCC (N) CC1<br>CCCCCCCCC=CC<br>CCCCCCCCC (=<br>=O) OCCCC=O   | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC=CC<br>CCCCCCCCC (=<br>=O) O (19),<br>OCCCCO (68)      |
|     | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>CCCCCCCC                          | purchase               | catalogue                           | n/a                                                                    |
|     | amine<br>aldehyde | NN1CCOCC1<br>CCCCCCCCC=CC<br>CCCCC (=O) O<br>CCCCCCCCC=O        | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC=CC<br>CCCCC (=O) O<br>(8),<br>OCCCCCCCCCO<br>(82)     |
|     | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>CCCCC                             | purchase               | catalogue                           | n/a                                                                    |
| C05 | amine<br>aldehyde | CCN1CCCC1CN<br>CCCCC=CCCC<br>CCCCCCCCC (=<br>=O) OCCCCC<br>CC=O | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCC=CCCC<br>CCCCCCCCC (=<br>=O) O (6),<br>OCCCCCCCCCO<br>(82) |
|     | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>C                                 | purchase               | catalogue                           | n/a                                                                    |
|     | amine<br>aldehyde | NCCN1CCNCC1<br>CCCCCCCCC<br>CCCCC (=O) O<br>CCCCC=O             | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a<br>CCCCCCCCC<br>CCCCC (=O) O<br>(92),<br>OCCCCCO<br>(93)           |
|     | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>C                                 | purchase               | catalogue                           | n/a                                                                    |
|     | amine<br>aldehyde | NCCN1CCNCC1<br>CCCCCCCCC<br>CCCCC (=O) O<br>CCCCC=O             | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a                                                                    |
| C06 | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>C                                 | purchase               | catalogue                           | n/a                                                                    |
|     | amine<br>aldehyde | NCCN1CCNCC1<br>CCCCCCCCC<br>CCCCC (=O) O<br>CCCCC=O             | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a                                                                    |
|     | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>C                                 | purchase               | catalogue                           | n/a                                                                    |
|     | amine<br>aldehyde | NCCN1CCNCC1<br>CCCCCCCCC<br>CCCCC (=O) O<br>CCCCC=O             | purchase<br>synthesise | catalogue<br>AGILE Tail A           | n/a                                                                    |
|     | isocyanide        | [C-] # [N+] C<br>CCCCCCCCC<br>C                                 | purchase               | catalogue                           | n/a                                                                    |

## J.12 TABLE S12 | BILL OF MATERIALS

Every distinct chemical that must be purchased to execute the panel, with the supplier count recorded in the procurement snapshot. Supplier records are dated and are not guarantees of stock, purity or lead time.

| SMILES                   | Role             | Suppliers | Candidates requiring it |
|--------------------------|------------------|-----------|-------------------------|
| OCCCCCO                  | alpha,omega-diol | 93        | 19                      |
| CCCCCCCCC=CCCCCCCCCN     | primary amine    | 19        | 15                      |
| OCCCCO                   | alpha,omega-diol | 68        | 12                      |
| CCCCCCCCC=CCCCCCCCCN     | primary amine    | 3         | 9                       |
| OCCCCCCCCO               | alpha,omega-diol | 82        | 9                       |
| [C-] # [N+] CCCCCCCCCCCC | isocyanide tail  | 3         | 8                       |
| CCCCCCCCC (=O) O         | carboxylic acid  | 107       | 6                       |
| CCCCC (C) CCC (=O) O     | carboxylic acid  | 66        | 5                       |
| CCCN (CCCC) CCN          | amine head       | 62        | 4                       |
| NCCCN1CCCC1              | amine head       | 75        | 4                       |
| NCCN1CCCC1               | amine head       | 82        | 4                       |
| NCCN1CCCC1               | amine head       | 75        | 4                       |
| C=CCCCCCCCC (=O) O       | carboxylic acid  | 115       | 3                       |
| CCCCCCCCC (=O) O         | carboxylic acid  | 101       | 3                       |

| SMILES                         | Role            | Suppliers | Candidates requiring it |
|--------------------------------|-----------------|-----------|-------------------------|
| CN (C) CCCN                    | amine head      | 72        | 3                       |
| NCCCN1CCCCC1                   | amine head      | 64        | 3                       |
| NN1CCCCC1                      | amine head      | 67        | 3                       |
| [C-] # [N+] CCCCCCCCCCCCCCCC   | isocyanide tail | 1         | 3                       |
| C#CCCCCCCCCCC (=O) O           | carboxylic acid | 74        | 2                       |
| CCCCCCCC=CC (=O) O             | carboxylic acid | 39        | 2                       |
| CCCCCCCC (=O) O                | carboxylic acid | 113       | 2                       |
| CCCCCCCC=CCCCC (=O) O          | carboxylic acid | 8         | 2                       |
| CCCCCCCCCCCCCCCC (=O) O        | carboxylic acid | 92        | 2                       |
| CCCCCCCCCCCCCCCCN              | primary amine   | 68        | 2                       |
| CCN (CC) CCCN                  | amine head      | 46        | 2                       |
| CCN (CC) CCN                   | amine head      | 64        | 2                       |
| CCN1CCCC1CN                    | amine head      | 126       | 2                       |
| CN1CCC (N) CC1                 | amine head      | 84        | 2                       |
| NCCN1CCOCC1                    | amine head      | 95        | 2                       |
| [C-] # [N+] CCCCCCCCCCCCCCCCCC | isocyanide tail | 1         | 2                       |
| C#CCCCCCCCCCCCC (=O) O         | carboxylic acid | 32        | 1                       |
| CCCCC=CCCCCCCCCCCCC (=O) O     | carboxylic acid | 6         | 1                       |
| CCCCCCC=CCCCCCCCCCC (=O) O     | carboxylic acid | 27        | 1                       |
| CCCCCCC=CCCCCCCCCCCC (=O) O    | carboxylic acid | 19        | 1                       |
| CCCCCCCC=CCCCCCCCCCCC (=O) O   | carboxylic acid | 6         | 1                       |
| CCCCCCCC=CC (=O) O             | carboxylic acid | 18        | 1                       |
| CCCCCCCC=CCCCCCC (=O) O        | carboxylic acid | 8         | 1                       |
| CCCCCCCC=CCCCCCCC (=O) O       | carboxylic acid | 2         | 1                       |
| CCCCCCCC=CCCCCCCCCCCC (=O) O   | carboxylic acid | 20        | 1                       |
| CCCCCCCCCCCC=CCCCCCC (=O) O    | carboxylic acid | 8         | 1                       |
| CCCCCCCCCCCCCCC (=O) O         | carboxylic acid | 88        | 1                       |
| CCCCCCCCCCCCCCCCCCCC (=O) O    | carboxylic acid | 103       | 1                       |
| CCCCCCCCCCCCCCCCCCCC (=O) O    | carboxylic acid | 158       | 1                       |
| CCCCCCCCCCCCCCCCCCCCCN         | primary amine   | 38        | 1                       |
| CCCCCN (CCN) CCCC              | amine head      | 1         | 1                       |
| CN (C) CCN                     | amine head      | 84        | 1                       |
| NCCCN1CCOCC1                   | amine head      | 78        | 1                       |
| NCCN1CCNCC1                    | amine head      | 64        | 1                       |
| NN1CCOCC1                      | amine head      | 75        | 1                       |

49 distinct materials in total.

### J.13 TABLE S13 | THE ROUTING ALGORITHM

Routing runs after generation and after the exact forward check, and never modifies a design. It answers one question per component: can this be bought, and if not, can it be reached from something that can. A design whose components cannot all be resolved is dropped rather than repaired.

```

for each generated product:
  decompose into (amine head, ester-linked aldehyde tail, isocyanide tail)
  for each component:
    if a supplier record exists in the procurement snapshot:
      mark purchase; resolved; 0 steps
    else if the component is an ester-linked aldehyde:
      disconnect by the AGILE Tail A route:
        carboxylic acid + alpha,omega-diol -> esterify -> oxidise
        resolved only if BOTH starting materials have supplier records
        2 steps
    else if the component is an isocyanide:
      disconnect to its primary amine:
        primary amine -> formylate -> dehydrate
        resolved only if the amine has a supplier record
        2 steps
    else:
      unresolved
  route_complete := every component resolved

```

```
total_steps := sum of component steps      # final Ugi assembly not counted
admit only if route_complete and total_steps <= 4
```

Amine heads have no disconnection rule in this instantiation, so an unseen head must be purchasable outright. Both tail routes are published chemistry rather than model proposals: the ester-linked aldehydes follow esterification of a carboxylic acid with an alpha,omega-diol and oxidation of the remaining primary alcohol, and the isocyanides follow formylation of a primary amine and dehydration of the resulting formamide.

Two limits bound what this establishes. Route completeness is computational, and predicts that a documented disconnection exists with purchasable starting materials; it is not evidence that any specific reaction will succeed at scale, in yield or in purity. And the eligible population is bounded by procurement coverage, so a design absent from the pool may be unroutable or may simply never have been assessed.

#### J.14 TABLE S14 | GENERATOR ARCHITECTURE AND PARAMETER BUDGET

The generator is a discrete flow-matching model over a sparse graph serialisation. It jointly denoises offspring topology, atom states, parent-bond states, sparse closure bonds and terminal decoration states through one shared role-aware backbone, rather than through separate topology and chemistry models.

| Component                | Value                                           |
|--------------------------|-------------------------------------------------|
| Total parameters         | 1,159,916                                       |
| Backbone                 | 4-layer bidirectional GRU, hidden dimension 192 |
| Dropout                  | 0.15                                            |
| Decoration conditioning  | bidirectional_anchor_local                      |
| Source probability floor | 1e-05                                           |
| Reverse sampling steps   | 8 discrete reverse-star transitions             |

| Parameter block                     | Parameters       | Share |
|-------------------------------------|------------------|-------|
| bidirectional GRU backbone          | 668,160          | 57.6% |
| output heads                        | 268,268          | 23.1% |
| shared projection and normalisation | 112,320          | 9.7%  |
| input embeddings                    | 111,168          | 9.6%  |
| <b>Total</b>                        | <b>1,159,916</b> |       |

The backbone carries most of the capacity. The model holds no component identifiers and no stored molecular fragments, so it cannot retrieve a head group or tail from a table; every component is constructed atom by atom under the declared support.

#### J.15 TABLE S15 | STATE SPACE AND DECLARED MOLECULAR SUPPORT

Generation is non-enumerative but bounded. The model emits categorical states over the vocabularies below, and the support bounds define the region within which the framework claims to generate. Designs outside these bounds are not produced rather than filtered afterwards.

| Categorical state         | Vocabulary size |
|---------------------------|-----------------|
| offspring branching state | 4               |
| atom state                | 14              |
| bond state                | 4               |
| precursor role            | 3               |
| within-origin position    | 32              |
| pending-frontier state    | 161             |
| decoration slot           | 7               |

| Support bound                                 | Value |
|-----------------------------------------------|-------|
| Maximum exterior atoms per product            | 80    |
| Maximum atoms per precursor-derived component | 32    |

| Support bound                         | Value |
|---------------------------------------|-------|
| Maximum children per node             | 3     |
| Maximum junction budget per component | 7     |
| Maximum cycle rank per component      | 2     |
| Maximum core attachments              | 2     |
| Maximum terminal decorations          | 7     |
| Bond classes                          | 4     |

Each product is encoded as a deterministic spanning tree plus its sparse residual closure edges, with a fixed five-atom Ugi core and three precursor-origin exterior regions. The morphology program supplied to the model carries only per-origin atom counts, junction budgets, cycle ranks and attachment counts. It carries no component identity, which is what makes the output generation rather than retrieval.

J.16 TABLE S16 | TRAINING AND CHECKPOINT SELECTION

| Setting                                      | Value                                                        |
|----------------------------------------------|--------------------------------------------------------------|
| Training corpus                              | 112,386 products, full corpus for the production refit       |
| Development training fold                    | 66,464 products                                              |
| Calibration fold, development early stopping | 15,800 products                                              |
| Production updates                           | 5,100, fixed, no early stopping                              |
| Stop rule                                    | fixed_final_step (maximum_steps)                             |
| Optimiser                                    | AdamW, batch size 128, learning rate 1e-4, weight decay 5e-5 |
| Gradient clipping                            | global norm 1.0                                              |
| Precision and hardware                       | deterministic float32, one NVIDIA L4 GPU                     |
| Flow time sampling                           | uniform, truncated to [0.02, 0.98]                           |

The production model was refit once from random initialisation for a fixed step budget with no early stopping and no post-refit checkpoint selection, so no held-out quantity chose the shipped weights. Architecture selection happened earlier, on a separate paired draw of 1,024 programs, under prespecified hard gates for validity, forward and inverse exactness, component-handle qualification, uniqueness and declared-support violations. Likelihood did not select the production model: the best calibration-loss checkpoint in development occurred at step 500, which was not the selected configuration.

Checkpoint pinned for these tables: results/phase1/ugi\_decoration\_coupling\_production\_refit\_v1/checkpoint\_latest.pt, step 5100.

Table 19: **Product structures of the frozen Library 0 pilot.** Canonical non-isomeric SMILES; generation is constitutional and proposes no stereochemistry. *Novel* marks absence from the enumerated reference library. Predicted activity and the retired calibrated bound are omitted, since this table records which molecules were proposed rather than how they were ranked. No member has been synthesized.

| ID  | Arm  | Steps | Novel | Product SMILES                                                                |
|-----|------|-------|-------|-------------------------------------------------------------------------------|
| P01 | HIGH | 4     | no    | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCCCCOC (=O) CCC (C) CCCCC) NCCN1CCCCC1      |
| P02 | HIGH | 2     | no    | CCCCCCCCCCCCCCCCCNC (=O) C (CCCCCCCCOC (=O) CCC (C) CCCCC) NCCN1CCCCC1        |
| P03 | HIGH | 4     | no    | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCCCCOC (=O) CCC (C) CCCCC) NCCN (C) C       |
| P04 | HIGH | 4     | yes   | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCCCCOC (=O) CCCCCCCC) NCCN1CCCCC1           |
| P05 | HIGH | 4     | no    | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCCCCOC (=O) CCC (C) CCCCC) NCCN1CCCCC1      |
| P06 | HIGH | 4     | no    | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCOC (=O) CCCCCCCCC) NCCN (CCCC) CCCC         |
| P07 | HIGH | 4     | yes   | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCCCCOC (=O) CCCCCCCCCC) NCCN (CCCC) CCCC    |
| P08 | HIGH | 4     | no    | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCOC (=O) CCCCCCCCC) NCCN (C) C               |
| P09 | HIGH | 4     | no    | CCCCCCCCCCCCCCCCCNC (=O) C (CCCCOC (=O) CCC (C) CCCCC) NCCN1CCCCC1            |
| P10 | HIGH | 4     | yes   | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCOC (=O) CCCCC=CCCCCCCCCCCC) NCCN1CCCCC1     |
| P11 | HIGH | 4     | yes   | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCCCCOC (=O) CCCC=CCCCCCCCC) NCCN1CCCCC1     |
| P12 | HIGH | 4     | no    | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCOC (=O) CCCCCCCCC) NCCN1CCOCC1              |
| P13 | HIGH | 4     | yes   | CCCCCCCCC=CC (=O) OCCCCC (NCCN1CCOCC1) C (=O) NCCCCCCCCC=CCCCCCCCC            |
| P14 | HIGH | 4     | yes   | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCCCCOC (=O) CCCCCCCCC) NN1CCCCC1            |
| P15 | HIGH | 2     | yes   | CCCCCCCC=CCCCCCCCC (=O) OCCCCCCCC (NCCN (CC) CC) C (=O) NCCCCCCCCCCCCC        |
| P16 | HIGH | 4     | yes   | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCCCCOC (=O) CCCCCCCCCCCCCCCCCC) NCCN1CCCCC1 |
| P17 | HIGH | 4     | yes   | C#CCCCCCCCC (=O) OCCCCC (NCCN1CCCCC1) C (=O) NCCCCCCCCC=CCCCCCCCC             |
| P18 | HIGH | 2     | yes   | CCCCCCCCC=CCCCCCCCCCCCC (=O) OCCCCC (NCCN (CC) CC) C (=O) NCCCCCCCCCCCCC      |
| P19 | HIGH | 4     | yes   | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCCCCOC (=O) CCCCCCCCCC) NCCN (CC) CC        |
| P20 | HIGH | 4     | yes   | CCCCCCCCC=CCCCCCCCC (=O) OCCCCC (NCCN (CC) CC) C (=O) NCCCCCCCCC=CCCCCCCCC    |
| P21 | HIGH | 4     | yes   | C#CCCCCCCCC (=O) OCCCCC (NCCN1CCCCC1) C (=O) NCCCCCCCCC=CCCCCCCCC             |
| P22 | HIGH | 4     | yes   | CCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCCCCOC (=O) CCCCCCCCC) NCCN (CCCC) CCCCC    |

| ID  | Arm  | Steps | Novel | Product SMILES                                                                |
|-----|------|-------|-------|-------------------------------------------------------------------------------|
| P23 | HIGH | 4     | yes   | CCCCCCCC=CCCCCCCCCNC (=O) C (CCCC (=O) CCCCCCCCC) NCCN1CCCC1                  |
| P24 | HIGH | 4     | yes   | CCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCCCC (=O) C=CCCCCCCC) NCCN1CCCC1             |
| P25 | HIGH | 2     | yes   | CCCCCCCCCCCCCCCCCNC (=O) C (CCCCCOC (=O) CCCCCCCCC) NCCN1CCOCC1               |
| P26 | HIGH | 4     | yes   | C=CCCCCCCCCCCC (=O) OCCCCCC (NCCCN1CCCC1) C (=O) NCCCCCCCCC=CCCCCCCC          |
| P27 | HIGH | 4     | yes   | C=CCCCCCCCCCCC (=O) OCCCCCC (NN1CCCC1) C (=O) NCCCCCCCCC=CCCCCCCC             |
| P28 | HIGH | 4     | yes   | C#CCCCCCCCCCCC (=O) OCCCC (NCCCN (CCCC) CCCC) C (=O) NCCCCCCCCC=CCCCCCCC      |
| P29 | HIGH | 2     | yes   | CCCCCCCCCCCCCCCCC (=O) OCCCC (NCCCN (CCCC) CCCC) C (=O) NCCCCCCCCCCCC         |
| P30 | HIGH | 2     | yes   | CCCCCCCCCCCCCCCCCNC (=O) C (CCCCCOC (=O) CCCCCCCCC) NCCCN1CCCC1               |
| P31 | HIGH | 2     | yes   | CCCCCCCCCCCCCCCCC (=O) OCCCCCC (NCCN1CCCC1) C (=O) NCCCCCCCCCCCC              |
| P32 | HIGH | 4     | yes   | C=CCCCCCCCCCCC (=O) OCCCCCC (NCCCN (C) C) C (=O) NCCCCCCCCCCCCCCCCCCCC        |
| P33 | HIGH | 4     | yes   | CCCCCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCOC (=O) CCCCCCCCCCCCC) NCCN (C) C       |
| P34 | HIGH | 2     | yes   | CCCCCCCCCCCC=CC (=O) OCCCCCC (NN1CCCC1) C (=O) NCCCCCCCCCCCC                  |
| C01 | LOW  | 2     | yes   | CCCCCCCC=CCCCCCCCC (=O) OCCCC (NCC1CCCN1CC) C (=O) NCCCCCCCCCCCC              |
| C02 | LOW  | 4     | yes   | CCCCCCCCCCCC=CCCCCCCCCNC (=O) C (CCCCCOC (=O) CCCCCC=CCCCCCCC) NC1CCN (C) CC1 |
| C03 | LOW  | 2     | yes   | CCCCCCC=CCCCCCCCCCCC (=O) OCCCC (NC1CCN (C) CC1) C (=O) NCCCCCCCCCCCCCCCCCCCC |
| C04 | LOW  | 2     | yes   | CCCCCCCCCCCC=CCCCC (=O) OCCCCCC (NN1CCOCC1) C (=O) NCCCCCCCCCCCCCCCC          |
| C05 | LOW  | 2     | yes   | CCCCC=CCCCCCCCCCCCC (=O) OCCCCCC (NCC1CCCN1CC) C (=O) NCCCCCCCCCCCCC          |
| C06 | LOW  | 2     | yes   | CCCCCCCCCCCCCCCCC (=O) OCCCCCC (NCCN1CCNCC1) C (=O) NCCCCCCCCCCCC             |

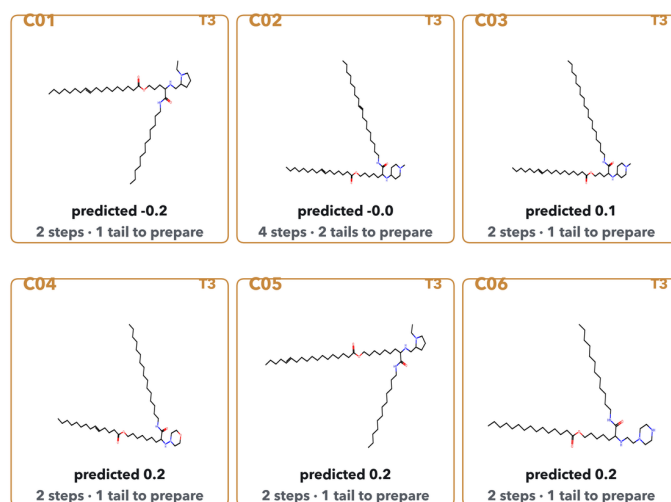
Table 20: **Components of every frozen candidate.** Each product is assembled from exactly these three precursor-derived graphs under the qualified forward Ugi transform.

| ID  | Amine head        | Aldehyde tail                       | Isocyanide tail                   |
|-----|-------------------|-------------------------------------|-----------------------------------|
| P01 | NCCN1CCCC1        | CCCCC (C) CCC (=O) OCCCCCCC=O       | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P02 | NCCN1CCCC1        | CCCCC (C) CCC (=O) OCCCCCCC=O       | [C-] # [N+] CCCCCCCCCCCCCCCCC     |
| P03 | CN (C) CCCN       | CCCCC (C) CCC (=O) OCCCCCCC=O       | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P04 | NCCCN1CCCC1       | CCCCCCCC (=O) OCCCCCC=O             | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P05 | NCCCN1CCCC1       | CCCCC (C) CCC (=O) OCCCCCCC=O       | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P06 | CCCN (CCCC) CCCN  | CCCCCCCCC (=O) OCCCC=O              | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P07 | CCCN (CCCC) CCCN  | CCCCCCCCC (=O) OCCCCCC=O            | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P08 | CN (C) CCCN       | CCCCCCCCC (=O) OCCCC=O              | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P09 | NCCN1CCCC1        | CCCCC (C) CCC (=O) OCCCC=O          | [C-] # [N+] CCCCCCCCCCCCCCCCC     |
| P10 | NCCCN1CCCC1       | CCCCCCCCC=CCCCC (=O) OCCCC=O        | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P11 | NCCCN1CCCC1       | CCCCCCCCC=CCCCC (=O) OCCCCCCC=O     | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P12 | NCCCN1CCOCC1      | CCCCCCCCC (=O) OCCCC=O              | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P13 | NCCN1CCOCC1       | CCCCCCCC=CC (=O) OCCCCCC=O          | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P14 | NN1CCCC1          | CCCCCCCCC (=O) OCCCCCC=O            | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P15 | CCN (CC) CCN      | CCCCCCC=CCCCCCCCC (=O) OCCCCCCC=O   | [C-] # [N+] CCCCCCCCCCCCC         |
| P16 | NCCN1CCCC1        | CCCCCCCCC=CCCCCCCCC (=O) OCCCCCC=O  | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P17 | NCCN1CCCC1        | C#CCCCCCCCC (=O) OCCCCCC=O          | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P18 | CCN (CC) CCCN     | CCCCCCCCC=CCCCCCCCCCCC (=O) OCCCC=O | [C-] # [N+] CCCCCCCCCCCCC         |
| P19 | CCN (CC) CCCN     | CCCCCCCCCCCC (=O) OCCCCCC=O         | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P20 | CCN (CC) CCN      | CCCCCCCCC=CCCCCCCC (=O) OCCCC=O     | [C-] # [N+] CCCCCCCCCCCCCCCCC     |
| P21 | NCCCN1CCCC1       | C#CCCCCCCCC (=O) OCCCCCC=O          | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P22 | CCCCCN (CCN) CCCC | CCCCCCCCC (=O) OCCCCCC=O            | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P23 | NCCN1CCCC1        | CCCCCCCCCCCC (=O) OCCCC=O           | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P24 | NCCN1CCCC1        | CCCCCCCCC=CC (=O) OCCCCCCC=O        | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P25 | NCCN1CCOCC1       | CCCCCCCCC (=O) OCCCCCC=O            | [C-] # [N+] CCCCCCCCCCCCCCCCC     |
| P26 | NCCCN1CCCC1       | C=CCCCCCCCC (=O) OCCCCCC=O          | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P27 | NN1CCCC1          | C=CCCCCCCCC (=O) OCCCCCC=O          | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P28 | CCCN (CCCC) CCCN  | C#CCCCCCCCCCCC (=O) OCCCC=O         | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P29 | CCCN (CCCC) CCCN  | CCCCCCCCCCCCCCCCCCCC (=O) OCCCC=O   | [C-] # [N+] CCCCCCCCCCCCCCCCC     |
| P30 | NCCCN1CCCC1       | CCCCCCCCCCCC (=O) OCCCCCC=O         | [C-] # [N+] CCCCCCCCCCCCCCCCC     |
| P31 | NCCN1CCCC1        | CCCCCCCCCCCCCCCCC (=O) OCCCCCC=O    | [C-] # [N+] CCCCCCCCCCCCC         |
| P32 | CN (C) CCCN       | C=CCCCCCCCC (=O) OCCCCCC=O          | [C-] # [N+] CCCCCCCCCCCCCCCCCCCCC |
| P33 | CN (C) CCN        | CCCCCCCCCCCCC (=O) OCCCCCC=O        | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| P34 | NN1CCCC1          | CCCCCCCCC=CC (=O) OCCCCCC=O         | [C-] # [N+] CCCCCCCCCCCCC         |
| C01 | CCN1CCCC1CN       | CCCCCCCCC=CCCCCCCCC (=O) OCCCC=O    | [C-] # [N+] CCCCCCCCCCCCC         |
| C02 | CN1CCC (N) CC1    | CCCCCCCCC=CCCCCCCC (=O) OCCCCCC=O   | [C-] # [N+] CCCCCCCCC=CCCCCCCC    |
| C03 | CN1CCC (N) CC1    | CCCCCCC=CCCCCCCCCCCC (=O) OCCCC=O   | [C-] # [N+] CCCCCCCCCCCCCCCCC     |
| C04 | NN1CCOCC1         | CCCCCCCCC=CCCCC (=O) OCCCCCCC=O     | [C-] # [N+] CCCCCCCCCCCCCCCCC     |
| C05 | CCN1CCCC1CN       | CCCCC=CCCCCCCCCCCCC (=O) OCCCCCCC=O | [C-] # [N+] CCCCCCCCCCCCC         |
| C06 | NCCN1CCNCC1       | CCCCCCCCCCCCCCCCC (=O) OCCCCCC=O    | [C-] # [N+] CCCCCCCCCCCCC         |

### The forty prospectively locked designs (4 of 4)

#### Low-ranked controls

lowest-ranked designs of the same evidence class, to test whether the ranking discriminates



Predicted values are on the measured transfection scale. Steps count the preparations required before the final Ugi assembly, common to every design. T1 unseen isocyanide only, T2 unseen amine head, T3 unseen aldehyde tail, T4 both tails unseen.

Figure 13: The frozen Library 0 pilot, structures 4 of 4. This page carries the remaining high-ranked designs together with the six low-ranked controls, so it is landscape where the first three pages are portrait.